

ETAS INCA V7.1

Getting Started



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1 Introduction

The customer in the automotive industry associates an advanced, "intelligent" vehicle with comprehensive safety, a high degree of driving comfort, low fuel consumption, and low pollutant emissions. For automobile manufacturers, this means anti-lock brake and traction control systems, adaptive drive programs for automatic transmissions, map-controlled gasoline and diesel injection with adaptive controls, etc.

For developers, this means implementing complex functions and control algorithms in microprocessor-based control units, fine-tuning and optimizing these systems for different engine and vehicle types, and the time and budget pressures involved in mass production.

INCA is a measuring, calibration, and diagnostic system that provides comprehensive measuring support, aids you in all essential tasks during control unit calibration, evaluates the measured data, and documents the calibration results.

INCA:

- can be used in the vehicle, test bench, and office/lab
- consists of modular hardware and software
- can be customized for specific projects and applications
- can be customized to fit both high-end and low-cost control units
- can be integrated in the development process for control unit software.

1.1 Safety Advice

**WARNING!**

Calibration activities influence the behavior of the ECU and the systems controlled by the ECU. This may result in unexpected behavior of the vehicle and thus can lead to safety critical situations. Only well trained personnel should be allowed to perform calibration activities.

**WARNING!**

Sending out CAN messages influences the behavior of the CAN bus network and the systems connected to it. This may result in unexpected behavior of the vehicle and thus can lead to safety critical situations. Only well trained personnel should be allowed to perform CAN message sending activities.

1.2 General Description

A variety of physical variables and factors determine a vehicle's overall behavior, which results in a large number of control parameters in the control unit software. The objective in calibrating is to find the optimal values for these parameters.

INCA lets you read measured data from the control unit and the engine in parallel. The program helps you determine measured engine data such as lambda, different temperatures and voltage values, etc. With INCA, you don't just get a tool that will adapt to a variety of different control units, but also a system that will optimize a wide range of different vehicle components.

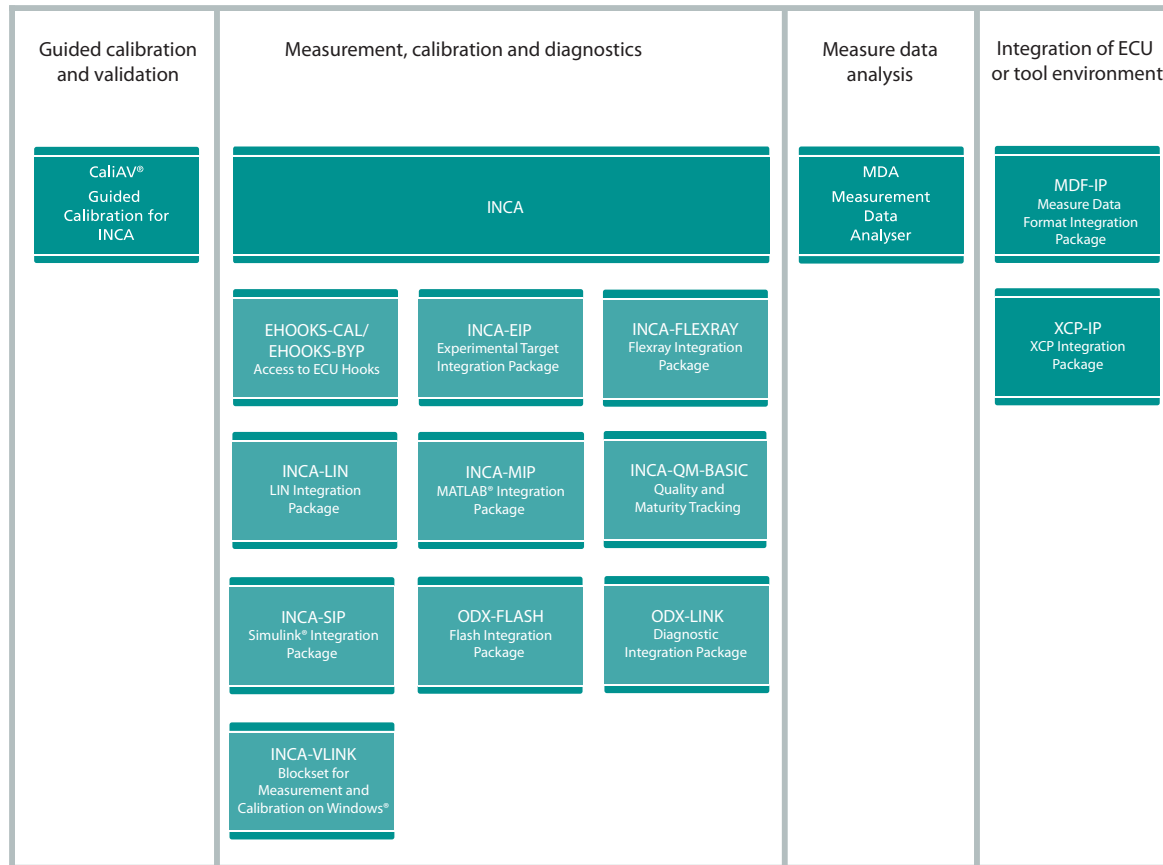
INCA is an "open system." Consistent implementation of the ASAM-MCD standard and support for data exchange formats that are established in this environment allow this program to be used for any ECU interfaces (can be customized to any manufacturer's control units) and to be integrated in existing data processing infrastructures.

INCA's modular configuration makes it possible to adapt to customer-specific requirements. The basic software provides the outer framework, which can be expanded using exactly and only those additional components that you actually need (see „Selecting the components for Installation“ on page 60). Moreover, a number of add-on modules for specific use cases is available (e.g. INCA-QM-BASIC, INCA-FLEXRAY, INCA-LIN).

In addition to that, INCA offers open interfaces which allow for the adaptation of INCA core capabilities as well as the remote control of INCA by other applications.

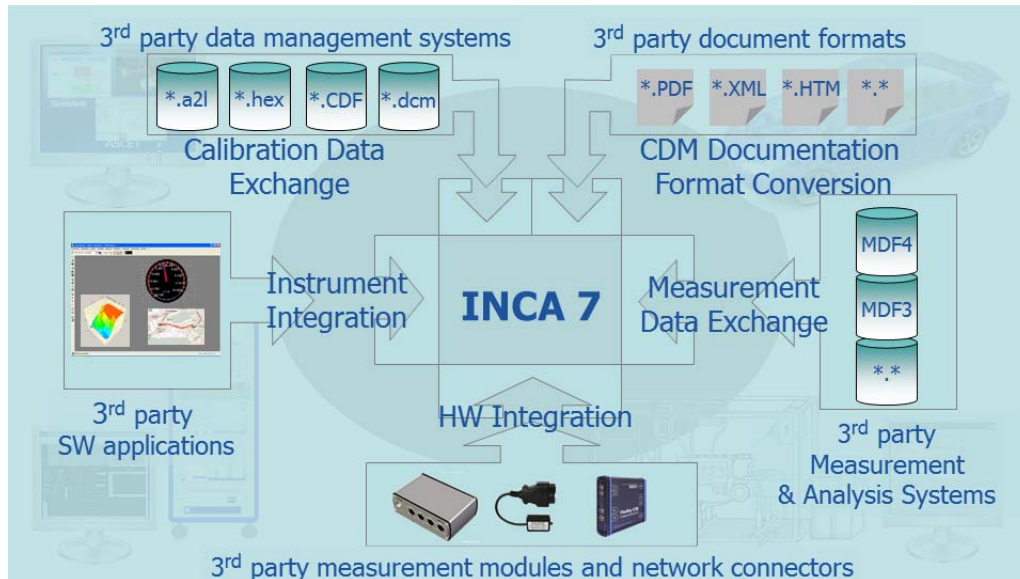
1.3 System Overview

INCA consists of a measurement and calibration core system which can be enhanced by various add-ons and customized extensions (e.g. INCA-MIP, INCA-QM-BASIC, INCA-FLEXRAY) that can be integrated in INCA. Further information is available on request.



1.4 Open Interfaces

In addition to automation system interfaces, the tools of the INCA product family provide a wide range of open and standardized interfaces for configuration, data exchange, documentation, automation, and integration with customer applications.



- INS.DK: Interface for the integration of instruments into the INCA Experiment Environment

INCA's open Experiment Environment provides an interface which allows for the integration of customer specific measure and calibration instruments for the most diverse use cases. Examples are animations, integrated web explorers, embedded model viewers, street map viewers visualizing measured courses of GPS coordinates, stimulation of parameters with generators, etc. As a consequence, customer specific data visualizations or data post processing does no longer require external applications connected via COM, MIP or ASAP3, but may be done directly in INCA.

Instrument Integration is supported through the INCA Instrument Integration Development Kit (INCA-INS.DK).

- MDF.IP: Library for measurement data exchange in standardized formats
Measure data that are recorded by INCA are written in measure data formats. For this purpose, INCA uses a component that is also available for 3rd party and inhouse tool providers to read and write measure data in standardized MDF4 format as well as in the formats MDF 3.0, MDF 3.3 and ETAS ASCII. Moreover, additional customized measure file formats can be added through a plug-in interface.

The MDF data exchange component and plug-in interface is provided through the MDF Integration Package.

- **HWI.DK: Interface for hardware integration**
INCA supports ETAS measure and calibration hardware. Moreover, it offers an open interface which enables 3rd party tool providers to add support of their hardware.
Hardware Integration is provided through the Open Hardware Integration Development Kit.
Hardware Integration is supported through the INCA Hardware Integration Development Kit and the ECU and Bus Interfaces Integration Package.
- **eCDM: Interface for the exchange of enterprise calibration data**
INCA's Enterprise CDM (eCDM) interface provides direct access to any enterprise calibration data management system from within INCA via a standardized API, thus making a manual data exchange via the file share obsolete.
eCDM functionality is provided through a separate eCDM add-on.
- **Plug-in interface for providing additional CDM documentation formats**
INCA CDM offers a number of output formats for the documentation of your calibration activities, e.g. ASCII, DCM, or HTML. By means of plug-ins, it is possible to add further converters supporting user-specific documentation formats, e.g. customized PDF output.
CDM provides the complete calibration data information in the standardized MSRSW ASAM XML intermediate format for further processing by the converters.
The plug-in interface is available with the INCA base package.

1.5 User Information

1.5.1 What You Need to Know

This manual addresses qualified personnel working in the fields of automobile control unit development and calibration. Specialized knowledge in the areas of measurement and control unit technology is required.

Basic knowledge in operating a PC and using the WINDOWS® operating system is a prerequisite. All users should be able to execute menu commands, to activate push buttons, etc. Users should also be familiar with the WINDOWS file storage system, especially the relationship between files and directories. Users have to know how to use the basic functions of WINDOWS Explorer. Moreover, users should be familiar with the "drag-and-drop" functionality.

1.5.2 How this Manual is Organized

This INCA manual consists of the following chapters:

- **Chapter 1: "Introduction"** (this chapter)
This chapter outlines the possible INCA (INtegrated Calibration and Acquisition Systems) applications. Furthermore, it contains general information such as user and system information.
- **Chapter 2: "New Features in INCA V7"**
This chapter contains a summary of the new features and changes.
You should read this section even if you are an experienced INCA user.
- **Chapter 3: "Installing the Program"**
The chapter titled "Installing the Program" is for all users who install, maintain or uninstall INCA on a PC or a network as well as system administrators who provide INCA on a file server so that the program can be installed via the network. It contains important information on the scope of delivery, hardware and software requirements for stand-alone and network installations and the preparation required for installation. The chapter also describes the procedures used to install and uninstall INCA.
- **Chapter 4: "ETAS Network Manager"**
This chapter explains the configuration of ETAS networks and the connection of such networks to your computer's Ethernet interface.
- **Chapter 5: "Getting Started"**
This chapter provides the user with a quick introduction to the program concept of INCA. You will receive an overview of the program functionality and working principles by means of practice-oriented work examples presented as flow diagrams.
- **Chapter 6: "Understanding INCA"**
This theoretic chapter provides an introduction to the calibration work and describes the use, structure, and functionality of INCA.
This chapter consists of the following subchapters:
 - Chapter 6.1: "Calibration Basics"
Explanation of the calibration types using ETK, CAN, and KWP2000/McMess as well as an overview of the ETAS calibration hardware.

- Chapter 6.2: "INCA Basics"
General information on the INCA functionality and description of the program concept.
- Chapter 6.3: "Product Structure from a User's Point of View"
This subchapter describes the modular structure of INCA, supported standards, and the resulting "openness of the system." Moreover, this subchapter explains the object-oriented database.

- **Chapter 7: "General INCA Operation"**

This chapter provides information on the window and menu structures, control options using the mouse and the keyboard, and the help system.

- **Chapter 8: "Glossary"**

The chapter titled "Glossary" explains all the technical terms used in the manual. The terms are listed in alphabetical order.

- **Chapter 9: "Appendix A: Troubleshooting INCA Problems"**

This troubleshooting chapter gives some information of what you can do when problems arise during your work with INCA.

- **Chapter 10: "Appendix B: Troubleshooting General Problems"**

This troubleshooting chapter gives some information of what you can do when problems arise that are not specific to an individual software or hardware product.

- **Chapter 11: "Appendix C: Reference Lists"**

The appendix entitled "Reference Lists" contains information on troubleshooting, the directory structure, and the reference files required. This chapter also includes a list of all keyboard commands sorted by working windows.

- **Chapter 12: "Appendix D: Further Reading"**

This appendix lists all additional documentation that is provided together with INCA and indicates where you can find further information.

1.5.3 Getting More Information

Together with INCA the INCA tutorial document is also installed; Precondition: You select the installation of the manuals during installation. The tutorial is available as PDF file and can be called up from INCA via the menu item **? → Manuals and Tutorials**. The tutorial is mainly intended for INCA beginners. On the basis of examples you will learn the operation of INCA. The total contents are divided up into short progressively structured teaching units. Before you start on the tutorial, we recommend studying the chapter "Understanding INCA" on page 114.

Subjects which can easily be shown instead of being described in words are presented as video tutorials. The videos are installed locally on your PC if you selected the installation of video tutorials during installation. They are available as MP4 files and can be called up from INCA via the menu item **? → Video Tutorials**. If you did not install them on your PC, you can either view them in the ETAS channel on YouTube, or you can download them from the ETAS Download Center.

In the INCA online help, you can find further detailed information on the INCA base program, INCA tools and on the configuration of your hardware. Information on using the online help can be found in Chapter "Help Functions" on page 147.

Further information on special subjects can be found in additional documents that are also provided with INCA. These documents are listed in "Appendix D: Further Reading" on page 186.

1.5.4 Using this Manual

Presentation of Information

All actions to be performed by the user are presented in a so-called "Use-Case" format. This means that the objective to be reached is first briefly defined in the title, and the steps required to reach the objective are then provided in a list. This presentation appears as follows:

Definition of Objective:

Any preliminary information...

- Step 1
Any explanation for Step 1...
- Step 2
Any explanation for Step 2...
- Step 3
Any explanation for Step 3...

Any concluding remarks...

Specific example:

To create a new file:

When creating a new file, no other file may be open.

- Choose **File** → **New**.
The "Create file" dialog box is displayed.
- Type the name of the new file in the "File name" field.
- Click **OK**.

The new file will be created and saved under the name you specified. You can now work with the file.

Typographic Conventions

The following typographic conventions are applied:

Choose File → Open .	Menu options are printed in bold, blue characters.
Click OK .	Button labels are printed in bold characters.
Press <ENTER>.	Key commands are printed in small capitals enclosed in angle brackets.
The "Open file" dialog box appears.	The names of program windows, dialog boxes, fields, etc. are enclosed in double quotes.
Select the <code>setup.exe</code> file.	Text strings in list boxes, in program code and in path and file names are printed using the <code>Courier</code> font.
A conversion between Logic and Arithmetic data types is <i>not</i> possible.	Emphasized text portions and newly introduced terms are printed in <i>italic</i> font face.

Important notes for the users are presented as follows:

Note

Important note for users.

List of Abbreviations

This manual uses the following abbreviations:

- **INCA** – **IN**tegrated **C**alibration and **A**cquisition Systems
- **ECU** – **E**lectronic **C**ontrol **U**nit
- **ETK** – **E**mulator-**T**astkopf (emulator test probe)
- **CAN** – Bus System for Data Communication (**C**ontroller **A**rea **N**etwork)
- **CCP** – **CAN** **C**alibration **P**rotocol, standard protocol based on MCD-1a
- **DPRAM** – **D**ual-**P**ort-**RAM** (also DPR)
- **CEBRA** – **C**ommon **ETAS** **B**ackplane for **R**emote Tool **A**ccess
- **CDM** – **C**alibration **D**ata **M**anager
- **WS** – **W**ork **S**pace
- **DBM** – **D**atabase **M**anager
- **EE** – **E**xperiment **E**nvironment
- **HWC Editor** – **H**ardware **C**onfiguration **E**ditor
- **MDA** – **M**easure **D**ata **A**nalyzer
- **EXP** – **E**xperiment
- **WP** – **W**orking **P**age
- **RP** – **R**eference **P**age
- **XCP** – **eX**tended **C**alibration **P**rotocol

2 New Features in INCA V7

This section contains a summary of the new features that have been introduced in INCA V7. You should read this section even if you are already an experienced INCA user. The latest changes are listed on top.

2.1 New Features in INCA V7.1

This section contains a summary of the new features that have been introduced with INCA V7.1.

2.1.1 Overview

The following functional extensions and improvements are contained in INCA V7.1:

- **General Changes**
 - Performance Improvements (2.1.2)
 - Standards: Support of New Versions and Features (2.1.3)
- **Experiment Environment**
 - New YT and XY Oscilloscope (2.1.4)
- **Calibration Data Manager (CDM)**
 - Using CDM and Experiment Environment in Parallel (2.1.5)
- **Hardware Support and Configuration**
 - Support of New Hardware (2.1.6)
- **Other New Features and Usability Improvements**
 - Support of ECUs Without Memory Segmentation (2.1.7)
 - Enhancements in the Testbed Interface (ASAM-MCD 3MC) (2.1.8)
 - Video Tutorials (2.1.9)
 - Other New Features and Usability Improvements (2.1.10)
- **Major Changes in Add-Ons and Software Development Kits (SDK)**
 - ODX-LINK Add-On: Simplified Configuration (2.1.11)
 - HWI.DK: New Interface for Hardware Integration (2.1.12)
 - New Features in Other INCA Add-Ons (2.1.13)

2.1.2 Performance Improvements

One of the key topics in the development of INCA V7.1 was the improvement of system performance as it relates to the handling of ECU projects with a large amount of measurement and calibration variables.

- **Optimized usage of resources:**
When acquiring data, INCA V7.1 utilizes all available processors of a multi-core PC. The display of the new INCA oscilloscope (see „New YT and XY Oscilloscope“ on page 21) is handled by the graphics card, which unburdens the PC's processor.
- **Experiment Environment:**
The Experiment Environment now opens sizeable projects decidedly faster. Moreover, resource management for experiments with several layers has been optimized which makes it possible to use a considerably larger number of variables in an experiment.
- **Variable Selection Dialog:**
Opening and working with the Variable Selection dialog has been noticeably expedited.
- **Calibration Data Manager:**
Any configuration step which will update the view (e.g. activating a filter, adding a dataset or selecting an action) is prepared in the background, and you can go on configuring without any waiting time.

This behavior can be enabled or disabled in the CDM settings under [General → Prepare in background](#).
- **Data Exchange with PaCo and CDF V2.0:**
You can select whether the data exchange files shall be optimized for readability or for performance.

The corresponding INCA user option can be set under [Data Exchange → PaCo / CDF V2.0 → XML Layout](#).
- **Data Analysis:**
During recording it is possible to index measurement files, thus making opening and working with the respective data in the MDA considerable faster.

File indexing can be enabled or disabled in the INCA user options under [Experiment → Measure → General → Write index to measure file](#).
- **MCD-3MC Configuration:**
The time for configuring ASAM-MCD 3MC sessions has been optimized, e.g. by loading only those database objects that the testbed needs, or offering the possibility to keep the INCA experiment open for manual interactions.

The desired behavior can be set in the [Options](#) dialog of the ASAM-MCD 3MC interface for INCA.

2.1.3 Standards: Support of New Versions and Features

INCA V7.1 supports the following new versions and/or features of standards:

- **FIBEX V3.x**

INCA-FLEXRAY V7.1 additionally supports the FIBEX versions V3.0 and V3.1.

- **XCP V1.1 / V1.2**

INCA V7.1 supports the following features of the XCP versions V1.1 and V1.2:

XCP V1.1:

- WRITE_DAQ_MULTIPLE
- EV_TIME_SYNC
- SLEEP
- TIMESTAMP_MODE
- ERR_RESOURCE_TEMPORARY_NOT_AVAILABLE
- Multiple Transport Layer Instances

XCP V1.2:

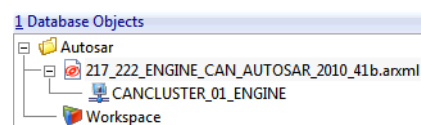
- DYNAMIC_DAQ raster check

- **Using AUTOSAR System Templates**

In terms of CAN monitoring, INCA now also supports, aside from DBC files, AUTOSAR system configuration templates (AUTOSAR V3.1 and V3.2).

AUTOSAR (AUTomotive Open System ARchitecture) files are description files in XML format (*.arxml). They describe the AUTOSAR system, e.g. the system topology including data busses and clustering, the software components, and the hardware. INCA uses only the CAN bus descriptions (CAN clusters) from the AUTOSAR system templates.

The original system template will be added to the Database Objects list of the INCA database. All CAN bus descriptions included in the AUTOSAR file will be listed separately under the AUTOSAR entry.



2.1.4 New YT and XY Oscilloscope

INCA V7.1 supports new YT and XY oscilloscopes. These replace the previous, outdated oscilloscopes.

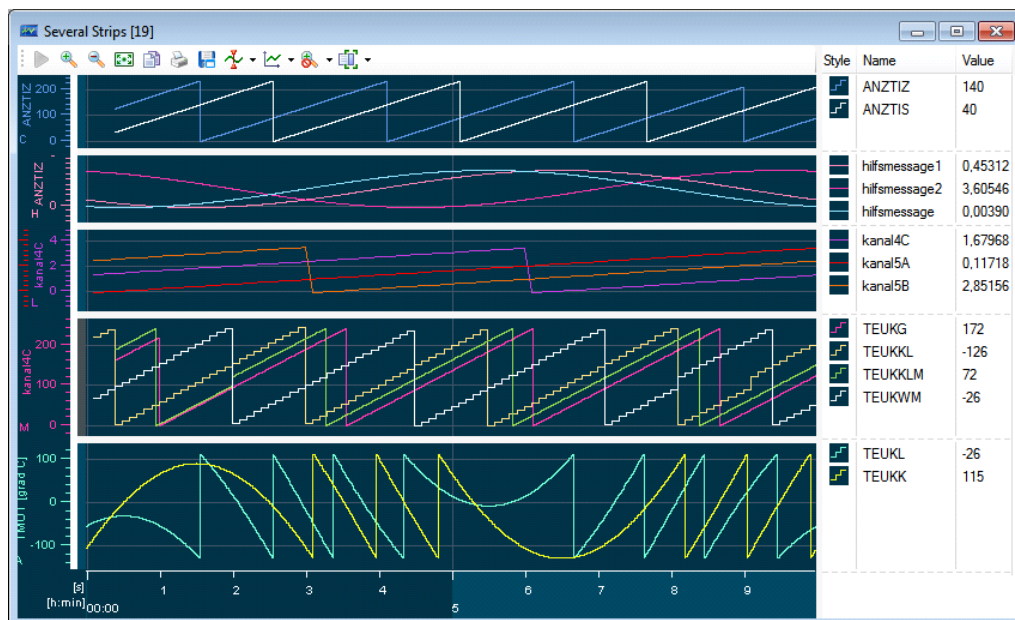
The new oscilloscopes support all use cases of the previous oscilloscopes, added by many further enhancements.

Note

The oscilloscope has been designed for maximum flexibility. It is within the user's responsibility to use this flexibility in an appropriate and purposeful way.

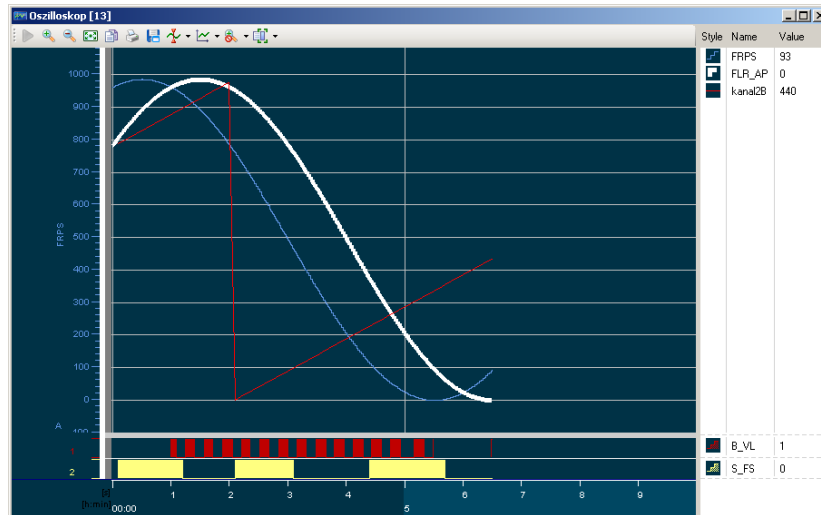
Highlights:

- The main advantages of the new oscilloscopes are better performance, the ability to display more variables in a single oscilloscope and the ability to include more oscilloscopes within a single experiment; the number of oscilloscopes and variables within one oscilloscope is mainly limited by the system resources.
- The oscilloscope has its own toolbar to provide easy access to the most important functions.
- The new oscilloscope makes it possible to dissect the display of analog measurement values into several horizontal strips, enabling vertical stacking of the display with a variety of signal groups. You can adjust the height of each analog strip. All strips are time-synchronized, i.e. they use the same time axis. The signals can be assigned to individual or common Y-axes as required.

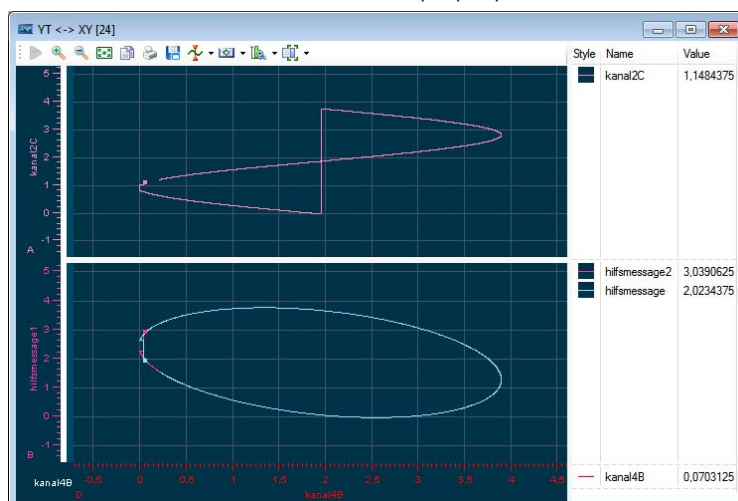


- You can either set the axes ranges manually or use dynamic axis adaptation. The latter automatically adapts the axis range to the current measured values so that the oscilloscope always shows the entire measurement range.

- The oscilloscope has different sections for analog and Boolean signals. When you add new signals, these will automatically be added to the corresponding section according to their definition in the A2L file, but you can explicitly move analog signals into the Boolean section or Boolean signals into the analog section.

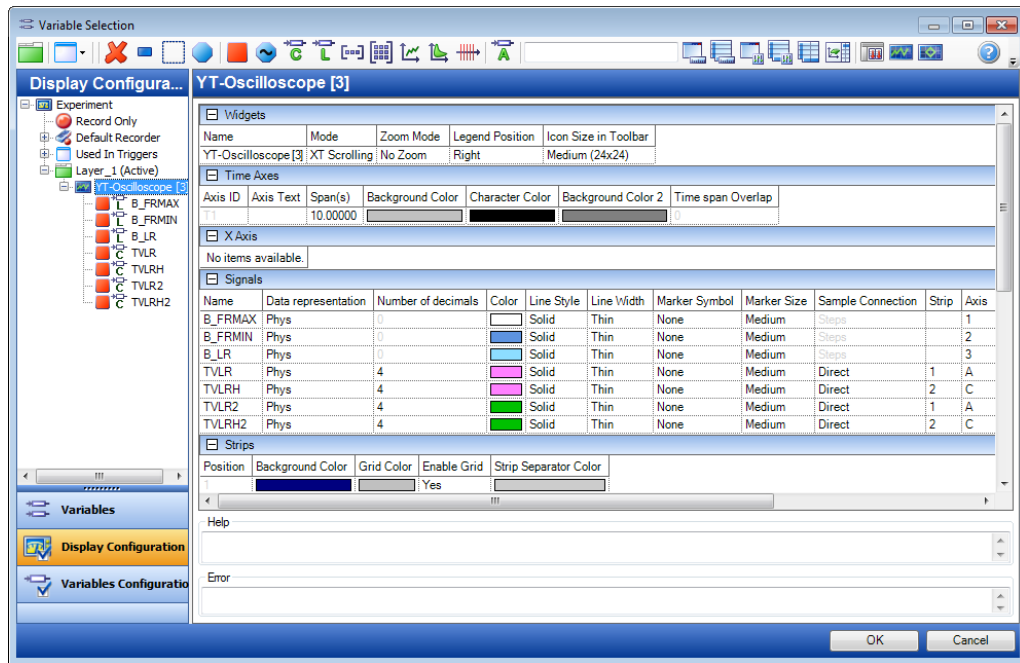


- To analyze measurement data, the oscilloscope's display can be stopped and scrolled back during an ongoing measurement session (for up to 125 s). In this way, cursors aid in the quick evaluation of earlier measurement values. Upon leaving the analysis mode, the current measurement values are again automatically displayed.
- The number of possible cursors is not limited. The values at each cursor and also the differences between the cursor values can be displayed in separate columns in the signal list next to the strips.
- For triggering the display in an oscilloscope, you can use any signal used in the oscilloscope including calculated signals; repetitive triggering is also possible.
- You can easily change a YT oscilloscope into an XY oscilloscope simply by dragging and dropping the respective signal to the X axis or by selecting XY mode in the toolbar or oscilloscope properties.



Configuration of Oscilloscopes

YT and XY oscilloscope properties can be centrally managed in the Display Configuration of the Variable Selection dialog. To open the Display Configuration, open the context menu of the oscilloscope and select **Properties**.

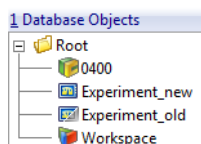


Compatibility and conversion

INCA V7.1 supports oscilloscopes of both the old types (outdated) and of the new types introduced with this version. In future versions of INCA, the outdated oscilloscopes will no longer be supported.

Within one experiment only one type of oscilloscopes can be used. Whether new experiments will support new or outdated oscilloscopes can be set in the user options under **Experiment → Measure → General → Oscilloscope type for new experiments**.

However, you can have different experiments in your database, where some contain exclusively oscilloscopes of the new type and others exclusively oscilloscopes of the outdated types.



You can tell which oscilloscope types are supported by an experiment by looking at the symbol:

 (color): The experiment only supports new oscilloscopes introduced with INCA V7.1.

 (half gray): The experiment only supports the outdated oscilloscopes.

When you open in INCA V7.1 an experiment for the first time which supports or even contains old oscilloscope types, you will be asked whether the experiment shall be converted to support new oscilloscope types exclusively.

Alternatively, you can convert experiments manually by selecting the experiment that you want to convert and then selecting one of the following menu options:

- **Edit → Convert Experiment → Convert to INCA V7.0 (replace oscilloscopes)**
- **Edit → Convert Experiment → Open experiment and update oscilloscopes.**

Note

The conversion of oscilloscopes contained in such an experiment is not a 1:1 conversion. For details see the online help in the Experiment Environment and search for "compatibility oscilloscope".

Note

Detailed information on the usage of the new oscilloscopes can be found in the INCA online help, e.g. by selecting an oscilloscope and clicking <F1>.

2.1.5 Using CDM and Experiment Environment in Parallel

In INCA V7.1 you can simultaneously open both Experiment Environment and Calibration Data Manager (CDM) to compare current calibration data in the CDM with the data from the reference page. Moreover, it is possible to compare and copy several sets of working data in the CDM concurrently with the open Experiment Environment. This has the following advantages:

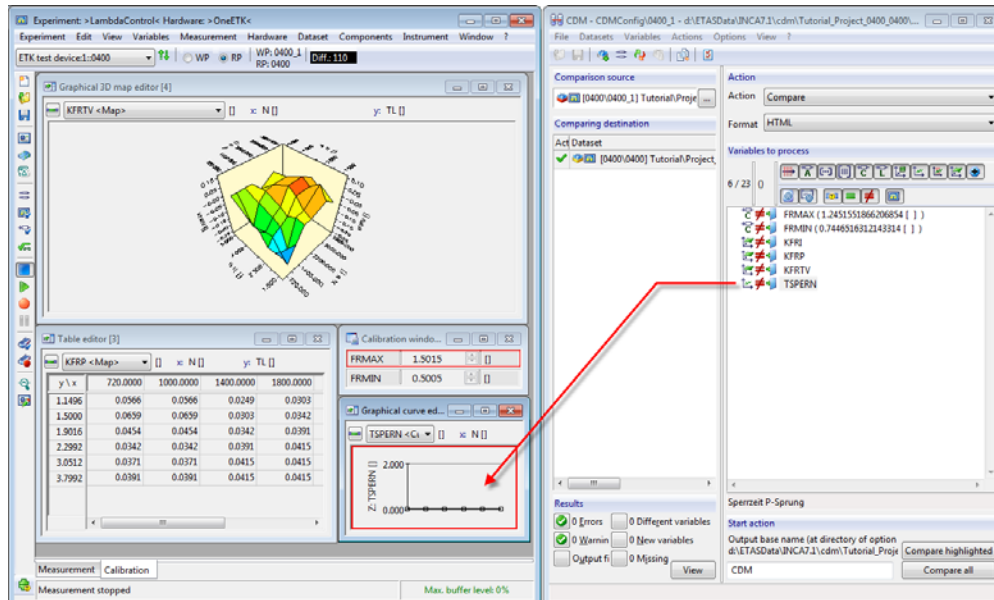
- In the CDM, you can compare working page and reference page on variable or function level instead of seeing only the amount of differences in Bytes.
- You can select concrete subsets of variables to write them into data exchange files, instead of having to write all variables.
- When you read in data exchange files, you can show the differences to the working page or reference page in advance in the CDM before loading them in the Experiment Environment.

Starting the CDM from the Experiment Environment

You can start the Calibration Data Manager directly from the experiment by selecting **Components → Calibration Data Manager**. Which variables will be used automatically in the CDM (e.g. the variables currently used in the experiment) can be set in the user options under **Experiment → Calibration → General → Variable Selection in CDM**.

If you open a calibration variable for editing by double-clicking it in the CDM, then the experiment window instead of the CDM Variable Viewer will be activated for calibration. If the variable is already used in the experiment, the corresponding layer and instrument will be activated. If the variable is not yet part of the experiment, a new layer named Calibration Data Manager with a new cali-

bration instrument containing that variable will be created. If you calibrate variables in the Experiment Environment, the changes will automatically be sent to CDM.



2.1.6 Support of New Hardware

INCA V7.1 supports the following new ETAS hardware:

- ES930 Multi-I/O Module
- ES415 A/D Module with Sensor Supply

Moreover, INCA V7.1 supports the following third party bus interface devices:

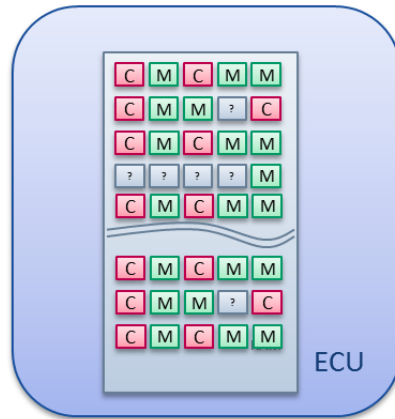
- Kvaser LEAF Professional
- Kvaser MEMORATOR Professional
- Kvaser USBCAN Professional
- Vector VN16xx modules: VN1610, VN1611, VN1630 and VN1640
- Vector VN8910 and the plug-in modules VN8950 and VN8970

Note

The operation of these devices with ETAS software must be enabled by ETAS. Please contact your local sales representative to get the enabling key.

2.1.7 Support of ECUs Without Memory Segmentation

INCA now supports ECUs where both measurements and calibrations are located in the same part of the RAM and where neither separate memory pages nor checksum calculation are supported.



Consequently, the A2L files describing such an ECU do not contain any information on memory segments. Project descriptions containing a mixture of both measurement and calibration variables without segmentation are represented in INCA by the following icon:

When adding an A2L file without information on memory segments to the INCA database, an empty dataset will be automatically created. The dataset(s) originating from this project description may contain fill bytes representing information that is not available in the A2L file. This is indicated by the following icon:

Download Behavior

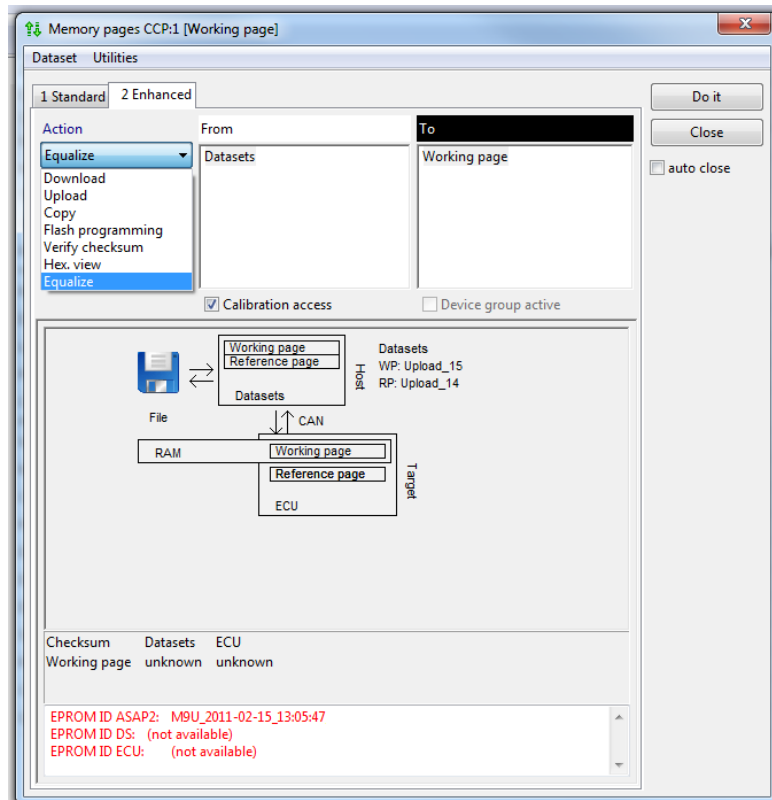
These frame conditions have an influence on the download behavior: If the dataset contains both valid and invalid calibrations (i.e. datasets with unknown calibration values), it is not possible to download the complete dataset to the control device memory as it is.

In the Memory Page Manager, you have the following options:

- You can download only all valid calibrations to the control device memory. This makes sure that your calibration results are available in the ECU; however, the dataset still contains invalid calibrations.
- You can upload all invalid calibrations from the control device memory, thus replacing both valid and invalid calibrations with the contents from the ECU memory. This makes sure that all calibrations in the dataset are valid; however, your calibration results are overwritten in the dataset.

- You can download the valid calibrations to the ECU and replace the invalid calibrations with calibrations from the ECU in one single step by selecting the action **Equalize**.

The Equalize action merges the contents from the dataset and the control device memory: It downloads the valid calibrations to the control device memory to make them active. At the same time, it replaces the invalid calibrations in the dataset with valid calibrations from the control device. As a result, all calibrations become valid.



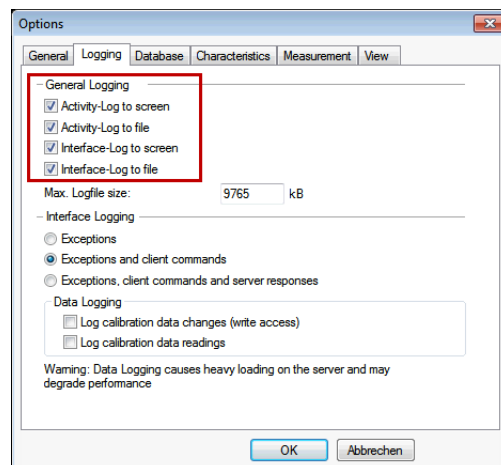
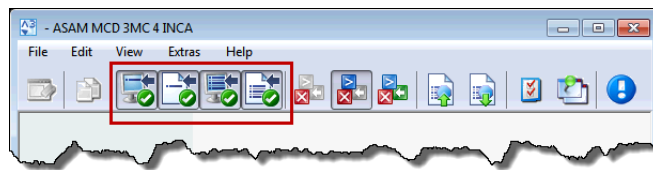
2.1.1.8 Enhancements in the Testbed Interface (ASAM-MCD 3MC)

The ASAM-MCD 3MC interface for INCA provides a number of features enhancing usability and performance:

- **Enhanced logging:**

You can select both the logging level and whether the log information shall be provided on screen and/or in log files. The activity log describes high-level protocol activities and enables testbed users to solve issues themselves. The interface log describes the low level protocol activity and represents the MCD3 COM interface traffic.

You can select the desired logging level in the toolbar, in the **View** menu, or in the Options dialog on the **General** tab.



- **Database Filter Extensions:**

The **Database** tab of the **Options** dialog optimizes the access of relevant database items:

- **Use only the current workspace**

Defines that only those projects and datasets are searched for that are defined in the workspace of the currently selected MCD project. This reduces the required database queries and thus the performance, and facilitates the selection of the relevant Logical Link.

- **Skip write protected datasets**

Defines that write-protected datasets will be ignored upon searching for projects and datasets. This ensures that only those datasets are returned that can be assigned as working page, thus avoiding uncontrolled growing of the database by creating writable copies of write-protected datasets.

- **Automatic download at Reconnection**

To support power fail scenarios on testbeds, the parameter **Connection Behavior** now supports a new option **Reinitialize automatically and download the working page**. This ensures that the hardware is reinitialized automatically after the connection is restored after a power failure. INCA checks whether the checksums in INCA and in the ECU are still equal. If the checksums differ, INCA automatically downloads the dataset to the ECU.

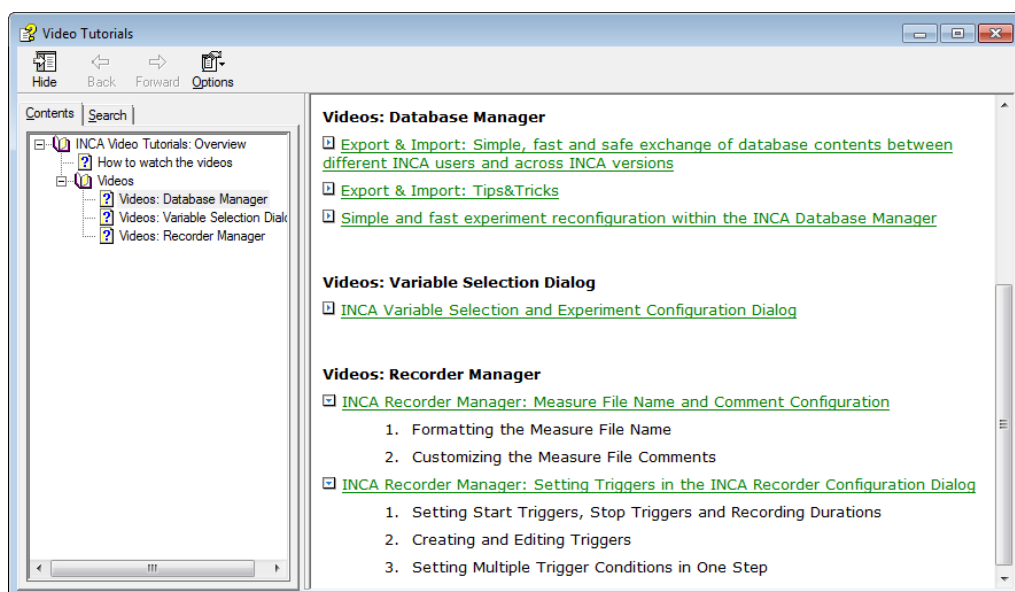
- **MCE optimization**

For a better MCE support, the user options provide on the **General** tab the new option **If available, always use FULI device**.

This option defines whether INCA shall attempt to configure all measurements and calibrations used by the MCD3 client automatically for INCA-MCE usage, thus maximising the reuse of existing testbench automation scripts. Using this mode the FULI LUNs are not available.

2.1.9 Video Tutorials

INCA provides short video tutorials which demonstrate specific features and tips & tricks. An overview leading to the individual videos can be opened from the Help menu by selecting **? → Video Tutorials** or from the product DVD.



Note

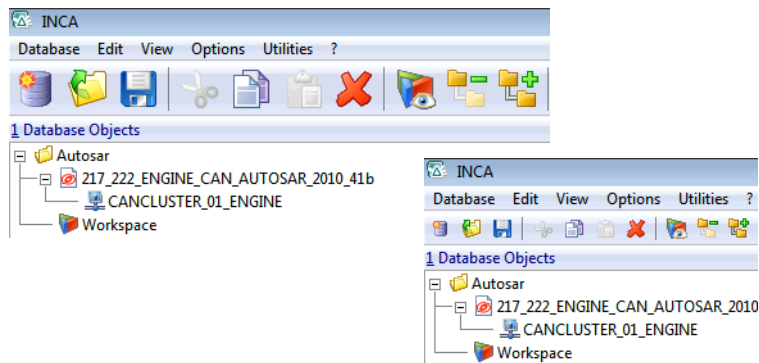
ETAS continuously creates further video tutorials, which can be downloaded from the [ETAS website](#), or viewed on the [ETAS YouTube channel](#).

2.1.10 Other New Features and Usability Improvements

INCA V7.1 includes additional useful features for improving the usability of major tasks:

- **Toolbar Icon Size**

You can select your preferred size of the toolbar icons in the INCA user options under **General → Toolbar Icon Size**. This setting affects the toolbars in the individual INCA windows (Database Manager, Experiment Environment, Hardware Configuration Editor, etc.) and the action icons in the Database Manager. The toolbar icon size of the new oscilloscopes can be selected in the corresponding oscilloscope settings.



- **Avoiding conflicts between MC (measurement and calibration) and prototyping**

If a bypass setup uses measurement rasters for prototyping purposes, conflicts in ECU access can arise between INCA and the prototyping application.

To avoid such conflicts, you can define in the user option **Hardware → ETK steal measurement rasters** whether measurement rasters are allocated for INCA measurement automatically, even if they are in use by another application.

- **Fast flashing for ECUs with limited emulation RAM**

When the ECU supports only limited emulation RAM, it is sometimes necessary to freeze the data (i.e. to flash the ETK data). INCA supports an ETK data flashing mechanism that works without a PROF configuration; instead the procedure works like a data download.

For security reasons, it is possible to define a Data Freeze safety mailbox in the A2L file. This safety mailbox makes INCA check if the ECU is in a safe state which allows flashing. In case no safety mailbox is used, INCA will only execute the operation if the user confirms the ECU safe state or the ECU reset state change.

You can set in the INCA user options under **Hardware → Show Data Freeze Message Boxes** if another message box (in addition to the Data Freeze safety mailbox) will appear asking you to confirm the ECU safe state or reset state change.

Note

This fast flashing mechanism is only available if the safety mailbox is configured in the A2L file and supported by the ECU. Otherwise the setting in this option will be ignored.

- **Coarse and fine steps for calibration**

For manual calibration, INCA supports the functions **Increment** (<F6>) and **Decrement** (<F7>). The basic step size for incrementing or decrementing the value of calibration variables is set in the INCA user options under **Experiment → Calibration → General → Float INC/DEC step** and **Integer INC/DEC step**, or in the properties of a calibration instrument. If you want to use a larger step size, you can use the option **Coarse INC/DEC factor** to set a multiplier and use the key combinations <SHIFT> + <F6> or <SHIFT> + <F7> for applying larger incrementation and decrementation steps.

- **Device information in MDF measure files**

You can save the available information on the device or system along with the measured data in the MDF measure file. Saving this information is supported only for measure files in the MDF 4 format. It can be enabled or disabled in the INCA user options under **Experiment → Hardware → Show hardware device information in MDF measure file**.

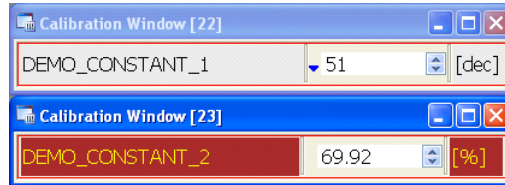
- **Showing / Hiding the Unit in Measure and Calibration Instruments**

As a space saving measure in the Experiment Environment, the display of physical units of measure and calibration variables can now be toggled on and off also in measure tables and scalar table editors.

The corresponding display settings are available in the global INCA user options for the corresponding instrument as well as in the instrument properties.

- **Emphasis for calibration variables**

INCA V7.1 now makes it also possible to display scalar calibration variables with color highlights (as already possible for scalar measurement variables). The colors can be set for each variable in the Display Configuration under **Signals** → **Background Color** and **Foreground Color**.



- **KWP2000 on CAN: 4-byte memory addresses and 2-byte length keys**

KWP2000 on CAN measurement in INCA makes use of the readMemoryByAddress service of the KWP2000 protocol. The service needs to be parameterized with an ECU memory address and the number of bytes to be read (address and length parameter).

Additionally to ECUs that need 3-byte memory addresses and 1 byte length key parameters, INCA now supports ECUs that require 4-byte memory addresses and 2-byte length keys.

You can switch between 3-byte and 4-byte measurement mode through the parameter KWP2000_DIA in the file

ETAS\Shared11\Devices\KWP2000\kwp2000.cnf:

- For 4-byte measurement:

```
KWP2000_DIA: ,          KWP_4BytesMemAddr;
```

- For 3 byte measurement (default):

```
KWPP2000_DIA: ,          KWP2000;
```

- **Virtual measurements**

ASAM MCD-2 MC describes with the keyword "VIRTUAL" a possibility to calculate measure values dependent on one or more inputs. The virtual result is only available in the MC tool (e.g. INCA), not in the ECU. INCA allows the user to use the virtual measurement variable in the same way as "normal" measurement variables for recording or input in other calculations.

- **Extended Logging and Troubleshooting**

If you encounter a specific problem and need help from ETAS for troubleshooting, exact log files might be required for reproducing and analyzing the situation properly. However, extended logging is disabled by default, because it might slow down system performance.

Selecting **? → Log Files → Extended Logging** in the Database Manager opens a dialog which provides further information on how to enable extended logging for better cause analysis in case of problems.

2.1.11 ODX-LINK Add-On: Simplified Configuration

In ODX-LINK, the INCA diagnostics add-on, the required configuration effort was significantly reduced. INCA automatically recognizes and configures the OBD interface of the connected ECUs. Because ODX-LINK takes the required communications parameters from the ODX file, it no longer needs an A2L file. In the Experiment Environment, all OBD diagnostic values can be read from the ECU with the push of a button or automatically prior to the generation of a snapshot.

For details on the ODX add-on see ODX-LINK / ODX-FLASH User's Guide.

2.1.12 HWI.DK: New Interface for Hardware Integration

The new Open Hardware Integration Development Kit supports 3rd party developers in integration of 3rd party measurement and calibration hardware into INCA.

The SDK supports the following use cases:

- Development of bus controller drivers for accessing a CAN/LIN/FlexRay bus access hardware
- Development of ASAP1b drivers for accessing a bus device hardware
- Development of ASAP1b drivers for accessing general measurement hardware

Note

The new Open Hardware Integration Development Kit replaces the existing OHI (Open Hardware Integration) SDK.

2.1.13 New Features in Other INCA Add-Ons

The new features for the INCA add-ons INCA-FLEXRAY, INCA-EIP and INCA-MIP are described in the user's guides of the corresponding add-on.

2.2 New Features in INCA V7.0

This section contains a summary of the new features that have been introduced with INCA V7.0.

2.2.1 Overview

The following functional extensions and improvements are contained in INCA V7.0:

- **General Changes**
 - Support of Windows 7 (2.2.2)
 - Increased Database Capacity (2.2.3)
 - Enhanced Usability Through New Icons (2.2.4)
 - Support of ASAM MCD3 V2.2 (2.2.5)
 - ASAM MCD 2MC V1.6: Support of New Keywords (2.2.6)
 - Support of the New MDF V4.0 Standard (2.2.7)
- **Database Manager**
 - Import and Export of INCA Option Settings (2.2.8)
 - Introduction of Virtual Devices for Efficient Offline Preparation of the Experiment (2.2.9)
- **Experiment Environment**
 - Enhanced Dialog for Experiment Configuration (2.2.10)
 - Enhanced Recording Functionality (2.2.11)
 - Grouping of ECUs in the Database Manager, Experiment Environment and Variable Selection Dialog (2.2.12)
- **Calibration Data Manager (CDM)**
 - Plug-in Interface for CDM Documentation Formats (2.2.13)
- **Hardware Support and Configuration**
 - Automatic Firmware Check Directly from Within INCA (2.2.14)
 - Extended MCD Support for the ES9xx Family (2.2.15)
 - Measurement of GPS Data Through Support of GPS Devices (2.2.16)
 - Support of New Hardware (2.2.17)
- **Other New Features and Usability Improvements**
 - Other New Features and Usability Improvements (2.2.18)
- **Major Changes in Add-Ons and Software Development Kits (SDK)**
 - New SDK: Measurement Data Format Integration Package (2.2.19)
 - New SDK: Instruments Integration Package (2.2.20)
 - New Add-On eCDM for Enterprise Calibration Data Management (2.2.21)
 - New Add-On INCA-SIP (INCA Simulink Integration Package) (2.2.22)
 - INCA-EIP: Working Page / Reference Page Concept for ES910 (2.2.23)

2.2.2 Support of Windows 7

INCA V7.0 supports the following variants of the Windows®7 operating system:

- Windows 7 — 32 bit
- Windows 7 — 64 bit in 32 bit compliance mode

Note

The following hardware is not supported if INCA 7 is installed under Windows 7:

- *hardware connected over the parallel port (ES585/KIC, SIC, MAC, and ES1000.1)*
- *ES300 devices*

2.2.3 Increased Database Capacity

The increasing number of measure and calibration variables in A2L projects as well as the addition of quality meta data leads to increasing database sizes. For that reason, INCA V7 now allows for database sizes of more than 4 GByte.

- Databases from previous INCA versions can either be used in the INCA V5/6 type or converted to the new INCA V7 large-capacity database type.
- New databases can be created either in the V5/6 or the V7 database type. The type for new databases can be set in the INCA user options on the **Database** tab.
- The database currently opened in INCA V7 can be converted from INCA V5/6 to INCA V7 type and vice versa by selecting **Save As** from the **Data-base** menu and selecting the corresponding database type. Saving a database in the INCA V5/6 type is only possible if the database size does not exceed the 4 GB limit.

2.2.4 Enhanced Usability Through New Icons

The usability of INCA V7 has been further enhanced through the introduction of new desktop and tool icons. The new icons

- are Windows compliant,
- provide an improved high recognition value, due to consistency between use case and symbol,
- and they provide a common look&feel with other ETAS solutions for prototyping and test automation (MDA, ASCET, INTECRIO, LABCAR).

INCA can now be started via the following desktop icon:



The following comparisons illustrate the enhancements in the INCA toolbar icons:

Toolbar icons in the INCA Database Manager:

INCA V6:

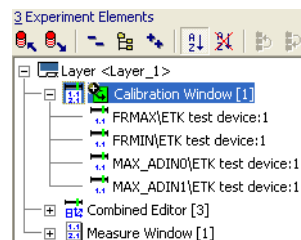


INCA V7:

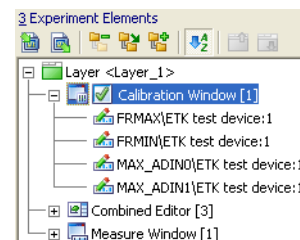


Tree view of experiment elements in the INCA Database Manager

INCA V6:

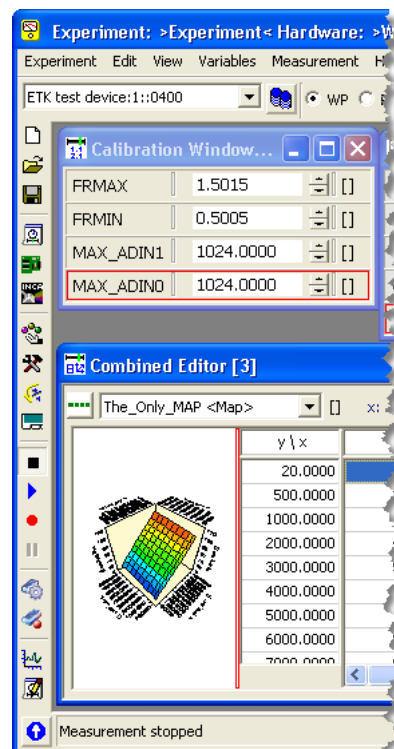


INCA V7:

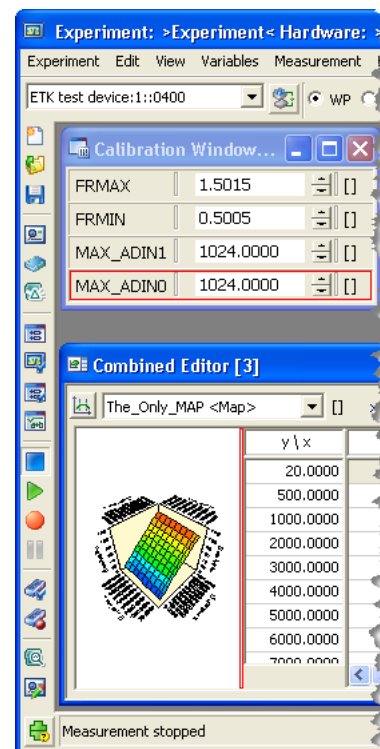


INCA Experiment Environment:

INCA V6:



INCA V7:



To ease the transition to the new icons, INCA comes with a printable list of the most important new icons, which you can find in the Documentation folder on the product DVD as well as in the Manuals folder of your INCA installation.

For more complete information on the icons used in the different INCA windows, please see the INCA online help.

2.2.5 Support of ASAM MCD3 V2.2

INCA V7 allows multi-client access via the ASAM MCD3 V2.2 interface. The benefit is increased efficiency, thanks to the ECU access via INCA V7 with an automation- and optimization tool in parallel.

To enable the new version of the standard in INCA, you first have to set it in the INCA user option **ASAM-MCD 3MC interface version** on the **General** tab.

2.2.6 ASAM MCD 2MC V1.6: Support of New Keywords

INCA V7 supports new keywords of the standard ASAM MCD 2MC V1.6, which offer significant usability improvements for the daily work:

- Transparent visualization of the sensor status in INCA V7 with the introduction of Enumeration as keyword;
- Eased work with float values through improved increment/decrement keyboard behavior thanks to enhanced step size functionality;
- Improved international A2L and CDF file exchange thanks to correct encoding of other language characters (UTF8, UTF16);
- Higher flexibility for the INCA user with the new monotony attributes.

INCA V7 supports the extensions released with the ASAP2 version, which ease the A2L file generation and modification for the SW development process:

- Easier conversion exchange between different standards, due to support of ANSI-C and MSRSW compliant operators;
- Easier modification of conversions, due to support of system constants as part of conversions;
- Less A2L file content overhead, due to the decoupling of conversion and unit;
- Increased flexibility for record layout alignment with new static record layout;
- Improved handling of microcontrollers using 64 bit data types, due to 64 bit calibrations support

2.2.7 Support of the New MDF V4.0 Standard

INCA V7 and MDA V7 support the new ASAM MDF V4.0 standard.

MDF V4.0 has been defined to resolve limitations of the currently used MDF V3.x industry standard.

The main benefits of the new MDF V4 standard are the following:

- Extension of the maximum file size (by now limited to 4 GByte)
- Support of meta-data information (e.g. the user having created the file, timestamp of the recording, etc.)
- Higher accuracy of the global start time (nano seconds) with additional information on time zone offsets and summer time offsets (UTC)

- Introduction of angle, distance or index for synchronization in addition to time
- Extension of comment length, record length, array size, and number of signals per group.

To ease the introduction of MDF4 in customer and 3rd party tool chains, ETAS offers a Measurement Data Format Integration Package (MDF-IP).

INCA V7 can write measure files in the formats MDF V3.0, MDF V3.3, and MDF V4.0; in addition to that, conversion into additional formats is supported, e.g. ETAS ASCII and MATLAB.

2.2.8 Import and Export of INCA Option Settings

INCA V7 provides an import/export mechanism for INCA option settings:

- **Exchanging INCA option settings with other persons (installation of the same INCA version on different PCs):**

In the Database Manager, you can exchange user option settings with persons by exporting the settings of the current user into a zip file (**Options → User Options → Export**) and importing them on another PC (**Options → User Options → Import**). The zip file contains files with the option settings for INCA and CDM.

This functionality is useful for defining and distributing default settings for divisions, working groups, or projects.

- **Taking over settings from previous INCA installations on the same PC:**

You can copy option settings from a previous INCA version and use them as default settings in a new INCA installation.

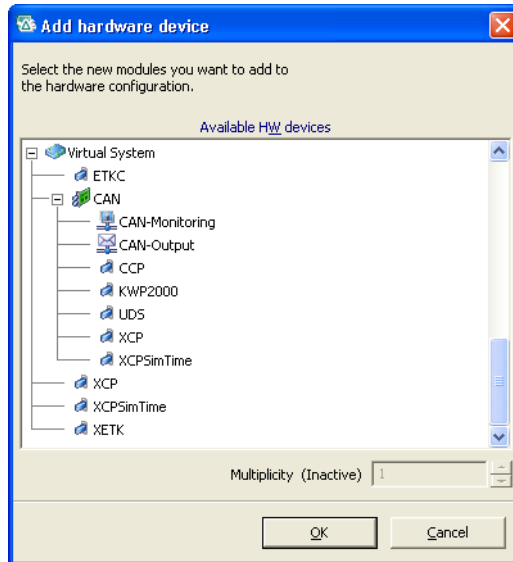
In this case, there is no explicit menu option for the copy action. Instead, a dialog box is displayed offering you different copy options when you start a new INCA version for the first time.

Before overwriting the settings with the settings from the import file, INCA generates a backup file and saves it in the User folder under your ETASData location. If you need to retrieve the original settings, you can import the saved settings from the backup file.

2.2.9 Introduction of Virtual Devices for Efficient Offline Preparation of the Experiment

In previous INCA versions, if you did not know which device would be used with the selected project, you could add a non-specific offline device to the workspace for preparing the experiment. However, this device could only be used for very limited offline-configuration; there are no rasters defined for offline devices, and therefore there is no raster check supported. As a result, connecting real hardware later on usually required additional work for mapping variables to real rasters and reducing the number of selected variables to the available raster filling level.

From INCA V7.0, you can use a virtual system instead of an offline device. One main advantage is the support of rasters and raster checks. Moreover, all main interfaces are supported.



Note

FlexRay and LIN interfaces are supported by the virtual system if the INCA-LIN and INCA-FLEXRAY add-on are installed as well.

This makes the use of a virtual system meaningful whenever the use case is clear, but the specific device used on the vehicle is not known. You can use the virtual system for efficiently preparing the experiment offline, without having to reconfigure the experiment upon connecting the actual system used.

2.2.10 Enhanced Dialog for Experiment Configuration

The **Variable Selection** dialog has been extended by a new tab for the display configuration of the experiment. You can use the **Display Configuration** tab to configure the display of instruments and variables once they have been added to the experiment via the **Variable Selection** dialog. Consequently, the **Variable Selection** dialog now contains the following 3 tabs:

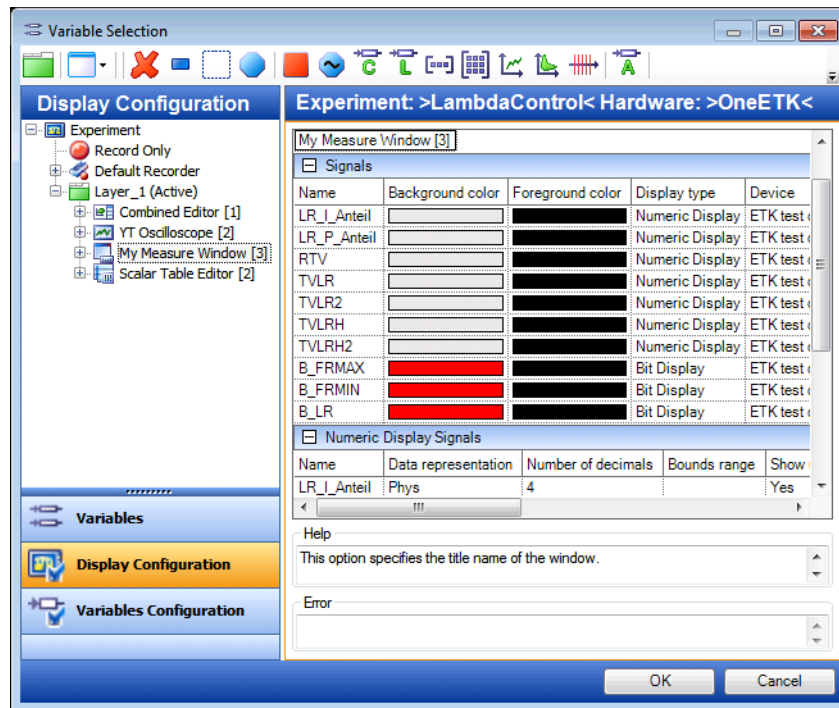
- **Variables** tab for variable selection
- **Display Configuration** tab for setting display properties of instruments and variables (e.g. display type, number of decimals, coloring)
- **Variables Configuration** tab for setting variable properties (e.g. raster assignment, assignment to recorders)

Note

The basic structure and use of the enhanced Variable Selection dialog is demonstrated in a video which you can find in the folder Manuals/Video in your INCA program directory.

Display Configuration

You can open the display configuration by clicking , pressing <F4>, or selecting **Variables → Display Configuration**.



To set the display properties, mark an instrument or a variable in the tree view on the left, then define specific display settings for that element in the table on the right.

Example use cases:

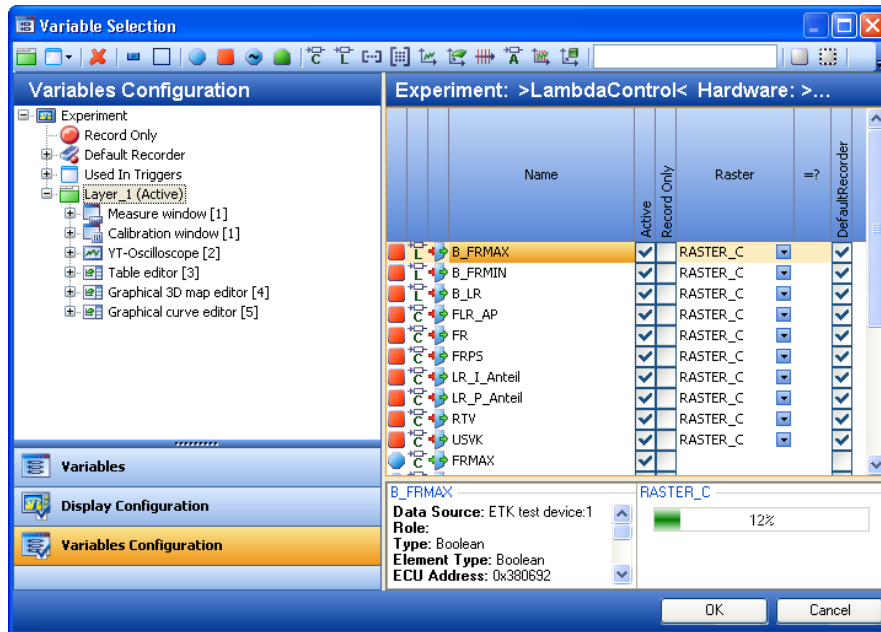
- Select the same line color for a specific variable in all oscilloscopes:
 - Filter for the variable name by entering the name in the Search field.
 - In the tree view, select all oscilloscopes.
 - In the **Signals** table, mark all signals. Then select the desired color and confirm your selection.
- Set all variables of type Boolean in measure windows to Bit:
 - Use the toolbar to filter for Booleans.
 - In the tree view, select all measure windows.
 - In the **Signals** table, mark all signals. Then set the display type to Bit.

Note

You can also move variables from one instrument to another by drag and drop or cut and paste in the tree view. Moreover, you can create layers and use drag and drop or cut and paste to move instruments between layers.

Variable Configuration

You can open the variable configuration by clicking , pressing <ALT> + <T>, or selecting **Variables** → **Variable Configuration**.



The table on the right provides an overview of the raster assignment and their assignment to recorders. The raster filling level information is also displayed in the dialog.

Example use case:

- Change the display and recording state for multiple variables at once:
 - Filter for the state of the variables, e.g. by selecting the **Record Only** item in the tree view on the left.
 - In the table, mark all variables and change their state to the desired value.

2.2.11 Enhanced Recording Functionality

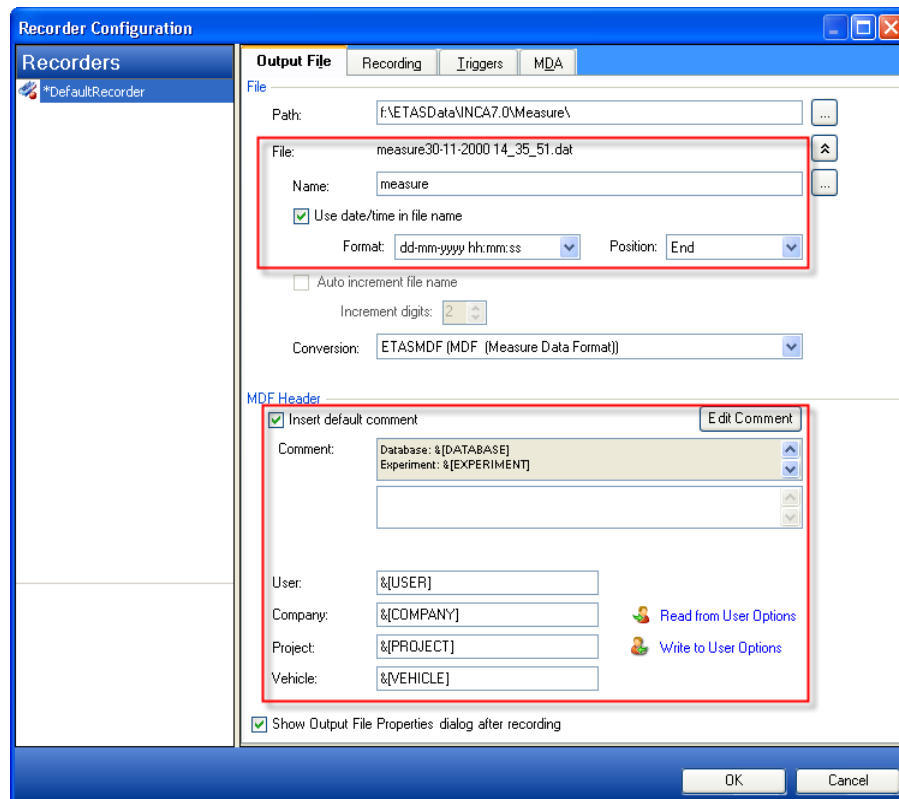
INCA V7.0 includes the following enhancements in the recording functionality, which can be configured in the **Recorder Configuration**:

- **Output File** tab: Adding time and date to the measure file name
- **Output File** tab: Using variables for meta data and default comments
- **Recording** tab: Using manual measurement event markers

Adding time and date to the measure file name

With the checkbox **Use date/time in file name**, you can determine whether the measure file name shall include the corresponding information. This allows for a better overview and for easy chronological sorting in the file explorer. In addition to the date/time format used, you can also determine whether the time/date information shall be appended at the beginning or at the end of the file name.

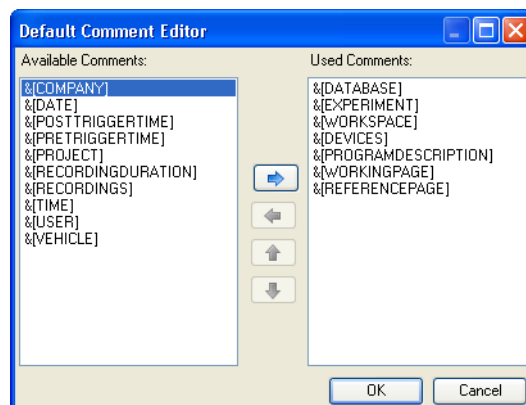
The **File** field above the checkbox will display a preview of the measure file name that results from the settings on the **Output File** tab.



The presets for time and date in measure file names can also be set in the INCA user options under **Experiment → Measure → General**.

Using variables for meta data and default comments

INCA V7.0 offers a number of variables which you can add to the default comment by clicking the **Edit Comment** button. The variables that can be used in the default comment refer to system information (time, date), settings from the INCA user options (e.g. user, vehicle), the INCA database (e.g. database, experiment, working page) and to the Recorder Configuration (e.g. posttrigger time, recording duration).



In the fields below the comment field, you can either enter the data manually, or you can use the corresponding variables (&[USER], &[COMPANY], &[PROJECT] or &[VEHICLE]). Using variables, the corresponding information will be taken from the INCA user options and written to the measure file when the recording stops. This is especially useful if the experiment is shared by several users.

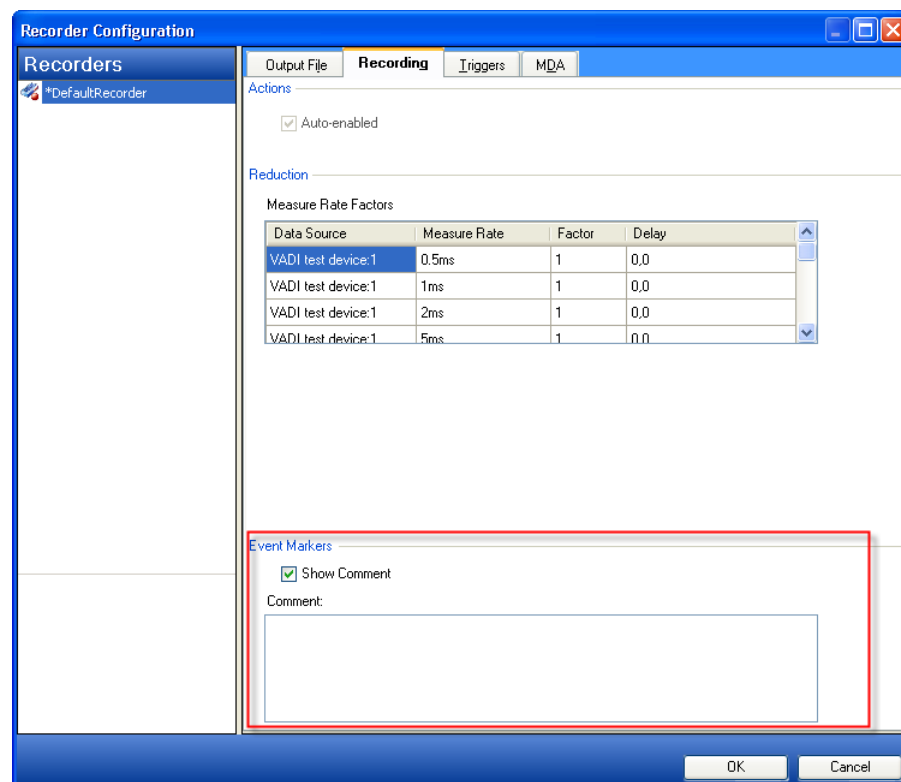
The actual values represented by a variable is displayed in the tooltip when the mouse hovers over the variable.

The entry fields **Project** and **Vehicle** store the last ten entries of a user. If you enter letters and they correspond to one of the last entries, a drop-down menu opens from which one of the last ten entries can be selected.

Using manual measurement event markers

The INCA V7 Recorder Configuration provides a field on the **Recording** tab for entering comments for event markers. The comment in this field will be used as default comment for measurement event markers.

When you disable the **Show Comment** checkbox, the **Insert User Comment** dialog box used for inserting measurement event markers will be suppressed, thus avoiding user interactions during measurement. In this case, the default comment provided in the **Comment** field will be assigned to the measurement event marker.

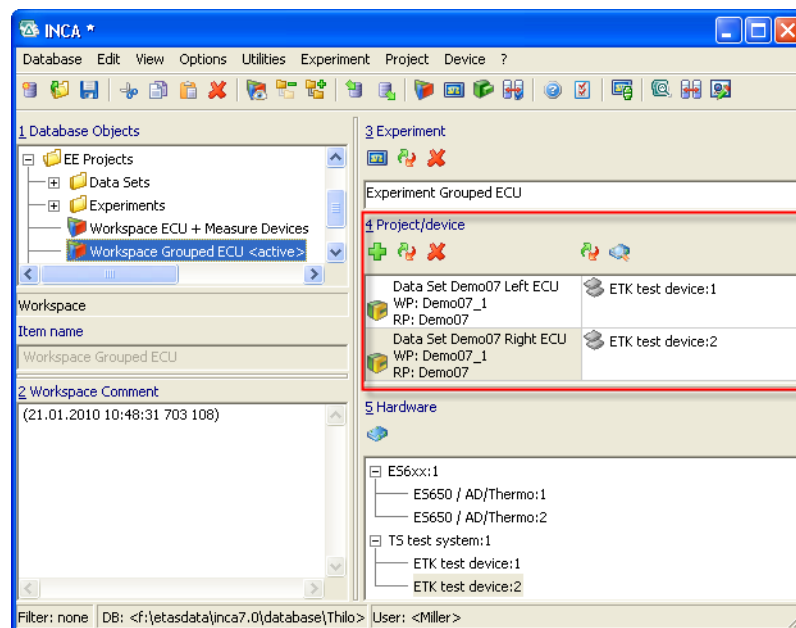


2.2.12 Grouping of ECUs in the Database Manager, Experiment Environment and Variable Selection Dialog

Grouping of ECUs enables joint calibration of all control devices in an ECU device group. The functionality for the grouping of ECUs has been enhanced in the following aspects:

- **Grouping of ECUs in the Database Manager**

It is now possible to enable and disable grouping functionality also in the Database Manager. To add an ECU device to a group, you must mark it in the **Project/device** list of the Database Manager and select **Group Device** from the context menu. This step must be repeated for all other ECU devices that shall belong to the group. All grouped devices will belong to the same device group.



- **Grouping of ECUs in the Experiment Environment**

Calibrations made on a calibration variable of a grouped ECU are applied to all ECUs of the device group.

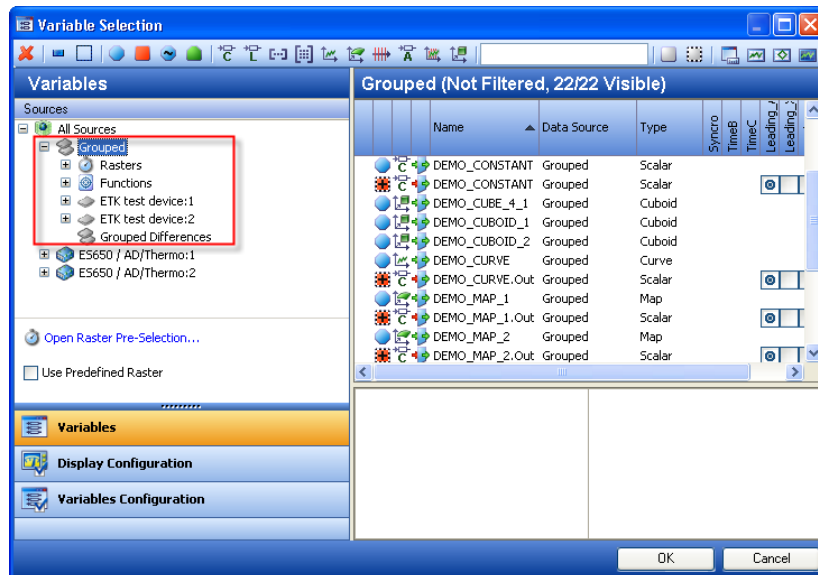
Starting with INCA V7.0, this is also possible when copying data from a data exchange file (e.g. in DCM format) to grouped variables in the Experiment Environment. In this case a copy action on a variable in the Experiment Environment automatically applies the changes on the other grouped variables as well.

Note

If the values of a variable are different for the different grouped devices, the changes are only applied for the device of the selected variable.

- **Grouping of ECUs in the Variable Selection dialog**

For grouped devices, the **Sources** list in the Variable Selection dialog contains one entry named **Grouped**. When the item **Grouped** is marked, all variables belonging to the grouped devices are listed just once. Any changes such as changing the raster assignment, are applied to the variables of all grouped devices.



When a measure variable from a grouped ECU is selected in order to add it to a measure instrument, it is automatically selected for each of the grouped devices. If these variables shall be distributed to different windows, you can set this behavior in the **Variable Selection Preferences** dialog box, by ticking the checkbox **Add Grouped Variables to separate display widgets**.

When a calibration variable from a grouped ECU is selected in order to add it to a calibration instrument, it is selected only once. Calibrations made on this calibration variable will be applied to all grouped ECUs.

2.2.13 Plug-in Interface for CDM Documentation Formats

CDM offers a number of predefined documentation formats for documenting the results of the calibration work.

A new converter plug-in interface provided with INCA V7.0 enables you to customize your documentation by adding your own documentation formats.

The complete calibration data information is provided by CDM in the standardized MSRSW ASAM XML intermediate format, which is used for further processing by the converters.

The plug-in interface is available with the INCA base package. A ready-to-use PDF converter is already integrated into INCA.

Further information on the plug-in interface for CDM documentation formats is provided in the CDM documentation [7].

2.2.14 Automatic Firmware Check Directly from Within INCA

INCA is often used in a number of different environments, e.g. for testing different vehicles or test beds. INCA V7.0 can automatically check the firmware version of the connected ETAS hardware and display the results graphically.

It can be set in the INCA user options under **Hardware → Automatic Firmware Check** whether INCA shall check the firmware version each time a new hardware is initialized during the current INCA session (e.g. upon opening an experiment). The overall result of the firmware check will be indicated in the status bar of the INCA Experiment Environment by a small icon displayed on top of the icon for the connection status.



The icons have the following meaning:

-  The newest suitable firmware is running on the hardware.
-  A newer suitable firmware version exists, and therefore a firmware update is recommended.
-  The firmware status is ambiguous or could not be detected; however, it is possible to use the hardware with this firmware version.
-  The hardware uses a firmware version which must not be used with this INCA version; it is not possible to use this firmware version.
-  HSP is available, but the firmware check has not been performed.
-  (no icon)
HSP is not installed on your system, or the installed firmware version is not compatible with this INCA version. Therefore no firmware check is possible.

Note

*You can check and update the firmware status manually from the Hardware Configuration Editor or the Experiment Environment.
Both manual and automatic firmware checks are only possible for hardware that is fully supported by HSP.*

Automatic firmware checks require HSP V9.0 or higher.

2.2.15 Extended MCD Support for the ES9xx Family

The ES910.x Compact Rapid Prototyping Module now also supports LIN Monitoring. FlexRay functionality can be added through the new ES920 FlexRay Module: this module can be plugged into an ES910.x and provides a FlexRay interface for FlexRay monitoring and XCP on FlexRay.

Note

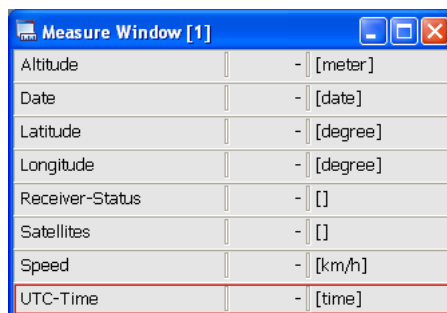
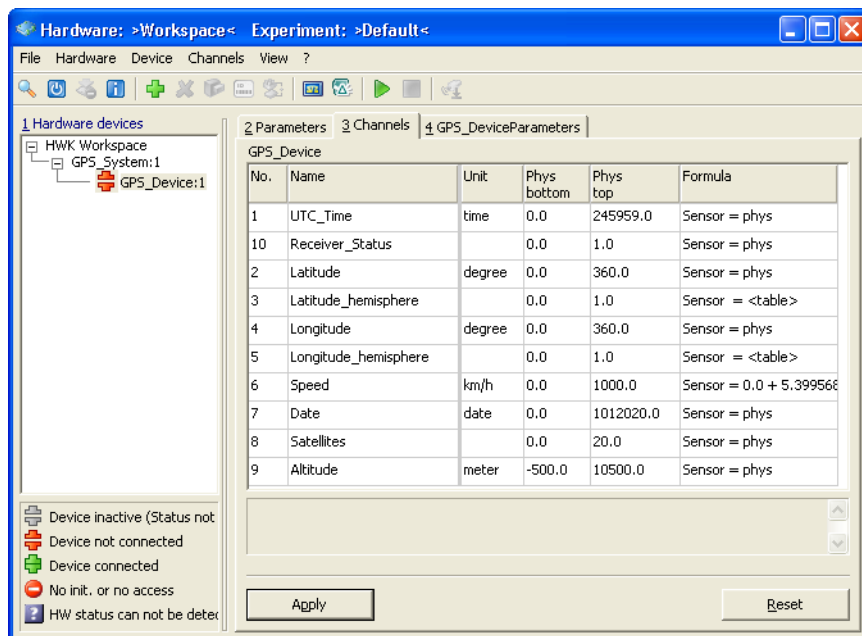
The configuration of the FlexRay and LIN interfaces in INCA is possible only if the corresponding add-on (INCA-FLEXRAY, INCA-LIN) was also installed.

In addition to its predecessor ES910, the ES910.3 supports hardware synchronization. Moreover, INCA is prepared for the support of the INCA-MCE add-on, which will be provided after INCA V7.0 and enable faster ECU measurement and calibration access.

2.2.16 Measurement of GPS Data Through Support of GPS Devices

INCA V7.0 supports GPS signal recording with GPS hardware. Highlights:

- Flexible integration and automatic HW search of any GPS mouse with virtual COM port driver (USB, CAM or Bluetooth) or NMEA-183 protocol
- Visualization of basic GPS signals in INCA Experiment standard windows (Longitude, Latitude, Speed, Satellites, Altitude, UTC time, Receiver, Status and Date)
- Synchronized recording with ECU, bus and sensor signals. Parallel recording as GPX file for visualization in, for example, GoogleEarth.



Note

The geographical visualization of the GPS data in MDA will be supported in a later MDA version.

2.2.17 Support of New Hardware

INCA V7.0 supports the following new ETAS hardware:

- Third party GPS devices (see „Measurement of GPS Data Through Support of GPS Devices“ on page 47)
- ETAS ECU and bus access devices:
 - ES592 Interface Module (ETK, CAN and LIN)
 - ES593-D Interface Module (ETK and CAN)
 - ES595 Interface Module (ETK, CAN, FlexRay and LIN)

Note

These ES5xx modules are supported through the ES5xx add-on for INCA, which is available for free and provided with the INCA installation.

- ES581.3 USB-on-CAN Module
With the new ES581.3, it is possible to handle multiple ES581.3 modules in parallel, thanks to the new serial number mapping. Moreover, the ES581.3 has a second CAN port, and the new CCP speed-up allows for faster ECU reprogramming.
- ES720 Drive Recorder
The ES720 module is a universal Drive Recorder which can be used for a variety of measurement tasks in the development, calibration, and validation of automotive electronic systems.
For measurements, the ES720 replaces the laptop with INCA software. The selection of measurement signals, the setting of data acquisition rates and the configuration of triggers, communication and diagnostic protocols are directly transferred from INCA. The ES720 module provides for unattended data recording. Once configured, operation of the unit is straightforward, even for untrained personnel.
For more information on the ES720 Drive Recorder please see the ES720 User's Guide.

Note

The support of the ES720 Drive Recorder is made available through a separate INCA add-on which is provided together with the hardware device.

- Enhanced MCD support for ES910.x Compact Rapid Prototyping Module:
 - LIN Monitoring for ES910.x
 - FlexRay Monitoring and XCP-on-FlexRay support for ES910.x with ES920 plug-in module

- The new ES910.3 Compact Rapid Prototyping Module supports hardware synchronization. Moreover, it is prepared for the support of the INCA-MCE add-on, which will be provided after INCA V7.0 and enable faster ECU access.

Note

These ES9xx modules are supported through the ES9xx add-on for INCA, which is available for free and provided with the INCA installation. The FlexRay support requires the INCA-FLEXRAY add-on. The LIN support requires the INCA-LIN add-on.

Moreover, INCA V7.0 supports the following third party bus interface devices:

- Vector CANCardXL
- Vector CANCaseXL
- Vector VN7600

Note

The operation of these devices with ETAS software must be enabled by ETAS. Please contact your local sales representative to get the enabling key.

2.2.18 Other New Features and Usability Improvements

INCA V7.0 includes some useful features for improving the usability of major tasks. Highlights:

- **Advanced keyboard support in Calibration Scenario Editor (CSE)**

Keyboard operation in the Calibration Scenario Editor has been enhanced by the following functionality:

- Moving between different CSE areas by means of the <TAB> key
- Extension of variable selection by means of the arrow keys
- Toggling between scenarios by <Ctrl> + <9>
- Cut, copy, paste, delete by
<CTRL> + <X>, <CTRL> + <C>, <CTRL> + <V>,

- **Logging of Calibration Scenario meta data to the measure file**

INCA now writes the scenario name and change information as event comments to the measure file, thus documenting automatic parameter modifications and logging all scenario switches during recording. These scenario meta data can be visualized in the data analysis with MDA.

- **Replacement of table editor by combined editor**

With INCA V7.0, the table editor was replaced with the combined editor. When you open an experiment with table editors from previous INCA versions, the table editors are automatically converted to combined editors.

Note

Table editors are still used for axes and arrays. Scalar table editors are used for scalars.

- **Invalid variables remember display windows**

In the INCA Variable Selection dialog on the Variables Configuration tab, you can set individual variables to Inactive. If a variable is disabled, it is no longer displayed in the experiment environment, not recorded and not included in the raster check, which is useful in cases of raster overflow.

Inactive variables now remember the properties they had before having been disabled. If a variable is enabled again, it is now displayed in the same measure/calibration window, in the same position, and with the same properties (e.g. colors, monitoring boundaries) as before.

- **Serial number mapping**

The new serial number mapping mechanisms for USB, PCMCIA & PClexpress hardware enables the user to work with multiple third party devices of the same type in parallel. The serial number is read when searching for hardware, and it is displayed for the corresponding device in the Hardware Configuration Editor.

- **Synchronization of CAN Controller Settings**

If more than one device is connected to the same CAN controller, it can happen that the different devices, having different projects associated, have different CAN settings. To avoid problems being caused by conflicting CAN settings, it is possible to define which of the connected CAN devices will be used as master device for the setting of the CAN controller.

Please note that only CAN devices of the types KWP2000, UDS, CCP, and XCP with projects being assigned can be selected as CAN master devices; devices of the types CAN-Monitoring and CAN-Output can not be selected as CAN master device.

By default, the first master capable device that is shown under the controller item in the hardware tree is selected. If this device gets removed, the next device under the CAN controller will automatically be selected.

You can select from all master capable devices that are listed under the CAN controller item in the hardware tree.

- **New user option for CDF20 — BOM encoding information**

INCA provides a new user option for CDF V2.0 BOM encoding information. This option is located under [Data exchange → CDF V2.0 → BOM](#). It determines whether INCA shall write and accept CDF V2.0 files with a BOM (Byte Order Mark); BOM is a special mechanism that adds 3 bytes in front of a text file to define the encoding used in the text file.

- **Combined Editor enhancements**

To ease visual comparison, you can now display several curves in a combined editor as overlays. To enable overlay mode, select **Properties** from the context menu of the combined editor, go to the **Overlay** tab and set the parameter **Overlay Mode** to the value **Curves only**. All curves are now displayed at once in the graphical display of the combined editor.

You can further optimize the display by means of the **Overlay selection** dialog, which can be opened by selecting **Overlay selection** from the context menu. In this dialog, you can disable the overlay functionality for individual curves as well as set the line color for each curve.

You can select any curve in the graphical display and edit it; the curve does not need to be explicitly selected in the combo-box at the top of the combined editor.

- **Indication of boundary violations**

Violations of user defined boundaries for measurements can be indicated in different ways: in addition to a symbol in the measure window or measure table, a marker next to the measure value or a sound can be used to indicate boundary violations. In INCA V7.0, it is also possible to change the background color of the field showing the measure value, thus making the boundary violation clearly visible.

The properties for indicating boundary violations can be set in the properties window for the corresponding measure window or measure table, or on the **Display Configuration** tab of the Variable Selection dialog.

- **New document: INCA Functionality Overview**

INCA V7.0 includes a new document which maps the main INCA features to the INCA versions supporting that feature, thus showing you at a glance which major changes have been introduced with a new INCA version.

You can find this document on the INCA product DVD and in the INCA Manuals folder in your INCA installation on your PC.

2.2.19 New SDK: Measurement Data Format Integration Package

For the validation of automotive systems INCA is used in combination with various third party tools for data acquisition and analysis.

One main challenge for system validation experts is the handling of different measurement data formats used by the different tools.

INCA 7 introduces a standard measurement data exchange component, which enables third party and inhouse tool providers to use the same data formats as INCA and MDA (MDF V3, MDF V4, ETAS ASCII), as well as to integrate own data formats into INCA

The MDF data exchange component and plug-in interface is provided through the MDF Integration Package.

Benefits:

- Standardized data exchange based on the recently released standard ASAM MDF V4
- High flexibility due to the support of the former MDF V3 and further formats

- Powerful platform independent software component for fast data access
- Open interface for own data format filters
- Easy third party application integration with comfortable development kit
- Standardized software component for consistent and fast data exchange with other applications
- Open plug-in frame work for new data formats
- Faster integration of the new MDF V4 standard in your own application
- Platform independent component for integration in other environments

2.2.20 New SDK: Instruments Integration Package

INCA's open Experiment Environment provides an interface which allows for the integration of customer specific measure and calibration instruments for the most diverse use cases. Examples are animations, integrated web explorers, embedded model viewers, street map viewers visualizing measured courses of GPS coordinates, stimulation of parameters with generators, etc. As a consequence, customer specific data visualizations or data post processing does no longer require external applications connected via COM, INCA-MIP or ASAP3, but may be done directly in INCA.

Instrument Integration is supported through the INCA Instrument Integration Development Kit (INCA-INS.DK), which is available for free from ETAS.

2.2.21 New Add-On eCDM for Enterprise Calibration Data Management

INCA's Enterprise CDM (eCDM) interface provides direct access to any enterprise calibration data management system from within INCA via a standardized API, thus making a manual data exchange via the file share obsolete.

Access to enterprise calibration data management from INCA is provided in the Database Manager, the Experiment Environment and the Calibration Data Manager.

eCDM functionality in INCA is provided through a separate eCDM add-on.

Note

To be able to access your enterprise calibration data management tool via the INCA interface, you need to install a plug-in for the corresponding tool. Please check with your tool provider for the support of the INCA interface.

2.2.22 New Add-On INCA-SIP (INCA Simulink Integration Package)

INCA-SIP (INCA Simulink Integration Package) is an INCA add-on that provides measurement and calibration access to MATLAB/Simulink-modeled ECU functions via INCA.

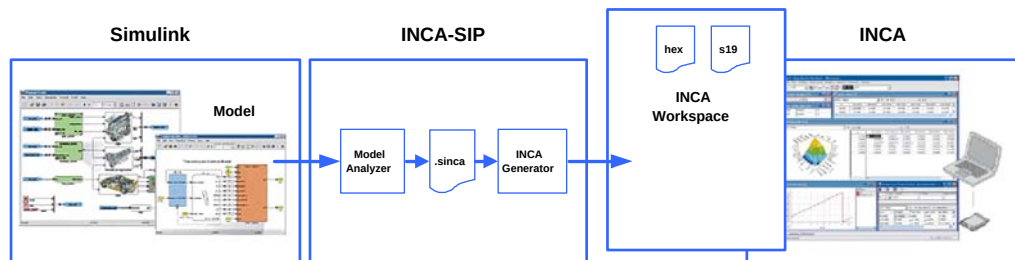
Effectively it is a Simulink tool box – after installation a new menu-item is available in Simulink, with which a connection between INCA and the model (opened in the Simulink Model Editor) can be established.

INCA-SIP converts the Simulink math model into automotive INCA objects (A2L, Hex, INCA workspace). Most of the models can be converted out-of-the-box; customer-specific modeling rules may require a customization.

Connecting Simulink to INCA

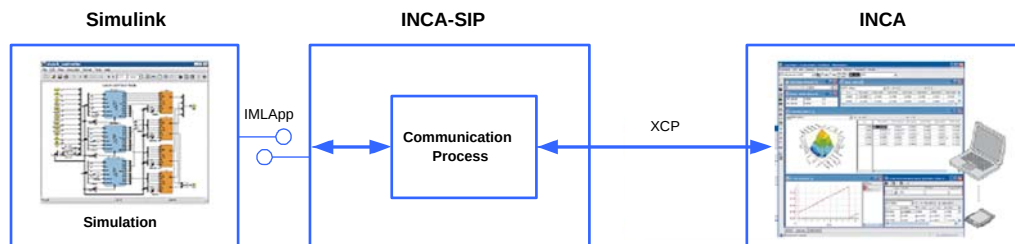
When connecting to the model, INCA-SIP starts an analysis of the model blocks. This analysis can be customized to deal with customerspecific modeling guidelines. This customization is only required once, when INCA-SIP is introduced.

All parameters required for M/C access are determined and all artefacts required by INCA are generated.



Measurement and calibration for Simulink

For measurement and calibration, INCA connects to an ordinary XCP device – the INCA-SIP exchanges data via MATLAB's IMLApp interface. Measurement and calibration can be done while Simulink runs the model.



2.2.23 INCA-EIP: Working Page / Reference Page Concept for ES910

New Working and Reference Page Concept for ES910 Rapid Prototyping System

In previous versions of INCA-EIP, E-Targets for the simulation system always had only one memory page that was always accessed as the working page from within the INCA experiment environment. When conducting experiments on the E-Target, you could not switch back and forth between the working page and reference page in the INCA experiment environment.

With INCA-EIP V7.0, dataset handling of an ES910 is now similar to that of an ETk: When using an ES910 as E-Target, both a working and reference page are available, which enables you to switch back to an approved dataset whenever required. Copying datasets from one page to another is also possible.

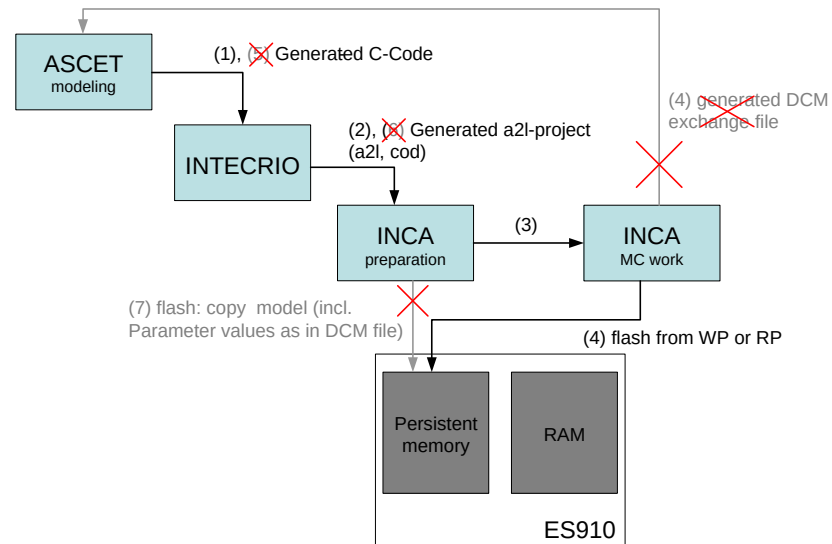
Note

The support of the working page / reference page concept in INCA requires that the concept is supported by the ES910 model. For this purpose, the model must be created with INTECRIO V3.2 or ASCET-RP 6.1 or a later version.

Direct Flashing of Calibrated Data for ES910 Rapid Prototyping System

In previous versions of INCA-EIP, flashing of calibrated data into an ES910 Rapid Prototyping system required a complicated sequence of working steps involving several tools (ASCET, INTECRIO, and INCA).

With INCA-EIP V7.0, calibrated data can now be flashed directly to the ES910 Rapid Prototyping system, thus shortening the workflow considerably.



3 Installing the Program

The chapter titled "Installing the Program" is for all users who install, maintain or uninstall INCA on a PC or a network as well as system administrators who provide INCA on a file server so that the program can be installed via the network. This chapter also contains information on the scope of delivery, hardware and software requirements for stand-alone and network installations and the preparation required for installation. The chapter also describes the procedures used to install and uninstall INCA.

3.1 Preparing to Install

Check to make sure all the items have been delivered and that your workstation complies with the system requirements. Make sure you have the necessary user privileges for the operating system and the network connection being used. You must have administrator privileges to install INCA.

3.1.1 Package Contents

Essentially the INCA-DVD-ROM has the following contents:

- Program files for INCA
- Free of charge INCA Add-Ons
- Hardware Service Pack (HSP)
- Documentation: Online help, INCA manual and tutorial in PDF format (Acrobat Reader), video tutorials, and additional PDF documents

3.1.2 INCA License

You require a valid license for the use of INCA. You can obtain the license file required for licensing either from your tool coordinator or through a self service portal on the ETAS Internet Site under <http://www.etas.com/support/licensing>. To request the license file you have to enter the activation number which you received from ETAS during the ordering process.

For further information on licensing please refer to "Licensing the Software" on page 73.

3.1.3 System Requirements

The minimum requirements for the INCA PC ensure that INCA will run smoothly with smaller projects, and the recommended PC configuration means that it will operate very efficiently. Please consider that large ECU projects and experiments require more memory.

Added performance boost depends on the hard disk's average access time; SSD hard disks have shown to be particularly advantageous. Also the power-saving modes that have been enabled for the computer components (processor, hard disk, etc.) play an important role. Moreover, more memory improves execution time of repetitive operations.

For detailed information on the causes influencing the INCA performance, please see [6]: "Performance Tips and Tricks", which you will find in the **Manuals** folder of your INCA installation.

Note

Please see the Readme file for the latest information on the INCA system requirements.

System Requirements: Minimum PC Requirements:

- Required hardware:
 - 2 GHz processor
 - 2 GB RAM
 - DVD-ROM drive for installation
 - Network adapter
 - Graphics card with a resolution of at least 1024 x 768, 256 MB RAM, 16 bit colors and DirectX 9
- Required operating system:
 - WINDOWS® XP SP3 (32 bit)
 - WINDOWS® Vista SP1 (32 bit)
 - WINDOWS® 7 SP1 (32 or 64 bit. On a 64 bit operating system INCA uses the 32 bit compatibility mode)

Note

English, French, Japanese, Chinese and German OS versions are supported.

- Required free disk space:
 - 1 GB (not including the size for user data)

System Requirements: Recommended PC Requirements:

- Required hardware:
 - 3 GHz quad core processor or equivalent
 - 4 GB RAM
 - DVD-ROM drive for installation
 - Network adapter

- Graphics card with a resolution of at least 1280 x 1024, 1 GB RAM, 32 bit color and DirectX 9.
- Required operating system:
 - WINDOWS® 7 SP1 64 Bit (on a 64 bit operating system INCA uses the 32 bit compatibility mode)

Note

English, French, Japanese, Chinese and German OS versions are supported.

- Required free disk space:
 - Minimum of 10 GB

3.1.4 User Privileges Required for Installation and Operation

User Privileges Required for Installation:

In order to install INCA on a PC, you need administrator authorization access. Please contact your system administrator, if necessary.

User Privileges Required for side-by-side installations of different INCA versions:

Operating different INCA versions on the same PC requires administrator authorization access for COM registration. Please contact your system administrator, if necessary.

User Privileges Required for Operation:

In order to work with INCA under WINDOWS, each user must obtain the following rights from the administrator:

- Read and write access to the registry folder
HKEY_LOCAL_MACHINE\Software\ETAS and all subfolders
- Read and write access to the TEMP directory used by INCA; the TEMP-directory is determined during the INCA installation.
- Read and write access to the INCA installation directory (e.g.
c:\ETAS\INCA7.1)
- Read and write access to shared components in ETASShared11 (e.g.
c:\ETAS\ETASShared11).
- Read and write access to the directory for the INCA data (e.g.
c:\ETASData) and all subdirectories
- Read and write access to the directory for the ETAS log files (e.g.
c:\ETAS\LogFiles) and all subdirectories.

Unlimited access is required for all mentioned directories.

3.2 Installing from DVD

The dialog boxes for installing from the DVD and installing from the network drive are identical. The chapter "Installing from a Network Drive" on page 62 *only* describes additional steps.

3.2.1 Initial Installation

To install INCA:

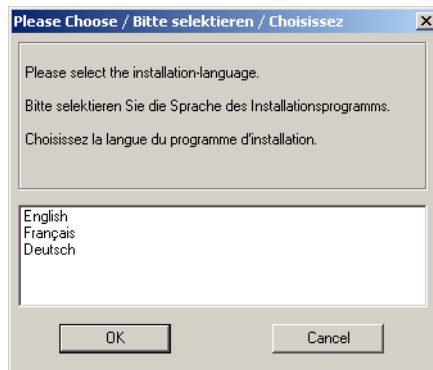
- Insert the DVD with the installation program into the DVD drive of your computer.
- If the installation routine does not start automatically, execute the **autostart.exe** program on the DVD manually.
The DVD containing the installation program for INCA, tools and utilities, documentation, and further information, is started.
- Click **Main**.
- Select the INCA version you would like to install.
The installation program is started.
- Follow the instructions displayed on the screen and confirm each screen by clicking **Next**.

Most of the screens are completely self-explanatory. The following sections explain only those parts of the installation procedure where additional information might be helpful:

- "Setting the Language for the Installation Program" on page 59
- "Specifying the INCA Destination Directories" on page 59
- "Selecting the components for Installation" on page 60
- "Licensing of INCA" on page 61

Setting the Language for the Installation Program

If the Regional Setting under Windows is neither English (USA) nor German (Germany), the following dialog box appears so you can select the language. This setting applies to the language the installation program will use. The language for the ETAS application is selected at a later time in the installation procedure.



- Select the desired language.
- Confirm by clicking **OK**.

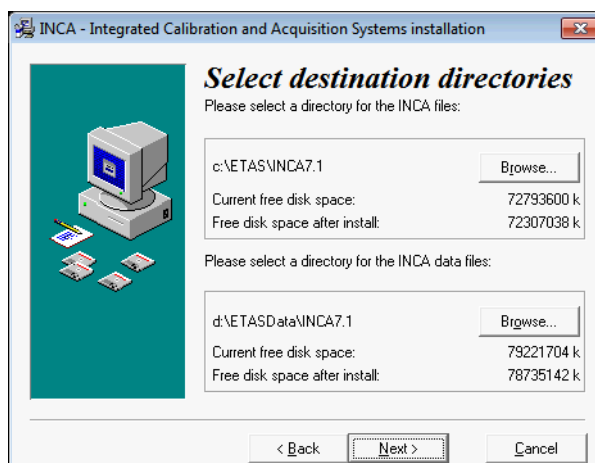
If one of the above regional settings is already active, the installation program will be executed automatically in that language and the dialog box will not appear.

Note

The language can be set in the Windows Control Panel under Regional Settings.

Specifying the INCA Destination Directories

In the following screen, you will be prompted to specify destination directories for program and data files:



The program **files** and program **data** are stored in different directories. If you uninstall or update the program later, only the program files will be deleted or overwritten. The program data will continue to be available to you.



WARNING!

INCA data files must not be saved in the Program Files directory. Selecting the Program Files directory for INCA data files may lead to problems in INCA as the access to the program folder depends on the Windows user rights. Select a folder in a data area where all users have read and write access rights.

The program data include the following:

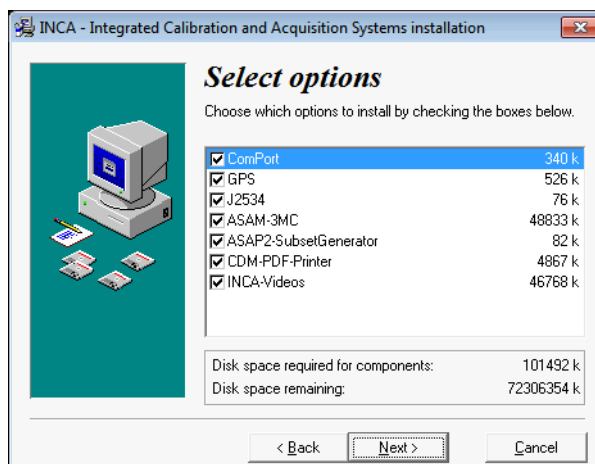
- Databases
 - User interfaces
 - Demo files
 - Measure files
 - User profiles
- To change the default folders, click the **Browse** button.
 - In the dialog box, select the desired directory.
If you specify a directory that does not exist, the installation routine will automatically create it.

Note

Make sure that users have unlimited access to the set directories. If not problems can occur while working with INCA.

Selecting the components for Installation

- In the "Select Components" window, select the components to be installed.



The installation of the following components is optional in INCA V7.1:

Note

The more components are selected, the longer is the loading time in INCA. For that reason you should select only those components which you actually would like to use. It is always possible to install further components afterwards.

Component	Description
ComPort	Using the COM Interface for Outputting Measured Data
GPS	Support of 3 rd party GPS devices
J2534	Integrating SAE J2534 compliant hardware for control unit reprogramming and diagnosis
ASAM-3MC	Remote control interface for the calibration system, e.g., via a test bench computer
ASAP2 Subset Generator	Support for developing sub-projects for ASAM-MCD-2MC projects
CDM-PDF-Printer	Print engine for generating CDM variable listings in PDF format
INCA-Videos	Short video tutorials demonstrating useful INCA features

You can also install additional components later on (see „Special Installation Steps and Dialogs“ on page 61).

Licensing of INCA

You require a valid license for the use of INCA. For information on licensing please refer to "Licensing the Software" on page 73.

3.2.2 Special Installation Steps and Dialogs

To install components afterwards:

- Restart the installation program.
- In the "Select Components" window, highlight the modules you want to add to the installation.
- Finish the installation as usual.

Note

It is not possible to uninstall individual components!

3.3 Installing from a Network Drive

A network installation has the advantage of allowing you to adjust the installation files even before actually installing the program on the workstation. This allows you to set company-specific defaults.

The installation options that can be customized are described under "Customizing the Network Installation – install.ini" on page 62.

3.3.1 Installing INCA from a Network Drive

The sequence of dialogs when installing from a network drive is the same as when installing from a DVD (see „Installing from DVD“ on page 58).

To provide data on the network server:

You have to copy the installation files from the DVD to the desired network drive.

- [Create a source directory on the desired drive.](#)
- [Copy all data from the DVD to the source directory.](#)

To start the INCA installation program:

- [Search for the file INCA.exe.](#)
- [Double-click the file.](#)
- [Follow the instructions in the installation routine.](#)

3.3.2 Customizing the Network Installation – install.ini

You can change certain default settings before conducting the network installation. This is done by adjusting the configuration file **install.ini**. This file is in the installation directory.

The following customizing options are available for network installation:

- You can customize the installation dialogs, e.g., modify the default settings for directories, etc. (see „Customizing installation dialogs“ on page 65).
- When one installation is finished, you can perform up to 10 additional installations (see „Concatenating Installations“ on page 66).
- You can specify the storage location for the network installation log file (see „Logging the Installation Information“ on page 66).
- You can perform the INCA installation fully automatically and transparently in the background without any user intervention (see „Using Command Line Parameters“ on page 69).
- You can overwrite the files provided in the product data directory (default setting, e.g., D:\ETASdata\INCA7.1\) using your customized files, and/or add files to the existing directories (see „Installing User-Specific Files“ on page 70).

Before the different types of possible customizations are described, see the example for this type of configuration file (serving as a template for all ETAS installations):

Note

A semicolon at the beginning of a line denotes it as a comment line. To enable a line, delete the initial semicolon and change the line contents according to your installation requirements.

Example:

```
;In this file, you can configure the defaults of the installation.
;For any changes remove the ';' in the first coloum of the keyword.

[Preferences]
;Adds or removes the INCA preloader application from system autostart
;DoPreload=False

;Sets the menu group in the start menu under programs
;StartMenuGroup=ETAS\INCA

;Sets the main directory of INCA (no version number will be appended
;MainDir=c:\ETAS\INCA

;Sets the main data directroy for INCA (no version number will be
;appended)
;MainDataDir=d:\ETASData\INCA

;ETAS shares several software modules among different software
;products and versions. You can choose how to install these shared
;modules.
;A) share modules between products (required if you want to
;use several ETAS programs concurrently)
;B) use a local copy of all modules
;Using shared modules may lead to side effects on existing ETAS
;product installations.
;PrivateShared=A

;Sets the directory for the shared ETAS files (compatibilit-
;ID of the ETASShared will be appended: specifying e.g.
;c:\ETAS\ETASShared will lead to an installation-directory
;of e.g. c:\ETAS\ETASShared1 !!!)
;ETASShared=c:\ETAS\ETASShared

;Sets the directory for the log files
;ETASLogDir=c:\ETAS\LogFiles

;Sets the directory for the temporary files
;ETASTempDir=c:\temp

;specify which optional component (as seen in options-dialog)
;should be preselected. For a silent installation only the
;preselected options will be installed.
;The options are represented by upper-case letters, e.g. 'A'
;for 1st option, 'B' for 2nd and so on. If you want to have none of
;the options installed, specify "-".
;
;list of optional components in INCA V7.1.0
;
;COMPONENTS=ABCDEFGHJKLMNOPQRSTUVWXYZ
;  A ComPort
```

```

; B GPS
; C J2534
; D ASAM-3MC
; E ASAP2-SubsetGenerator
; F CDM-PDF-Printer
; G INCA/Videos
;

;User-License information
;FirstName=
;LastName=
;Company=
;Department=
;CountryCode=
;AreaCode=
;Phone=
;EMail=
;Street=
;ZIPCode=
;City=
;Country=

;Many ETAS-applications support different languages. Please select
;preferred language (global setting for all ETAS-applications):
;set variable to EnglishUS, German, French, Japanese or Chinese
;Language=EnglishUS

;The Manuals may be available in different languages. Please select
;which languages are to be installed. Use a comma separated list
;of languages (EnglishUS, German, French or Japanese) with no spaces.
;If for some manual the specified language is not available a
;substitute (usually english) will be installed.
;ManualLanguages=EnglishUS,German

;Specify up to 10 additional installations to be run automatically
;after this installation is finished.
; * Use a relative path for the executables.
; * you cannot have those installation in the same directory as the
;   first installation.
; * Command-line args may be specified in corresponding
;   AutoCmdLineArgs0X
; * you may launch links (extension .lnk)
;
;AutoInstall01=..\MDA_V7.1.0\mda.EXE
;AutoCmdLineArgs01=/silent
;AutoInstall02=
;AutoCmdLineArgs02=

[Network-Preferences]
;Sets the path for the user registration during installation over
;a network
;You may specify a prefix or a suffix to the NetUserLog-Name, which
;is the product-name. Prefix and Suffix are concatenated with product
;name with no additional intermediate characters, so please include
;a hyphen or other filename-valid character as a separator if
;desired.
;
;NetUserLogPath=i:\User
;NetUserLogPrefix=
;NetUserLogSuffix=

;If you prefer to disable the logging functionality, set following

```

```

;constant to false.
;NetUserLogging=true

[SilentInstallation]
;prevents a silent installation from rebooting when set to "Suppress"
;Restart=Suppress

[Licensing]
; format: absolute file path to license file
;LicenseFileName = 'c:\temp\xxx.lic'
; format: LicenseName1 LicenseName2 use blanks to separate licenses
;LicensesToBorrow = 'INCA'
; format: Modes 'Interval' and 'Date' are possible
;BorrowExpiryMode = 'Interval'
; format: yyyy-mm-dd
;BorrowExpiryDate = '2007-04-01'
; format: number of days
;BorrowExpiryInterval = '100'
; format: number of days
;BorrowAutomaticExtensionInterval = '100'
; format: number of days when automatic borrow extension is
performed.
;ExecuteBorrowAutomaticExtensionInterval = '1'

[IPMServer]
; auto configuration flags
; format: true for active, false for inactive
;AutoConfigNIC=true
;AutoIPRange=true

[ES720]
;Enables the ES720 Mode, which is necessary to allow an installation
; on the ES720.
; ES720_Installation= true

```

Customizing installation dialogs

The following example will show you how to modify the default settings in the installation dialogs.

To customize installation dialogs:

- Open the **install.ini** file with a text editor.
The following is a typical example of an entry in this INI file:

```

;Sets the main directory of INCA (no version number
;will be appended)
;MainDir=c:\ETAS\INCA

```

- To modify the default setting, delete ";" (Comment) on the line with the **MainDir** keyword.
- Change the path to, e.g.,
D:\programs\ETAS\INCA.

The entry should now look like this:

```

;Sets the main directory of INCA (no version number

```

;will be appended)

MainDir=C:\programs\ETAS\INCA

- In the same way, modify all other entries in **install.ini** as desired (see below).
- Save your changes and then close the editor.

Now when you start the installation with **INCA.exe**, the dialog boxes will show the new settings as defaults.

Concatenating Installations

Using the **AutoInstallXX** parameter in **install.ini**, it is possible to access up to 10 additional installations after finishing one installation.

To concatenate installations:

- Open the **install.ini** file with a text editor.
The following entries pertain to concatenating installations:

```
;AutoInstall01=..\Installation2\Install.EXE
```

```
;AutoInstall02=
```

- To enable this function, delete the ";" (Comment) that is on the same line as the keyword **AutoInstall01**.
- Change the path to, e.g.,
..\..\HTMLInstall\setup.exe.

Be sure to enter a specific path that is different from the INCA installation file. Here, "..\" means one directory level upward.

```
AutoInstall01=..\..\HTMLInstall\setup.exe
```

- Repeat these steps, if necessary, for additional installation files (max. **AutoInstall10**).
- Save your changes.

Logging the Installation Information

When installing from a network drive, the information entered in the registration dialog is saved in a file (**INCA.usr**). It is used for your own reference only. By default, the file is stored on the installation drive from which the installation was started, under a dedicated directory named **User** (e.g., **F:\User\INCA.usr**).

As already mentioned during the registration for network installation, you can change the storage location specified in **install.ini**.

Note

*Make sure that all users have write-access to the log directory, i.e., the installation directory or the directory you specified in **install.ini** (see below).*

To specify the directory for registration information:

- Open the **install.ini** file with a text editor.
The following entry refers to this purpose:

```
;NetUserLogPath=i:\User
```

- To modify the default setting, delete ";" (Comment) on the line with the **NetUserLogPath** keyword.
- Change the path to, e.g.,
x:\UserLog.

```
NetUserLogPath=x:\UserLog
```

- Save your changes.

If the installation files are on a local hard disk, the installation process is not logged by default. If you also want to log the installation process for local installation, delete the comment character as described above and specify the desired path.

Setting the licensing behavior

In the [Licensing] section of the `install.ini`, you can define the way in which INCA and other ETAS software programs access the required licenses.

To define the access to the required licenses:

- Open the **install.ini** file with a text editor.
- Go to the [Licensing] section and modify the settings as desired. The parameters that can be included in this section, and their settings are described below.
- Save your changes.

The following parameters may be used:

- **LicenseFileName**
Defines the absolute path to the location of the license file which is to be added.
- **LicensesToBorrow**
You can use this setting if licenses can be borrowed from a license server. To enable the borrow mechanism, you must enter the name of the product or features license (e.g. INCA). If you enter more than one license, the license names must be separated by blanks.
- **BorrowExpiryMode**
Defines the way in which the expiration of the borrow status is given. Possible values are:
 - **Date**
If the **BorrowExpiryMode** is set to **Date**, the borrow period will expire at a certain date which is specified under **BorrowExpiryDate**.
 - **Interval**
If the **BorrowExpiryMode** is set to **Interval**, the borrow period will expire after a certain number of days which is specified under **BorrowExpiryInterval**.

- **BorrowExpiryDate**
If the **BorrowExpiryMode** is set to **Date**, this parameter specifies the date when the borrow period expires. The format is **yyyy-mm-dd**.
- **BorrowExpiryInterval**
If the **BorrowExpiryMode** is set to **Interval**, this parameter specifies the length of the borrow period in days.
- **ExecuteBorrowAutomaticExtensionInterval**
Defines at what point of time the borrow period will be automatically extended. This parameter specifies the number of days before the expiration of the current borrow period. When this time is reached, the borrow period is automatically extended to the interval specified under **BorrowAutomaticExtensionInterval**.
- **BorrowAutomaticExtensionInterval**
This parameter specifies the borrow interval in days that is applied when an automatic extension of the borrow period is executed (as defined under **ExecuteBorrowAutomaticExtensionInterval**).
- **ImmediateBorrow**
You can define that a license is automatically borrowed. Possible values are:
 - **True**
The license is borrowed automatically at installation time.

Note

ImmediateBorrow='True' works only for the user who performs the installation. Other users who work on the same computer do not own the borrowed license.

- **False**
The license will be borrowed at the first time when the program connects to the license server.
- **CustomLicenseFolder**
Due to the fact that the default location for added license files, i.e. **C:\Documents and Settings\All Users\Application Data\ETAS\FlexNet**, is only writeable for users with admin rights, a different path for the license file folder may be specified with this parameter.

The following example defines that borrowing is enabled for INCA. The license will be borrowed when INCA will be started for the first time; by default the license expires after 100 days.

```
[Licensing]
LicenseFileName = 'd:\licenses\MyLicense.lic'
LicensesToBorrow = 'INCA QMBASIC'
BorrowExpiryMode = 'Interval'
BorrowExpiryInterval = '100'
ImmediateBorrow = 'false'
```

Using Command Line Parameters

INCA.exe /? and **INCA.exe /h**

Displays the available command line criteria.

INCA.exe /s

Calling this up allows you to execute the INCA installation completely automatically and transparently in the background, i.e., without any user interaction required. All information that is otherwise prompted for in dialog boxes is entered with its defaults or with the values taken from `install.ini`.

If you create a batch file containing the "INCA.exe /s" command, and configure the required settings in `install.ini`, users can run the installation process themselves by executing this batch file without having to enter any further information.

The task bar displays an icon while the installation process is running. It is not possible, however, to interact with this process.

For you to find out whether the installation has been successful, a file called `silent.log` is created in the installation directory. The file is in the same format as the INI file. The first section contains the name of the product that has just been installed. When the installation process is just starting, the `install` entry is assigned the value "in progress". When the installation is finished, this line is replaced with "Setup is complete." If the workstation needs to be rebooted after the installation, the line "need reboot=YES" is inserted into the appropriate section, regardless whether the system was actually rebooted or not.

Normally, the workstation is automatically rebooted after the installation. You can prevent this by enabling the following entry in the `install.ini` file.

```
;Restart=Suppress
```

To suppress the system reboot, simply delete ";" that is on the same line as the keyword `Restart=Suppress`. This is appropriate if you want to install additional programs right after this one.

However, you are then responsible for carrying out the system reboot.

The `/s` parameter also uninstalls the application. Access it, e.g., by entering the following: `c:\ETAS\INCA7.1\unwise32.exe /s InstINCA.log`.

Note

When using the command line parameter `/s` in conjunction with other parameters, it must always be the first parameter when executing INCA.exe.

INCA.exe /ini="myPath\install.ini"

Generally, the `install.ini` file is used for configuring the installation in the current directory. If you want to serve several users via a central network drive, it is recommended to provide a separate `install.ini` file for each user. This could look like this: **INCA.exe /ini="F:\temp\install.ini"**

Suggestion: Create a specific `install.ini` file for each user in a directory and link it to the general installation directory containing the `INCA.exe` installation file.

Caution: To reference a file in the same directory as the installation, the installation program requires the following entry:

```
/ini=".\\MyIni.ini"
```

Note

*A strict syntax has to be followed when entering the command line criteria and their parameters: if a command line needs a value, it has to be entered in quotes, followed by an equal sign (**no spaces**).*

Installing User-Specific Files

During installation, the customization feature described below allows you to control the installation routine to the extent that your customized files will overwrite certain default files, or that other files will be included in the installation.

Doing this allows you to integrate your customized databases, user profiles, and user interfaces in the installation routine.

Create a subdirectory named **InstData**\\... in the installation directory and copy your customized files into it while maintaining the proper directory structures.

To create your customized files, install INCA on a test computer, and use this to create the files.

After completing the default INCA installation, the **ETASData\\INCA7.1**\\... directory contains several subdirectories with files. You can customize the files in the following subdirectories:

- **Database\\db**

The **db** directory contains the default database. Here you can, e.g., create another demo database under **Database\\DemoDB**\\.

Note

Note that the entire database directory must always exist. Individual database files may not be swapped out.

- **Data\\Screen Layouts**

The **Screen Layouts** directory contains user interfaces that can be used in the Experiment Environment.

- **User**

The directory contains the default user profile `UserDef.ini`. All configurable options are stored in this directory.

To customize data for network installation:

- Install INCA on your PC.
- Start INCA.
- Modify the user profile.
- Modify the database or add a new one.
- Modify the user interfaces.
- Exit INCA.

You have finished customizing and now want to integrate these files in the installation routine. You have two choices:

- Overwrite existing files that have the same name with your customized files. To do this, you must create a folder named **InstData\overwrite** in the installation directory.
- Rename your customized files and add them to the existing files. Existing files having the same name will not be overwritten. To rename your customized files, you must create a folder named **InstData\add-only** in the installation directory.

To include your customized files in the installation routine, be sure to copy the parent directories, too. Note that the **ETASData\INCA7.1** directory level must be the same as **InstData\overwrite** or **InstData\add-only**.

Examples:

```
InstData\overwrite\user\userDef.ini
```

```
InstData\add-only\database\additionalDB\
```

```
InstData\add-only\Data\Screen Layouts\my screen.vui
```

To integrate a customized user profile:

- Copy your customized **ETAS-Data\INCA7.1\user\[user name]\INCA.ini** directory to **InstData\overwrite\user**.
- Rename the **INCA.ini** file to **UserDef.ini**.

Doing this ensures that after the installation, this initialization file will be used for each new user.

To integrate a customized database:

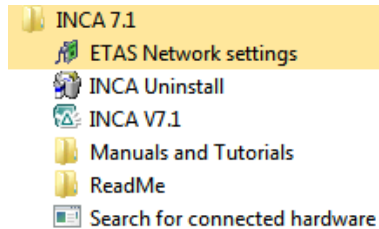
- Copy your customized database, i.e., the **\database\db** subdirectory into the **InstData\add-only\...** directory.
- Rename the **db** directory as desired, otherwise the database will not be copied.
Of course, you could also overwrite the **db** database by using the **InstData\overwrite** directory.

Integrating new or edited ***.vui** user interfaces is done in a similar manner.

Whenever you start the installation routine using **INCA.exe**, your customized files will replace the default files, and/or your new files will be added to the corresponding directories.

3.4 INCA V7.1 Program Group

After having installed INCA and restarted your PC, you will find the folder name you specified with the following entries in the Start menu (under **ETAS**):



- **ETAS Network settings**
Starts the ETAS Network Manager which is used to configure Ethernet hardware connections.
- **INCA Uninstall**
Starts the uninstall routine (see Chapter "Uninstall Procedure" on page 80).
- **INCA V7.1**
Starts the INCA program.
- **Manuals and Tutorials**
If you have installed the PDF user documentation, you can open the folder and view the desired documentation.
- **ReadMe**
Gives you the latest information on INCA V7.1.
- **Search for connected hardware**
The Target Server program is started. A search for connected hardware is conducted.

Another program group named **License Management** is added in the Start menu under **ETAS**. The program group contains the following entry:

- **ETAS License Manager**
Starts the ETAS License Manager, where you can view and manage licenses for your ETAS software products (e.g. adding licenses or borrowing and returning licenses).

3.5 Licensing the Software

To be able to work with an ETAS software product, you require a license. This section contains basic details on this subject.

- "ETAS License Models" on page 73
- "How to get a License" on page 74
- "The License File" on page 75
- "Grace Mode" on page 77
- "The "Expiration Warning" Window" on page 77
- "Borrowing a License" on page 78

Details concerning the scope of the licenses and other legal aspects can be found in "Terms and Conditions".

3.5.1 ETAS License Models

There are three different license models available for licensing your ETAS software:

Machine-Named License, Local

- A license of this type is managed by the user him/herself.
- As it is linked to a particular PC (better: to the MAC address of the Ethernet adapter), it is valid wherever the PC is used.
- When you change your PC, you require a new license.

User-Named License, Server-Based

- The licenses (of a department or company) are managed centrally on a server by a designated person.
- The license is linked to the user name with which the user is registered in the network and is available on every PC in the network.
- If the relevant PC is disconnected from the network, the license can be "borrowed."

Concurrent (or Floating) License, Server-Based

Most of what is true of the user-named license applies to this type of license. The difference is that here several users share a limited number of licenses.

3.5.2 How to get a License

If your company has a tool coordinator and server-based license management for ETAS software, contact this person. Otherwise (in the case of a machine-named license) you obtain your license from the ETAS license portal (the URL is shown on your Entitlement Certificate).

There are three ways of logging in on the welcome page:

- **Activation ID**

Once you have logged in, a specific activation¹ is visible and can be managed – the activation ID is shown on your Entitlement Certificate.

- **Entitlement ID**

All activations of the entitlement² are visible and can be managed (e.g. for a company with just one entitlement).

- **E-mail and password**

All activations of the entitlements assigned to the user account are visible and can be managed (e.g. for a tool coordinator responsible for several entitlements).

If you need help in the portal, click the [Help](#) link.

What Information is required?

Information on the hosts must be entered to activate licenses:

- Machine-named license
The MAC address of the Ethernet adapter to which the license is to be bound is required here
- User-named license
Here, you need a server host or a server triad as well as a user name
- Concurrent (floating) license
Here, you need a server host or a server triad.

Note

If this data changes (e.g. due to changes in the hardware or a change of user), the license must be given a "rehost". This procedure is also described in the portal help file.

¹ The activations refer to a specific product, its license conditions, the available number of licenses and other details required for generating a license. Activations are identified uniquely with activation IDs.

² An entitlement shows the authorizations you have as a user; it stands for the right to own one or more licenses for a product. It is a kind of account of rights of use for software from which you can take licenses as you need to.

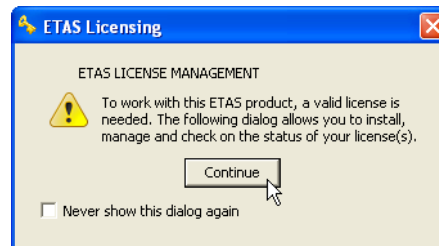
License File

The result of your activities is the provision of a file `<name>.lic` with which you can license your software in the ETAS License Manager.

3.5.3 The License File

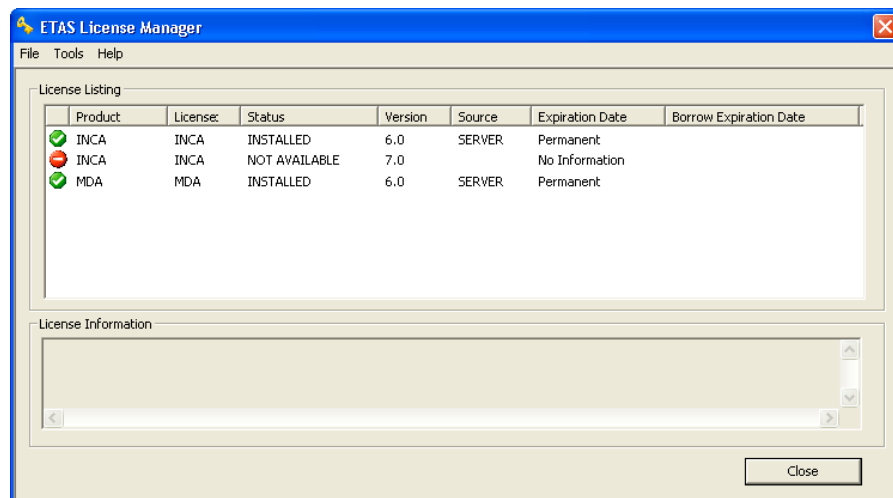
To check to license status

- In the Windows Start menu, select **Programs** → **ETAS** → **License Management** → **ETAS License Manager**.



- Click **Continue**.

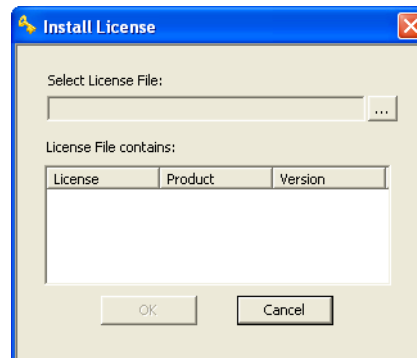
The ETAS License Manager contains one entry for each installed product. The symbol at the beginning of the entry and the "Status" column entry indicate whether a valid license has already been obtained or not.



To add a license file

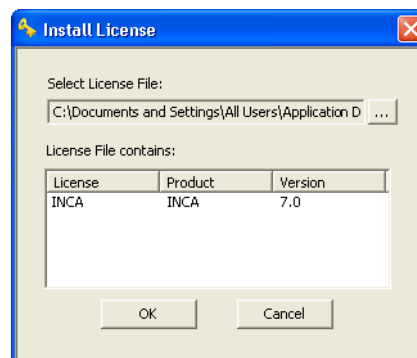
- Open the ETAS License Manager (cf. page 75) and select **File** → **Add Licensing File**.

The "Install License" dialog window opens.



- Next to the "Select License File" field click the ... button.
- In the file selection window, select the license file and click **Open**.

The "Install License" dialog window shows information on the selected license.



Note

The "Version" column shows the version number of the license, not the version number of the software.

- Confirm with **OK**.
The license just added is now listed in the ETAS License Manager. A green symbol before the entry shows that the license is valid.

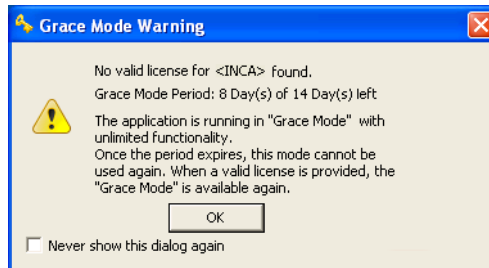
Note

If the green symbol is not displayed, there might be a problem with the license file or the license relates to another product. Additional information on the ETAS License Manager can be found in the online help of the ETAS License Manager.

- [Close the ETAS License Manager.](#)

3.5.4 Grace Mode

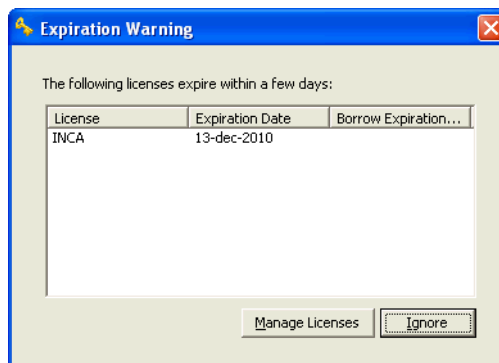
If you have not yet installed a license, you can still operate the software for a limited amount of time – it then runs in what is referred to as grace mode. The window shows you how long you can continue to operate the software in this mode.



3.5.5 The "Expiration Warning" Window

If your installed license runs out in the next 30 days, a warning is shown when you open the ETAS software.

The "Expiration Warning" window contains a list of licenses that expire in the next 30 days. The expiration date is shown for each license; in the case of borrowed licenses (see "Borrowing a License" on page 78) it also shows when the borrowing period expires (i.e. the date when borrow mode runs out).



Click [Manage Licenses](#) to open the ETAS License Manager and install a valid license file. As soon as you have installed a valid license, you can continue to operate the ETAS software in normal operating mode.

Click [Ignore](#) to close the dialog box and start the ETAS software. This is only possible during the grace period; as soon as the grace period has expired, you can only continue to use the ETAS software once you have installed a valid license file.

As soon as the expiration date has been reached, you can continue to use the ETAS software for a further 14 days in what is referred to as limited mode (see "Grace Mode" on page 77). Once this phase is over, the ETAS software can only be used when a new or updated license file has been installed.

3.5.6 Borrowing a License

The borrowing mechanism makes it possible to work offline even when using a server-based license (i.e. without being connected to the license server).

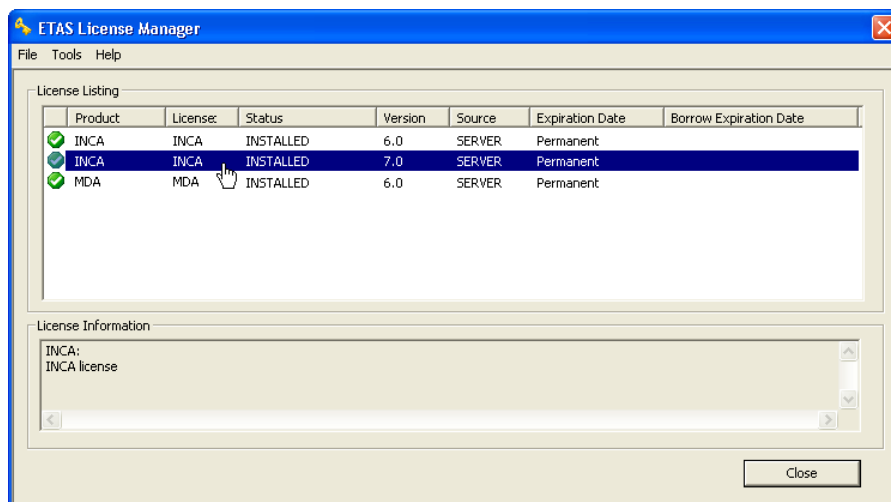
Note

You can only borrow a license if a server-based license is being used!

To borrow a license, proceed as follows:

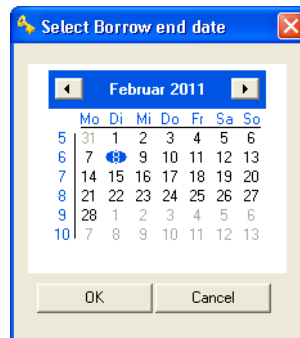
To borrow a license

- Make sure that the ETAS software the license of which you want to borrow is not open.
- Select the license you want to borrow in the "License Listing" table of the ETAS License Manager.

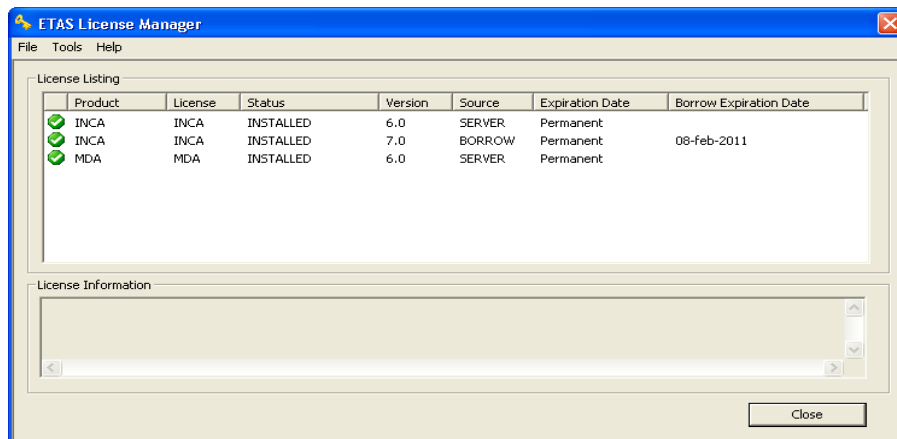


- Select **License** → **Borrow license**.
The "Select Borrow end date" dialog box opens.

- Select the date until which you want to borrow the license from the calendar displayed and click OK.



The text in the "Source" column of the ETAS License Manager changes from "SERVER" to "BORROW", and the expiration date of the borrowed license is displayed.



You can now use the relevant ETAS software offline until the expiration date of the borrowed license has been reached.

If you want to use the ETAS software longer than you had originally planned, you can borrow the license again. If you stop using the ETAS software earlier than planned, you can return the license to the license server early ([License](#) → [Return License Early](#)). A borrowed license can only be returned by the person that borrowed it; it cannot be returned by another person.

3.5.7 Customizing the Network Installation

You can change certain default settings before conducting the network installation; this also includes the access to the required licenses. The customization of the installation is done by adjusting the configuration file **install.ini**, which is located in the installation directory of the product to be installed.

3.6 Uninstall Procedure

3.6.1 Automatic Uninstall

If you would like to uninstall INCA, please always use the Uninstall routine that is installed together with INCA.

Note

Add-Ons are uninstalled automatically together with INCA. It is not possible to uninstall them separately.

To uninstall INCA:

- Select **INCA Uninstall** from the INCA **program group** and follow the instructions displayed on the screen.

4 ETAS Network Manager

The ETAS Network Manager is used for creating a configuration that will be used by the ETAS IP Manager. The IP Manager is responsible for dynamic IP addressing of the ETAS hardware used in your network (ETAS network).

4.1 Overview

ETAS software supports different configurations for hardware access via Ethernet:

- Using multiple network adapters:
 - one network adapter for the company network,
 - one or more network adapters for the ETAS hardware.
- Using one network adapter
 - automatic toggling between the company network and the ETAS hardware.

Note

You do not require a separate network adapter to connect the ETAS hardware to your PC. You can use the same network adapter both for the company network and the ETAS network.

The ETAS Network Manager supports you in selecting the network adapter for the ETAS hardware.

The ETAS Network Manager gives you an overview of the network adapters available for your PC and the type of IP address assignment. If more than one network adapter is available in the system, you can select the network adapter to use for connecting the ETAS hardware to your PC. You can also specify the address range for the IP assignment for the ETAS hardware.

You do not need administrator rights to select the network adapter and the network environment configuration for the ETAS hardware. You can toggle between the ETAS network and the company network without rebooting your PC.

Note

With the ETAS Network Manager, you cannot create or modify the configuration for the network adapter. Instead modify the network settings of your PC via the Control Panel (see the documentation for your operating system). Please note that this requires administrator rights.

4.2 ETAS Hardware Addressing

You can connect several ETAS devices (including those that are the same type) to your PC. The connected devices are identified in the local ETAS network by their unique IP address.

An IP Manager integrated in the ETAS software looks up which IP addresses are available in a pre-configured address pool and assigns available IP addresses to the connected ETAS hardware.

The address range for the address pool is specified using the ETAS Network Manager.

4.3 Network Adapter Addressing

4.3.1 Type of Network Adapter Addressing

The type of network adapter addressing done within the company network depends on the operating system being used and the network adapter configuration:

Operating System	Type of Network Adapter Addressing			
	Manual	DHCP	DHCP+APIPA	DHCP+ alternative IP address
Windows XP	yes	yes	yes	yes
Windows Vista	yes	yes	yes	yes
Windows 7	yes	yes	yes	yes

The ETAS network supports the following types of network adapter addressing:

Operating System	Type of Network Adapter Addressing			
	Manual	DHCP	DHCP+APIPA	DHCP+ alternative IP address
Windows XP	yes	no	yes	yes
Windows Vista	yes	no	yes	yes
Windows 7	yes	no	yes	yes

If you wish to use the network adapters both for the company network and the ETAS network, you cannot use the network adapters that exclusively support DHCP addressing for this dual operation.

Note

DHCP can be used only in combination with APIPA or an alternative IP address!

4.3.2 Addressing the Network Adapter Manually

Addressing a network adapter depends on the operating system.

For instructions on addressing your PC's network adapter, see the documentation for your operating system.

To address the network adapter manually, you need administrator rights. Please contact your system administrator, if necessary.

If the network adapter is addressed manually, i.e., it has a static IP address, it may happen that you accidentally end up searching for or initialize ETAS hardware, although the PC is connected to the company network. The Network Manager allows you to stipulate that if this happens, you are to receive a warning before an IP address is assigned to an ETAS hardware.

4.3.3 Addressing the Network Adapter via DHCP

Addressing via DHCP requires that the DHCP server be available. Should the DHCP server not be available, or if there is no DHCP server (as in the ETAS network), the network adapter has not been configured.

In this instance, each operating system has a feature that automatically assigns the network adapter an IP address:

Windows XP / Windows Vista / Windows 7

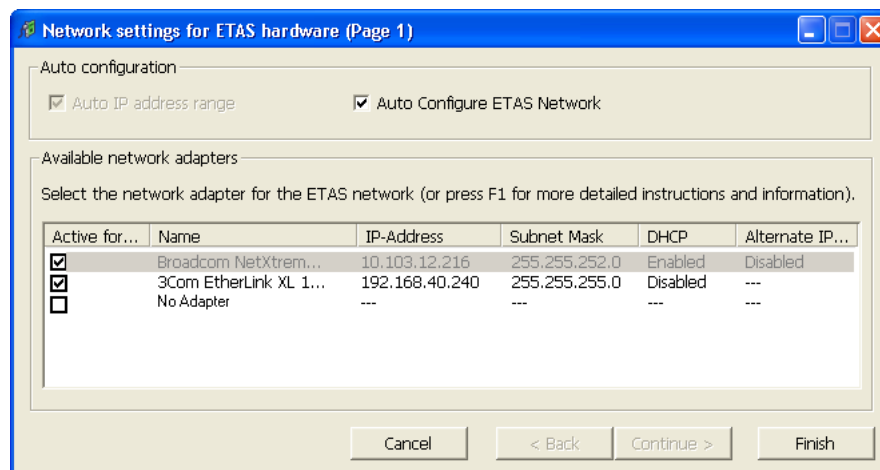
Windows XP, Windows Vista and Windows 7 automatically check whether there is a connection to the DHCP server. If there is none, it either assigns the IP address automatically via APIPA, or it uses the user-specified alternative IP address. The ETAS network always uses either the APIPA address or the alternative IP address.

When toggling between the DHCP network and ETAS hardware, make sure that the operating system is able to detect a connection failure because only then will reconfiguration be initiated. This may take up to 10 seconds. It takes the operating system 60 seconds to entirely reconfigure from a DHCP address to an APIPA address or to the alternative address. If the network adapter is once again connected to the DHCP network, configuring to a DHCP address takes place right after the connection has been detected.

Addressing a network adapter via DHCP without alternative addressing is not supported.

4.4 User Interface

4.4.1 Configuration Dialog Window 1 — Selection of Network Adapter ("Network settings for ETAS hardware (page 1)")



The following information on the available network adapters is displayed:

- **Auto IP Address Range** checkbox

If you tick this checkbox, the next configuration step is skipped, and the ETAS Network Manager automatically assigns default IP address ranges that will be used by the selected network adapter for addressing the ETAS hardware. If the IP address range is automatically changed by the IP Manager, a message is displayed in the system tray.

- **Auto Configure ETAS Network** checkbox

If you tick this checkbox, you can enable or disable several network adapters at once for auto-configuration through the IP Manager.

When you tick the checkbox, the "Auto IP Address Range" checkbox is activated. In the list of available network adapters, the "Active for ETAS Network" column is inserted, where you can determine which network adapters shall be available for auto-configuration through the ETAS Network Manager.

The ETAS Network Manager will go through the list (top-down) and use the first adapter which has a valid IP configuration for ETAS¹ and configure the IP address range automatically. If a configured network adapter fails, e.g. because the network adapter is being disabled or physically not available, the IP Manager will configure the next available network adapter automatically, and a message is displayed in the system tray, indicating the new configuration.

- **Active for ETAS Network** column

This column is only visible if the "Auto Configure ETAS network" checkbox is ticked. In the checkbox in this column you can determine which network adapters shall be enabled for autoconfiguration by the ETAS Network Manager (see „Auto-Configuring Network Adapter for ETAS Hardware“ on page 89).

- **Name** column

Name of the network adapter. This entry cannot be edited in this window.

This column also contains an entry with the name "No Adapter". If you want to disable the IP Manager, e.g. because there is no Ethernet hardware connected to the PC, you can tick this entry in the "Active for ETAS Network" column.

- **IP Address** column

IP address of the network adapter. This entry cannot be edited in this window.

- **Subnet Mask** column

Setting for the subnet mask. This entry cannot be edited in this window.

- **DHCP** column

Shows whether the network adapter is configured for DHCP:

- Enabled

The network adapter is configured for DHCP.

- Disabled

The network adapter is configured with a fixed IP address.

- **Alternate IP Configuration** column

Shows the alternative IP address of the network adapter if it is configured for DHCP. This indication depends on the operating system being used.

¹ An IP configuration is valid if the network adapter either uses a fix IP address, or if DHCP and APIPA are enabled.

- APIPA

Automatic Private IP Addressing: method for automating the IP configuration for network connections

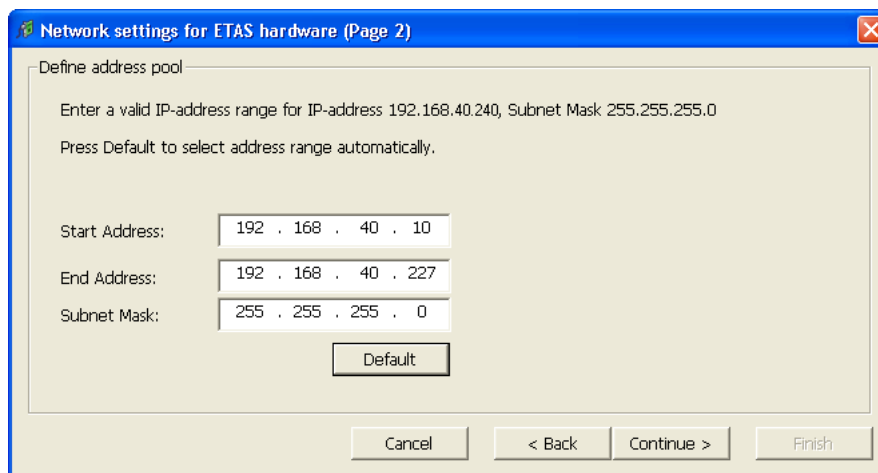
- ---

An alternative IP address does not exist.

- User defined

The user can define a user-specific alternative IP address.

4.4.2 Configuration Dialog Window 2 — Defining the Address Pool ("Network settings for ETAS hardware (page 2)")



In general, all values can be modified by directly typing them in the corresponding field, or by selecting the default setting from a list box.

The following network parameters can be set:

- **Start Address**

The first IP address in the IP address range for the ETAS hardware

- **End Address**

The last IP address in the IP address range for the ETAS hardware

- **Subnet Mask**

Associated Subnet Mask

Reserved IP Addresses

The following IP addresses are reserved for certain ETAS hardware in the IP address range that the ETAS hardware (192.168.40.1 - 192.168.40.254 with Subnet Mask 255.255.255.0) is currently using:

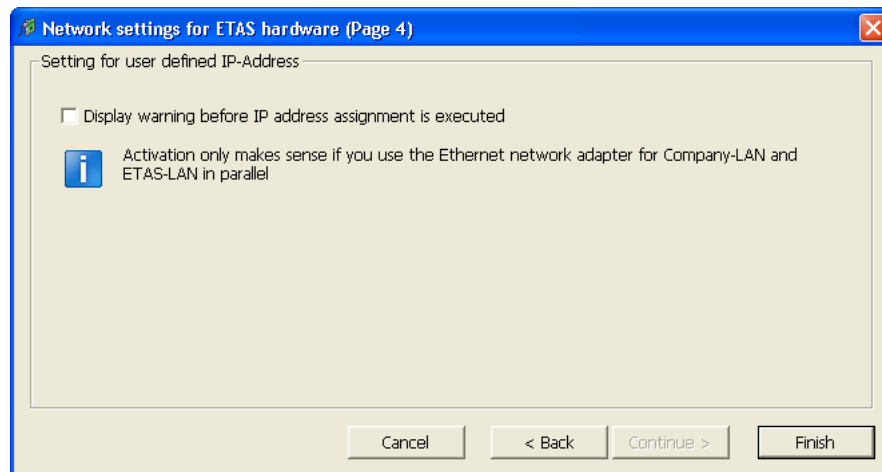
IP_Address	ETAS Hardware
192.168.40.10	ES1120
192.168.40.11	ES1130
192.168.40.12	ES780

IP_Address	ETAS Hardware
192.168.40.13	Reserved
192.168.40.14	LABCAR-RTPC
192.168.40.15	ES1135

These addresses are assigned exclusively to these devices and thus may not be used for other ETAS hardware. This has to be taken into consideration when defining the address pool.

4.4.3 Configuration Dialog Window 4 — Displaying Warning ("Network settings for ETAS hardware (Page 4)")

This dialog window appears only if the selected network adapter is addressed manually.



The following parameters can be set:

- **Display warning before IP address assignment is executed**
Use this checkbox to specify that a warning be displayed before an IP address is assigned to an ETAS hardware device.

Note

Enabling this warning is useful only if you want to run the PC both in the company network or on an ETAS measurement module in the ETAS network using this network adapter.

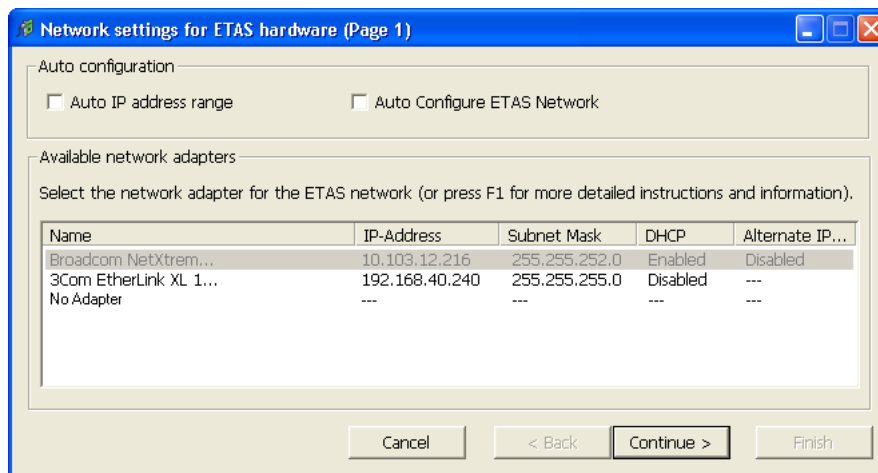
4.5 Configuring Network Addresses for ETAS Hardware

4.5.1 Manually configuring the Network Adapter

To configure the network adapter for ETAS hardware:

1. In the Windows Start menu, go into the program folder of your ETAS software (below **Start** → **Programs** → **ETAS**), and select **ETAS Network Settings**.

The "Network settings for ETAS hardware (Page 1)" dialog window is opened.



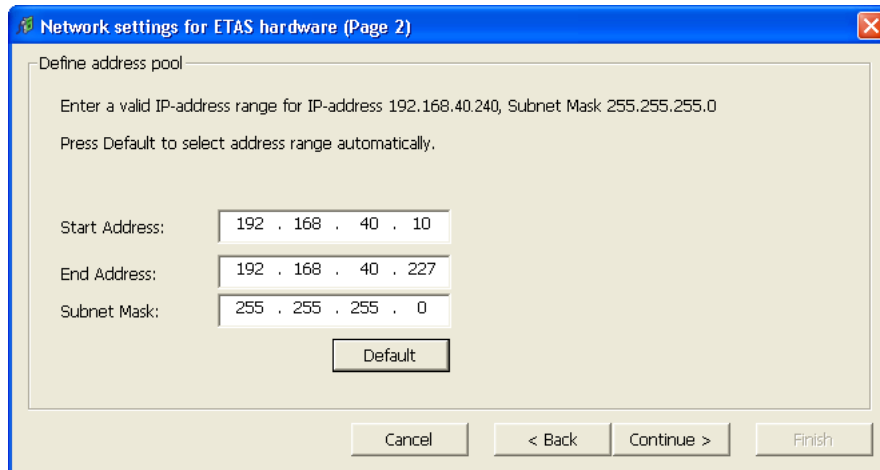
2. In the "Available Network Adapters" area, select the network adapter you want to use for the corporate network and the ETAS network. If no network adapter is installed on the PC, the dialog window contains no entries.

Note

The type of addressing of the network adapter must be supported by the ETAS network. Otherwise, you cannot select the network adapter.

3. Click **Continue**.

The "Network settings for ETAS hardware (Page 2)" dialog window opens.

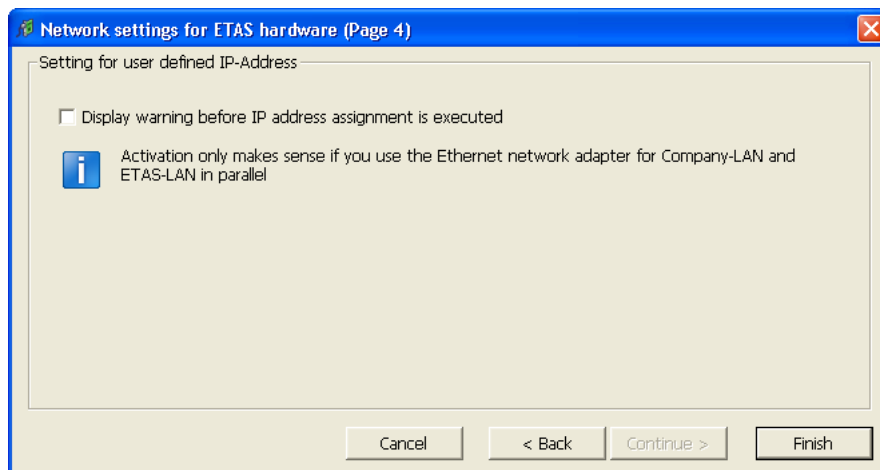


4. Enter the IP address range for the ETAS hardware and the subnet mask, or click **Default** to have the Network Manager automatically complete the IP address range and the subnet mask.

You may accept these settings or overwrite them.

5. Click **Continue**.

The "Network settings for ETAS Hardware (Page 4)" dialog window opens.



6. If you want to define that a warning is displayed before the ETAS hardware is assigned an IP address, enable "Display warning before IP address assignment is executed".

Note

Enabling this warning is useful only if you want to run the PC both in the company network or on an ETAS measurement module in the ETAS network using this network adapter.

7. Click **Finish**.
8. Restart the ETAS software to make the changes become effective. If the Network Manager was started automatically during a hardware search or initialization, you do not have to restart the ETAS software to apply the changes.

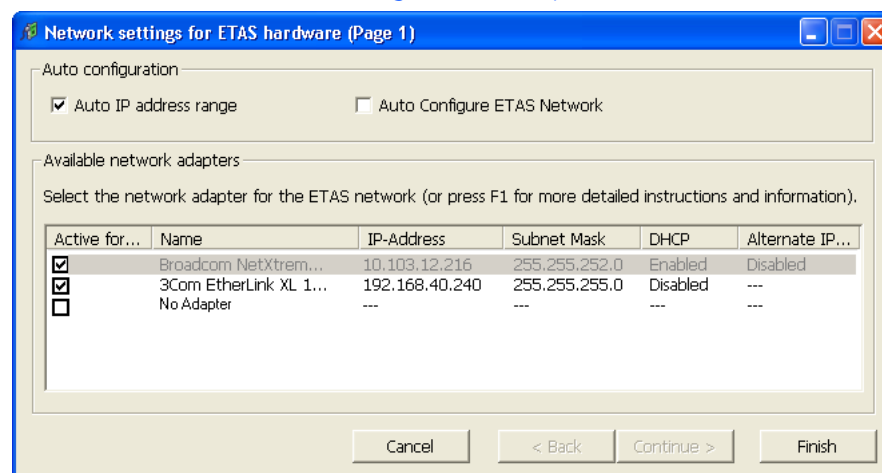
4.5.2 Auto-Configuring Network Adapter for ETAS Hardware

The ETAS Network Manager offers the automatic assignment of IP address ranges in the ETAS network through the IP Manager. You can either use auto-configuration of a selected network adapter, or, if more than one network adapter is available, let the IP Manager auto-configure the ETAS network, including the selection of the adapter to be used.

How to auto-configure a selected network adapter for ETAS hardware

1. In the Windows Start menu, go into the program folder of your ETAS software (below **Start** → **Programs** → **ETAS**), and select **ETAS Network Settings**.

The "Network settings for ETAS hardware (Page 1)" dialog window is opened.



2. In the "Available Network Adapters" area of the "Network settings for ETAS hardware (Page 1)" dialog window, select the network adapter you want

to use for the corporate network and the ETAS network. If no network adapter is installed on the PC, the dialog window contains no entries.

Note

The type of addressing of the network adapter must be supported by the ETAS network. Otherwise, you cannot select the network adapter.

3. Tick the "Auto IP Address Range" checkbox.
4. Click **Finish**.

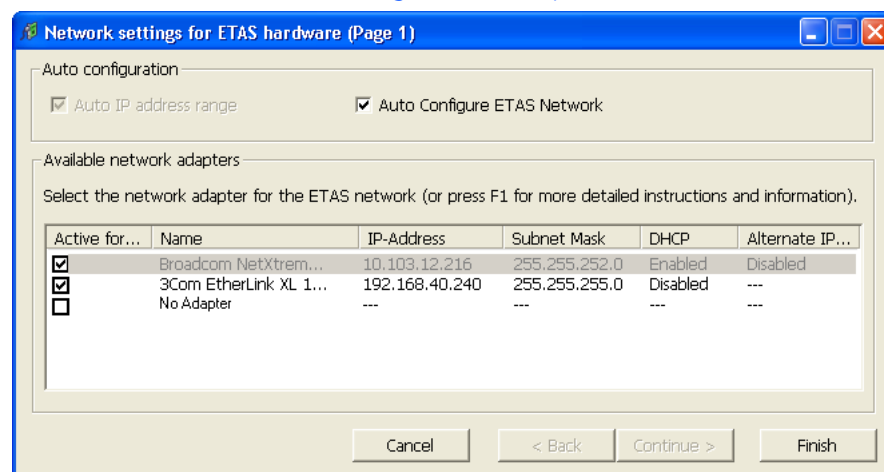
The ETAS Network Manager automatically assigns default IP address ranges that will be used by the selected network adapter for addressing the ETAS hardware.

If later on, the IP address range is automatically changed by the IP Manager, a message is displayed in the system tray.

How to auto-configure a network adapter, with multiple network adapters being available

1. In the Windows Start menu, go into the program folder of your ETAS software (below **Start** → **Programs** → **ETAS**), and select **ETAS Network Settings**.

The "Network settings for ETAS hardware (Page 1)" dialog window is opened.



2. In the "Network settings for ETAS hardware (Page 1)" dialog window, tick the "Auto Configure ETAS Network" checkbox.

The "Auto IP Address Range" checkbox is activated, and in the list of available network adapters, the "Active for ETAS Network" column is inserted.

3. Enable all network adapters that shall be available for auto-configuration through the IP Manager by ticking the corresponding checkbox in the "Active for ETAS Network" column.
4. Click **Finish**.

Starting with the network adapter that is currently selected in the dialog window, the ETAS Network Manager will go through the list (top-down), and use the first adapter which has a valid¹ IP configuration for ETAS and configure the IP address range automatically. If a configured network adapter fails, e.g. because the network adapter is being disabled or physically not available, the IP Manager will configure the next available network adapter automatically, and a message is displayed in the system tray, indicating the new configuration.

4.5.3 Enabling APIPA in the Registry

Note

You need administrator rights to make changes in the Registry.

To enable APIPA in the Registry:

1. Click **Start**, and then click **Run**.
2. Enter `regedit`, and click **OK**.
The Registry editor opens.
3. Click the entry for a network adapter in the Registry key:

Global key (refers to all network adapters)

`HKEY_LOCAL_MACHINE\SYSTEM\
CurrentControlSet\
Services\Tcpip\Parameters`

Interface or card-specific key

`HKEY_LOCAL_MACHINE\SYSTEM\
CurrentControlSet\
Services\Tcpip\Parameters\
Interfaces\adapter_name`

Note

If neither of these keys exists, the APIPA mechanism is enabled. If the global key exists but is not enabled, the setting of the global key is decisive. If the global key does not exist or is set to enabled, the card-specific key is decisive.

4. Set the value for `IPAutoconfigurationEnabled` to 1.
The APIPA mechanism is enabled.

¹ An IP configuration is valid if the network adapter either uses a fix IP address, or if DHCP and APIPA are enabled.

4.6 Troubleshooting Ethernet Hardware Access

In certain cases you might have problems accessing ETAS hardware via the Ethernet interface. For instance, if APIPA, a mechanism for IP addressing, has not been enabled on your system, you cannot select a network adapter in the Network Manager. You can find descriptions of potential problems and their solutions in "Appendix B: Troubleshooting General Problems" on page 163.

5 Getting Started

5.1 Introduction

This chapter is aimed at new users of INCA helping them to get started with INCA V7.1. You will receive an overview of the program functionality and working principles by means of practice-oriented work examples presented as flow diagrams. The essential working steps for calibration are presented in the framework of a concrete example, the lambda control for engine controllers.

The following briefly explains the major abbreviations used here.

- **CDM** – Calibration Data Manager
- **WS** – Workspace
- **DB** – Database
- **DBM** – Database Manager
- **EE** – Experiment Environment
- **EXP** – Experiment
- **HWC** – Hardware Configuration

The following sections give a brief overview of the individual working steps, separately by program user interfaces.

DBM (see „Working with the Database Manager – Preparations“ on page 94):

- Creating a New Database, Folder, Workspace, and Experiment
- Setting up a Project
- Setting up the Workspace: Assign project and dataset, select project hardware, allocate experiment

HWC Editor (see „Working in the Hardware Configuration Editor“ on page 98):

- Adding, Configuring and Initializing Hardware
- Matching Data Versions Between the PC and ECU

EE (see „Working in the Experiment Environment“ on page 100):

- Selecting Measure and Calibration Variables
- Configuring the Display
- Using Layers in an Experiment
- Configuring Measure Data Recording
- Performing a Measurement
- Performing Calibrations in the Editors
- Saving Datasets

CDM (see „Editing Datasets with the CDM“ on page 108):

- Selecting Datasets and Calibration Variables
- Executing the List, Compare, or Copy Action
- Analyzing Result and Output Files

DBM (see „Data Processing and Data Exchange in the DBM“ on page 111):

- Working with Database Items

- Exchanging Data Using the Import/Export Function
- Working with Databases

5.2 Working with the Database Manager – Preparations

The main purpose of the DBM is to systematically store all data that are created during the calibration process (workspaces, experiments, projects and datasets) in a database and manage them through a clear and well-organized user interface. Similar to the Windows Explorer, you can use the Database Manager to create directories and items as well as to move, copy, import, and export individual items and also to create entirely new databases.

You can organize all data centrally and independently from projects while always keeping a clear overall picture.

The following data are managed in the database:

- Workspaces (including hardware configuration)
- Experiments
- ECU projects (A2L) and datasets
- CDM configurations
- Calibration Scenario Configurations
- Measure catalogs
- ECU documentation or link to ECU documentation
- AUTOSAR system templates
- CAN DB descriptions
- CAN Message Lists
(available only if the component **CAN MessageSending** has been installed)
- FIBEX files with configurations for communication via the Flexray bus
(Flexray functionality requires the INCA-FLEXRAY add-on)
- LDF files with configurations for communication via the LIN bus
(LIN functionality requires the INCA-LIN add-on)
- ODX project files for diagnostic issues
(Diagnostic functionality requires the ODX-LINK add-on)
- ODX Flash project files with configurations for flashing via the ODX interface
(ODX flashing functionality requires the ODX-FLASH add-on)

The following types of data are **not** managed in the database:

- measure files (*.dat)
- Custom user interfaces (*.vui)
- *.a2l and *.dbc files
- *.hex and *.s19 files

Database items specific to an experiment mostly consist of a workspace and an experiment. When using an ECU interface, at least one project with associated working and reference datasets is also included. The following section teaches you how to set up a new database, create new database items and organize them in folders.

5.2.1 Creating a New Database, Folder, Workspace, and Experiment

INCA allows you to work with multiple databases. This results in better performance because the data volume is kept small and easy to handle. Before you generate data, you should set up a new database and create a folder structure that matches your specific requirements. This makes it easier later on to assign data to certain vehicles and experiment setups. The advantage of a new database is that you will not be confused by any other entries. Once the folder structure is in place, start creating the various database items, which are initially still "empty".

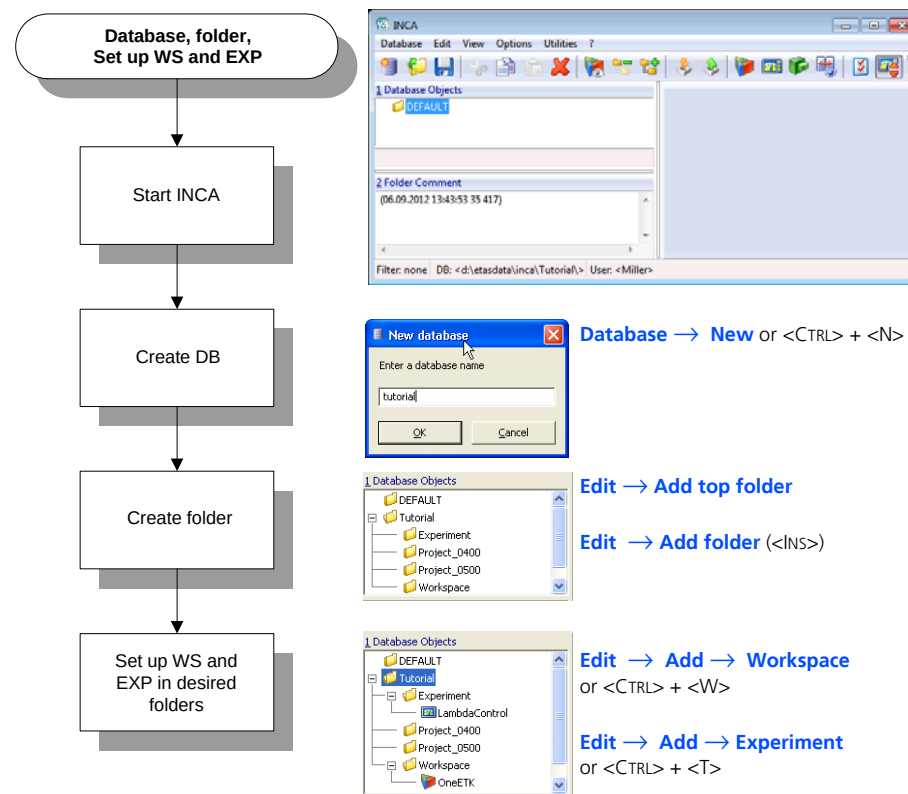


Fig. 5-1 Creating databases, folders and database items

5.2.2 Setting up a Project

Prerequisites for calibrating with INCA are creating a memory image of the control unit and storing the information in the form of datasets in INCA. This requires the following files: the project description file (*.a2l) with the physical description of the data and the hex file (*.hex, *.s19; Intel hex or Motorola format) containing the control unit program consisting of the code and the data. The different datasets of the working page and reference page are stored separately in INCA as the working dataset and the read-only reference dataset. Read-only datasets are identified by a red frame.

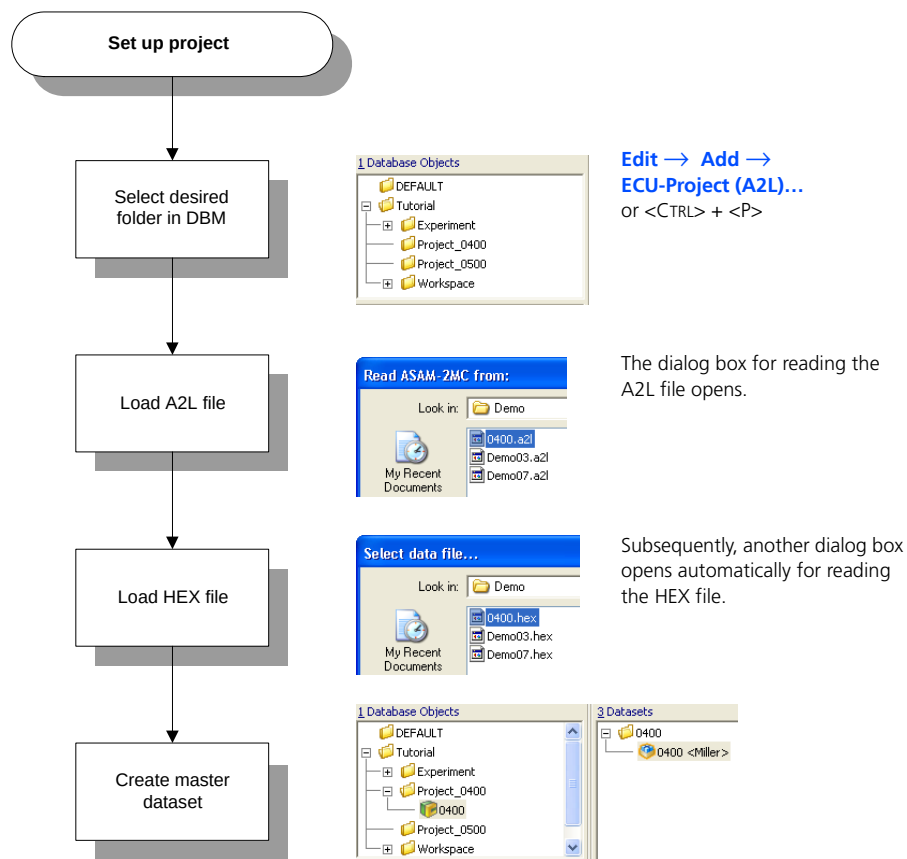


Fig. 5-2 Setting up a project

When loading the first HEX file, the code portion is mapped to the control unit project (transparent to the user). The data portion of this HEX file is stored as the so-called "Master" dataset. The master dataset is then used to also create the required working dataset.

5.2.3 Setting up the Workspace

When you create a workspace (WS) in the DBM, it is initially "empty", i.e., it still contains no references to other database items. To give life to this workspace, you have to assign a project to it, select the appropriate project hardware and allocate an experiment. When selecting a workspace, you can use the menus specific to each database item for the workspace to change or enter the entries in the "3 Experiment", "4 Project/device" and "5 Hardware" fields.

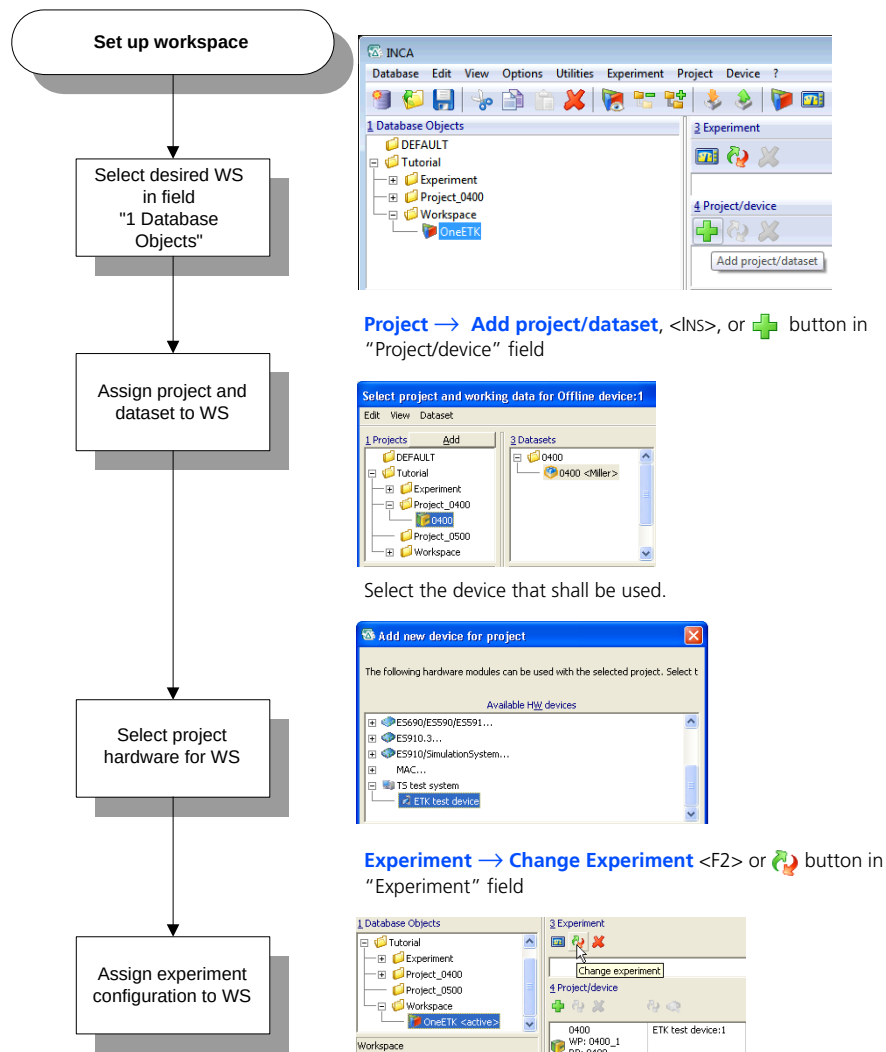


Fig. 5-3 Setting up the workspace

5.3 Working in the Hardware Configuration Editor

Essentially, the Hardware Configuration Editor enables you to manage and configure the hardware for the active workspace. Apart from the possibility just mentioned to inform the software about the hardware matching the project description file, the DBM also allows you to add further components from a list of possible modules (e.g., measuring hardware). You can use the Memory Page Manager integrated in the HWC Editor to match different data versions between the PC and ECU.

5.3.1 Adding, Configuring and Initializing Hardware

You can select different settings (e.g., module parameters, channel settings, etc.) for each hardware component using associated dialog boxes. When you add hardware components manually, the hardware still needs to be initialized.

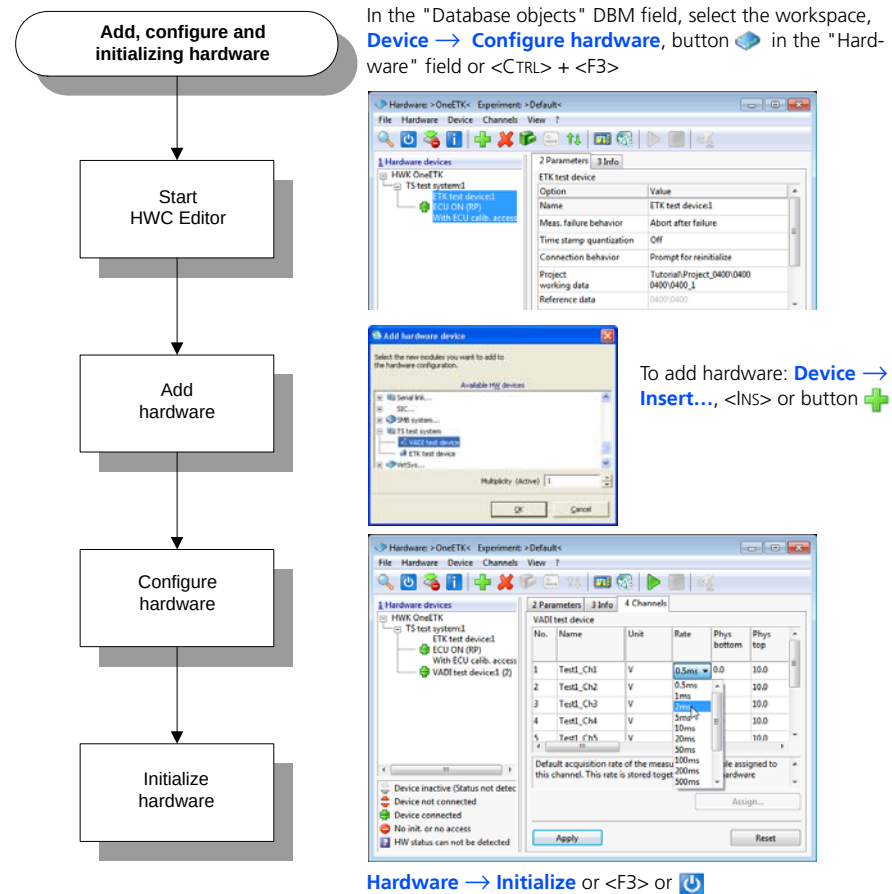


Fig. 5-4 Setting the hardware

5.3.2 Matching Data Versions Between the PC and ECU

INCA provides the Memory Page Manager to manage different datasets (working and reference datasets). This is a versatile tool which you can use to copy memory contents in any direction. For example, it allows you to read data versions into and from the control unit or copy data from the working data version to the reference data version or vice versa, save working datasets or write-protect them as an "intermediate result" and change reference datasets for the current project. The Memory Page Manager is opened automatically if different data versions are detected during the initialization of the control unit. You can call the Memory Page Manager both from the HWC Editor and the Experiment Environment by pressing <SHIFT> + <F8>.

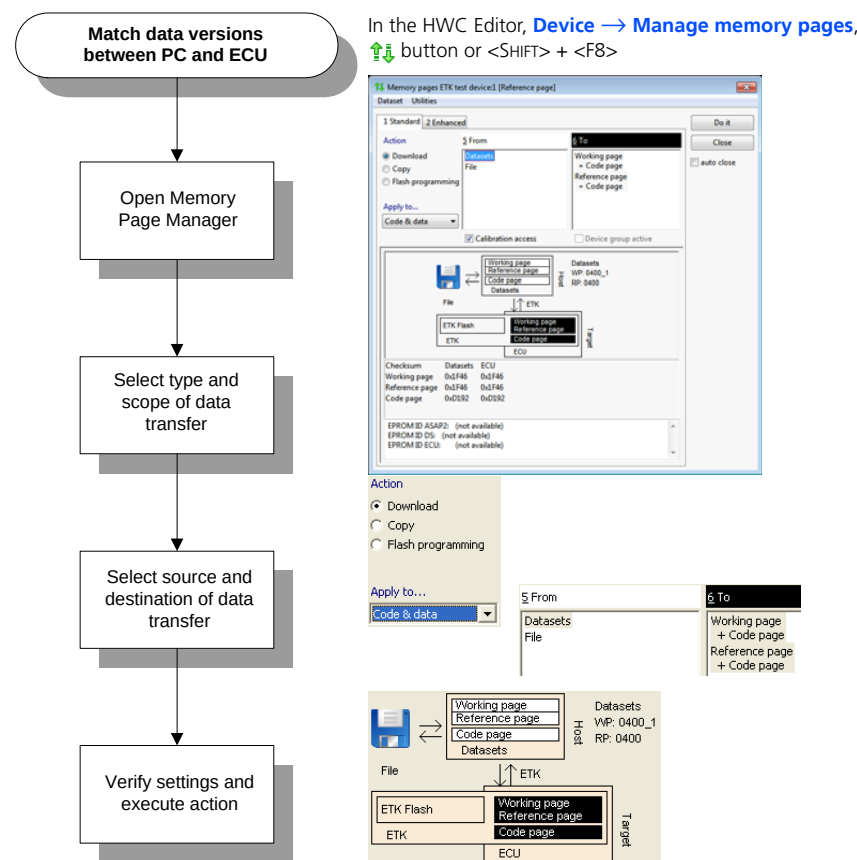


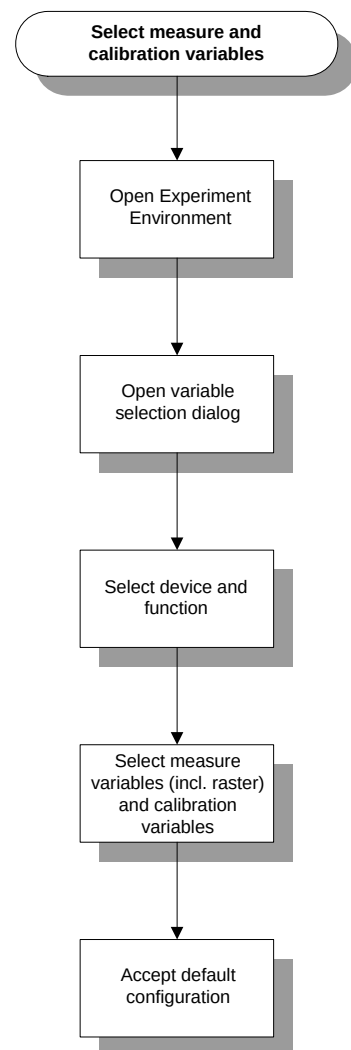
Fig. 5-5 Matching data versions

5.4 Working in the Experiment Environment

The Experiment Environment mainly contains the functionality required to perform a measurement and calibration task and used when preparing the individual partial tasks. It is used to select measure and calibration variables and their arrangement in various windows, such as oscilloscope, bar display, and table or graphical calibration editors.

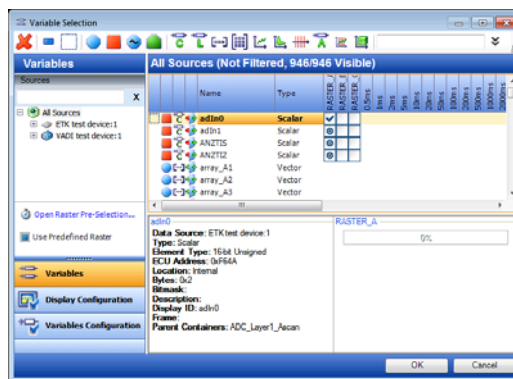
5.4.1 Selecting Measure and Calibration Variables

Depending on the hardware configuration used in the workspace, the experiment only provides the measure and calibration variables for selection which are defined in the connected hardware.

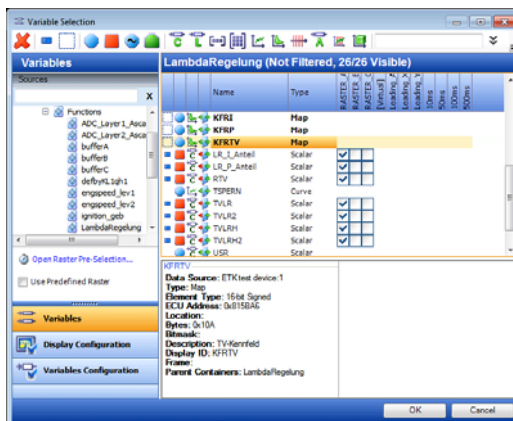


In DBM: in the "Database objects" field, double-click on the corresponding workspace, press <CTRL> + <F5> or select the workspace and in the "Experiment" field click

In EE: **Variables** → **Variable Selection** or <SHIFT> + <F4> or



In the "Sources" field select the hardware and, if necessary, function. If necessary use filter function in the Toolbar. In the tabular area make your selection and **OK**.

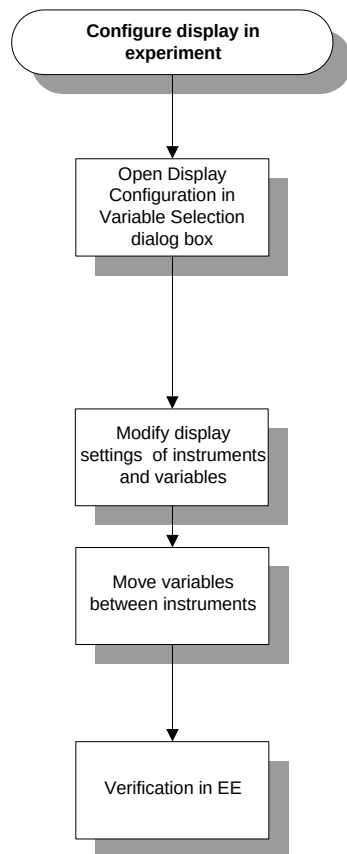


OK applies the selected variables and assigns the default measure and calibration windows for the variables.

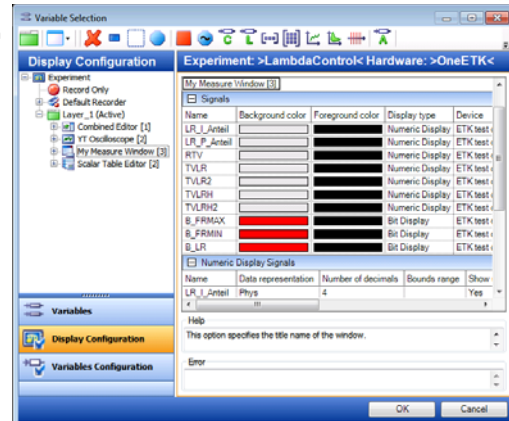
Fig. 5-6 Selecting the measure and calibration variables

5.4.2 Configuring the Display

For configuring the representation of measure and calibration variables, you have a range of measure and calibration instruments with the associated display options at your disposal. You can either select the adequate instrument for each variable individually or accept the instrument suggested by the program as a default. The following describes how to subsequently configure the representation.



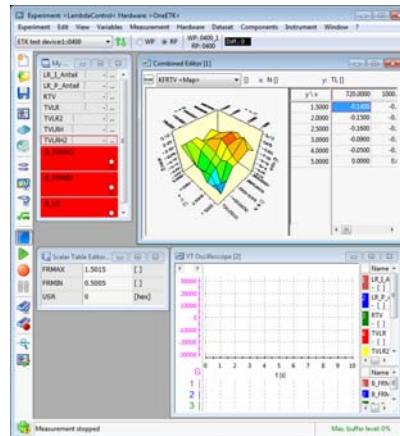
In EE: **Variables** → **Display Configuration** or <F4> or



In the tree view on the left, mark a window or a variable, then define specific display settings for that element in the table on the right.

Move variables from one window to another by drag & drop or cut & paste in the tree view.

Confirm your settings with **OK** and check the results in the Experiment Environment.



The display options can also be set directly from the shortcut menu of the instrument window (right mouse button or <SHIFT> + <F10>) → **Properties** or <ALT> + <RETURN>

Fig. 5-7 Configuring the experiment

The assignment of the variables, the arrangement of the individual instruments, and other settings such as color and display area are saved in the experiment.

5.4.3 Configuring the Usage of Variables

Apart from setting how the variables are displayed in the experiment, you can also set their raster, the recording status and whether they shall be used at all in the experiment.

It can be useful to set variables to inactive; they will still be saved in the experiment, but temporarily not used in the experiment, thus not filling the raster. They can easily be reactivated later, with the previous properties and instrument assignment being saved.

Moreover, you can modify the raster assignment and the assignment to the recorders that are configured in the experiment.

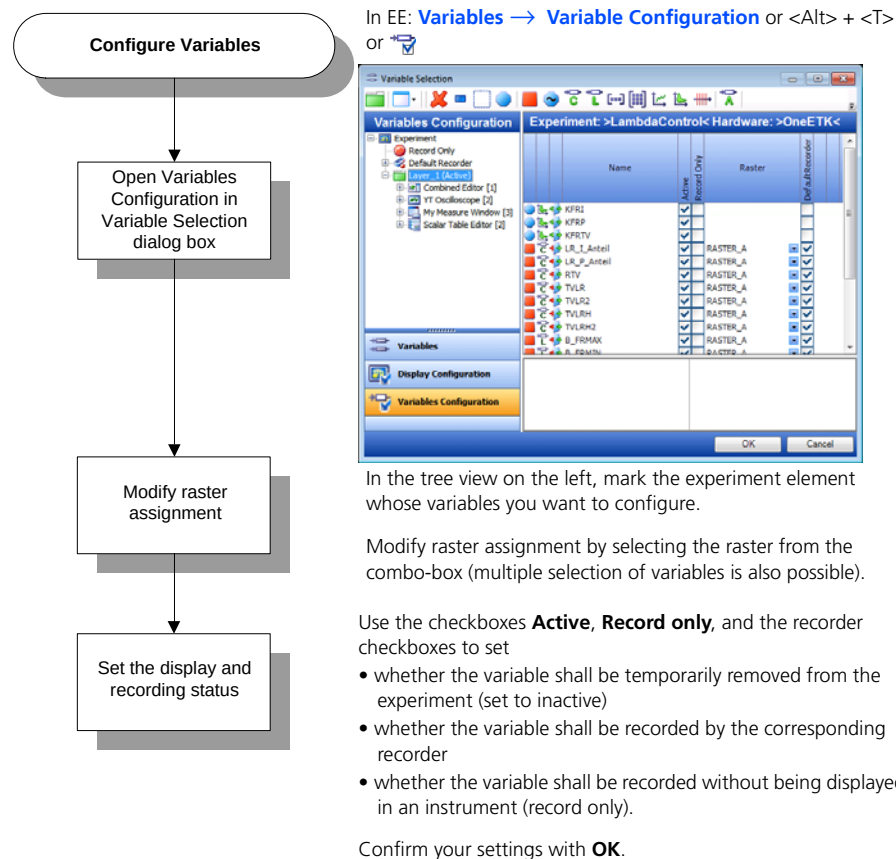


Fig. 5-8 Configuring variables

5.4.4 Using Layers in an Experiment

To improve the display of all measure and calibration instruments required in an experiment, INCA allows you to distribute the instruments to different layers. This is useful especially when working with small screens. You can create any number of layers and delete them as required. For example, you can define one layer for the measure instruments and another one for the calibration instruments. After creating the layers, you can distribute the instruments as required. Just click the mouse to move any layer to the foreground. Saving the experiment also saves all layers with their associated instruments.

A newly created experiment initially only contains one layer. After you have created additional layers, the footer of the experiment environment shows the associated tabs. To switch between the individual layers, just click on the corresponding tab or press <ALT> + <<> ("less than" character) or <ALT> + >>> ("greater than" character).

Layers can be configured in the Display Configuration or the Variables Configuration of the Variable Selection dialog.

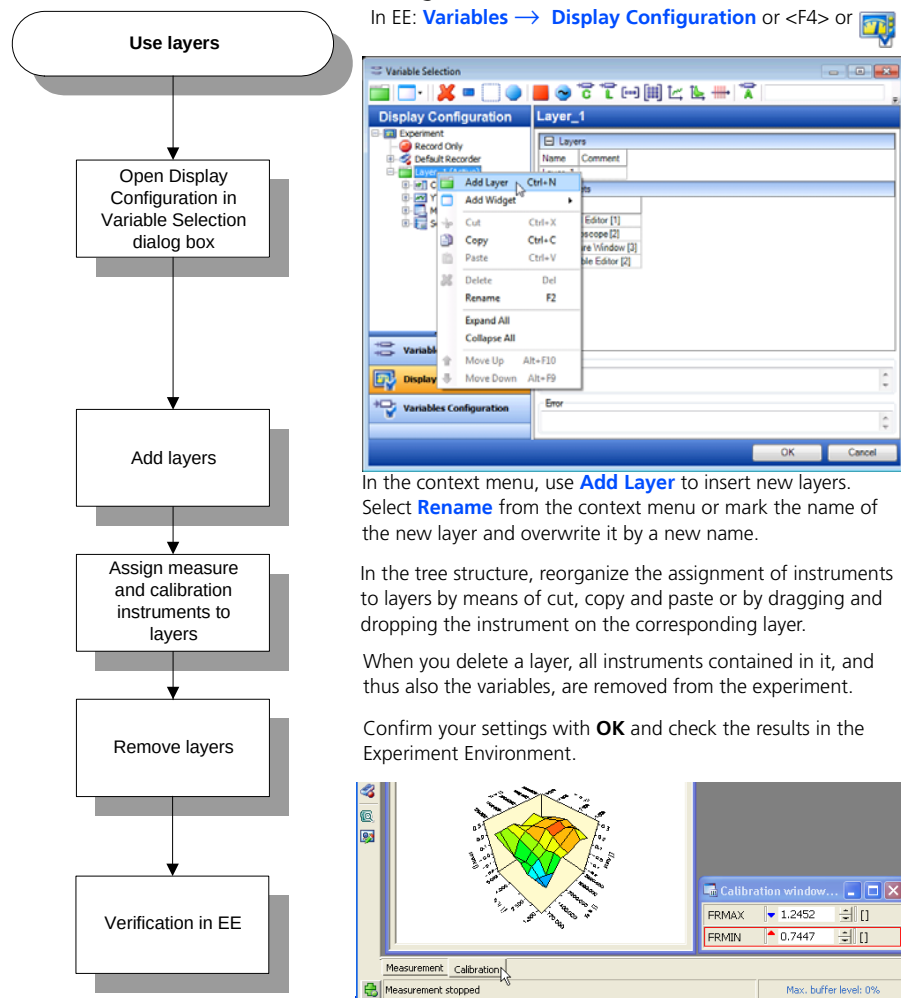


Fig. 5-9 Using layers in an experiment

5.4.5 Configuring Measure Data Recording

INCA provides extensive and convenient methods in a dedicated configuration window to prepare for and configure the storage of measured data. You can use them to specify the storage of the measure file by setup options such as storage location, storage name, format (ETAS binary file, ASCII, Matlab M-File, FAMOS, DIADEM and MDF) and other individual information.

To automate the start, repetition, and end of measure data recording, you can define trigger conditions (with pretrigger time and posttrigger time, if required) which trigger the desired event.

You can choose whether you just use one easy-to-use default recorder; as an alternative, you can individually configure different recorders that can be run separately from each other and that can be used for background recording. The following process describes the usage of the default recorder.

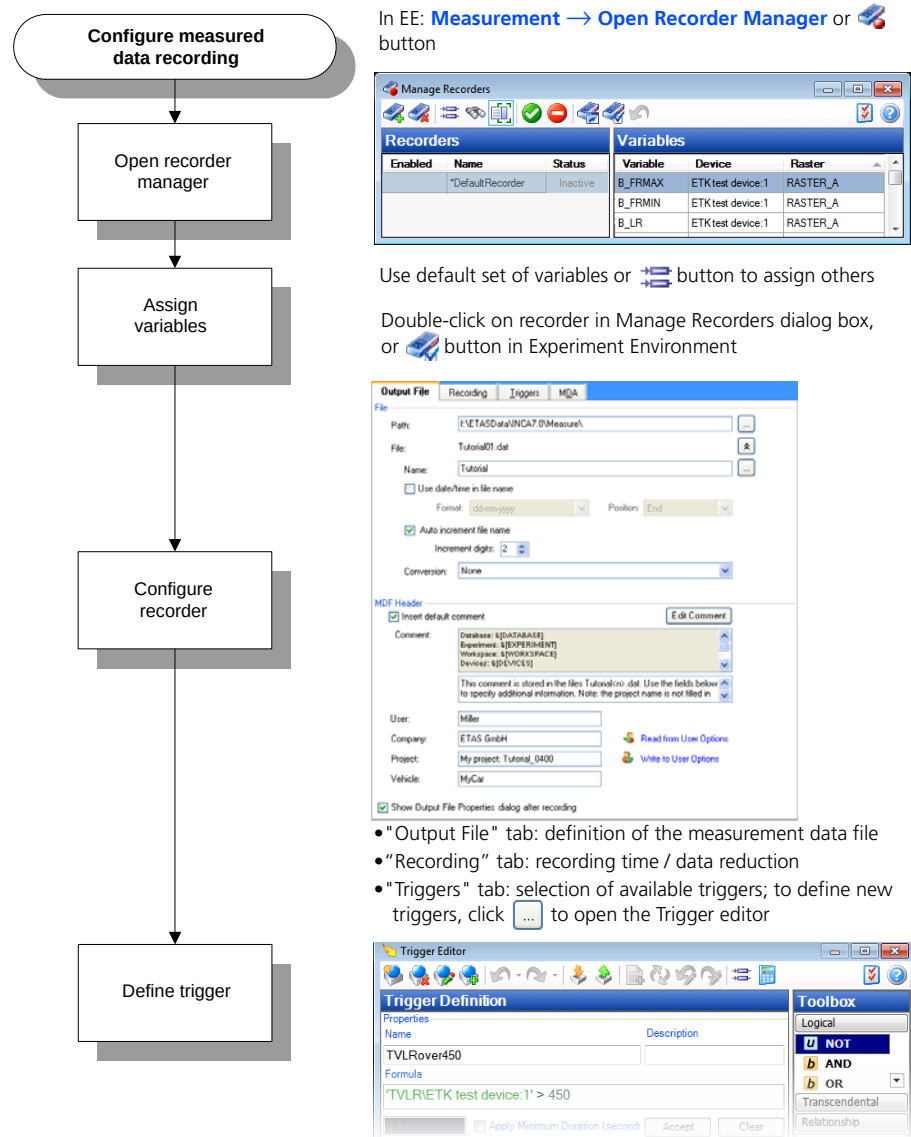


Fig. 5-10 Configuring measure data recording

5.4.6 Performing a Measurement

You have already finished all preparations. Now, you can start the actual measuring process. INCA offers you various measuring functionalities, which are supported by function keys. The measurement status is shown in the status bar. Please refer to the INCA online help for more detailed information.

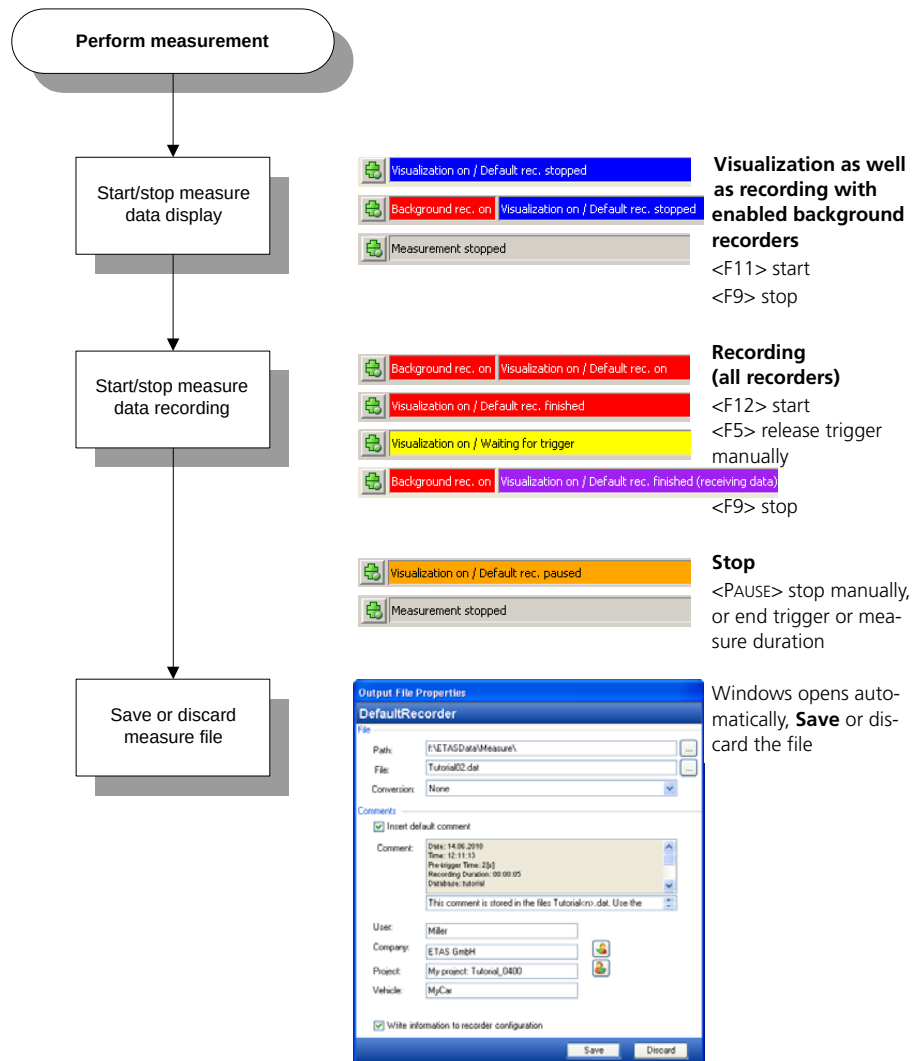


Fig. 5-11 Performing the measurement

5.4.7 Performing Calibrations in the Editors

The various calibration instruments display the calibration variables (scalars, curves, and maps) in the form of tables or graphics. The calibration instruments are also used as editors you can use to change the values of calibration variables directly in the visualization. INCA also assists you in operations such as offset, breakpoint and block calibrations. With curve and map editors, you can display the current process point of the engine.

The following editors are available: Calibration Window (numeric, slider or bit display), Calibration Table, Scalar Table Editor, Combined Editor.

All editors have one basic property in common: Since it is basically not possible to edit the values on the reference page, the editors change their background color to indicate whether the working or the reference page is currently being displayed. Differences in the values between the working dataset and reference dataset are also shown in the editors.

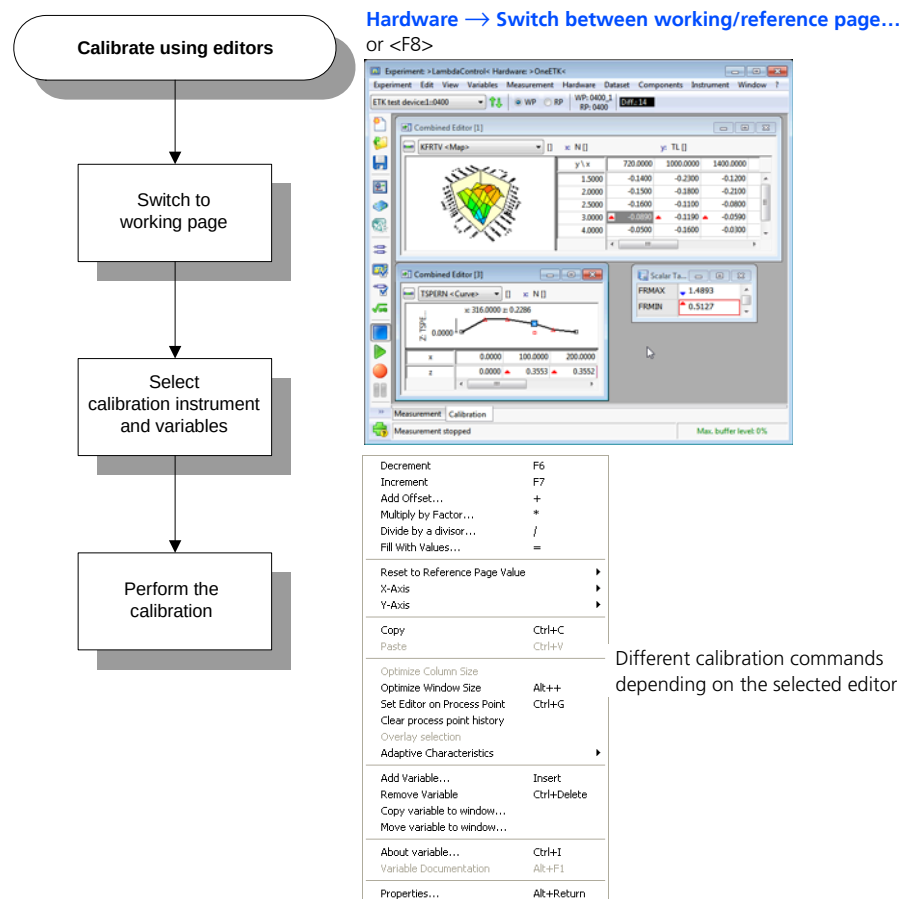


Fig. 5-12 Performing calibrations in the editors

The changes made in the editors become immediately effective.

5.4.8 Saving Datasets

INCA provides several means to save the current program version. To make your modified data available to other users, you can save the working dataset as a new HEX file in the file system. In INCA, you can save the current working dataset under a different name, to backup an intermediate version.

You can also copy the current data of the working dataset to a new reference dataset. The current reference dataset is write-protected and therefore cannot be replaced directly. Instead, you are prompted to enter a name for the new reference dataset.

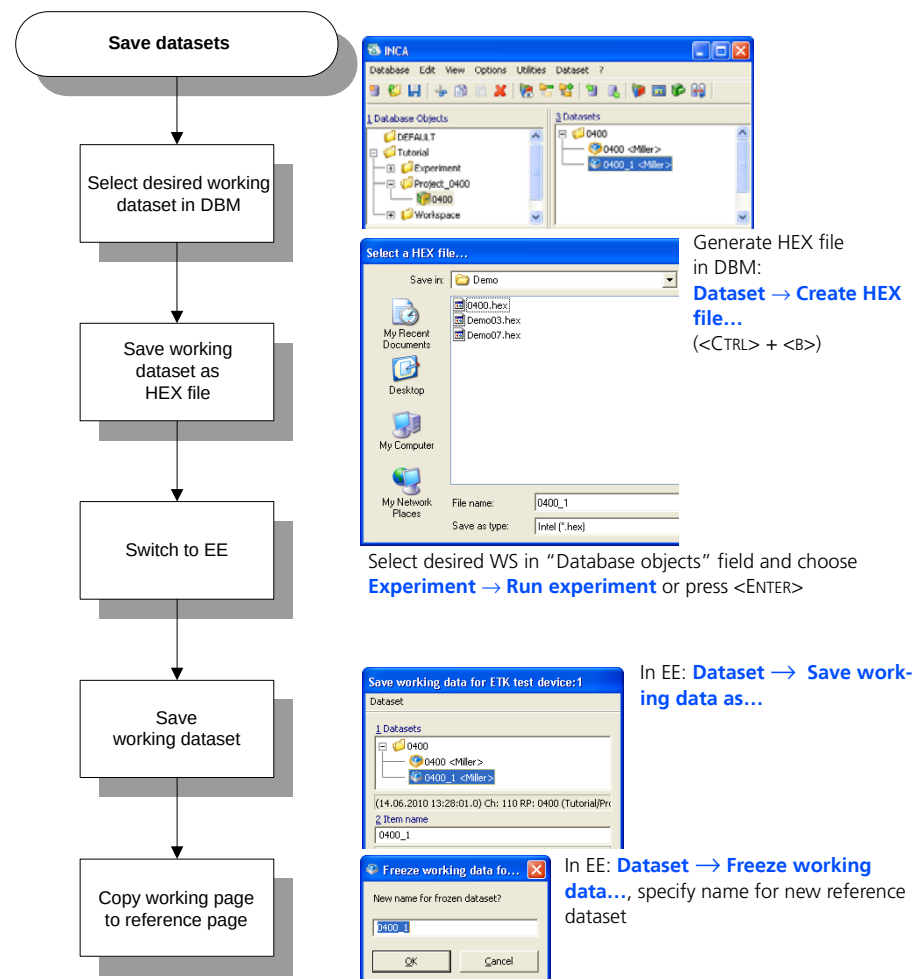


Fig. 5-13 Saving datasets

5.5 Editing Datasets with the CDM

You can use the Calibration Data Manager to analyze the datasets generated while running the test series. For a cross-project analysis, you select datasets from different projects. When selecting datasets of a project, you can analyze the reference and working dataset. You can list, compare and copy the data of the selected datasets.

5.5.1 Selecting Datasets and Calibration Variables

Before you can execute one of the **List**, **Copy** or **Compare** actions, you must first select a source dataset. For the **Copy** and **Compare** actions, you must select at least one more destination dataset. Selecting the calibration variables is required for each of the three actions. The CDM offers you various selection methods.

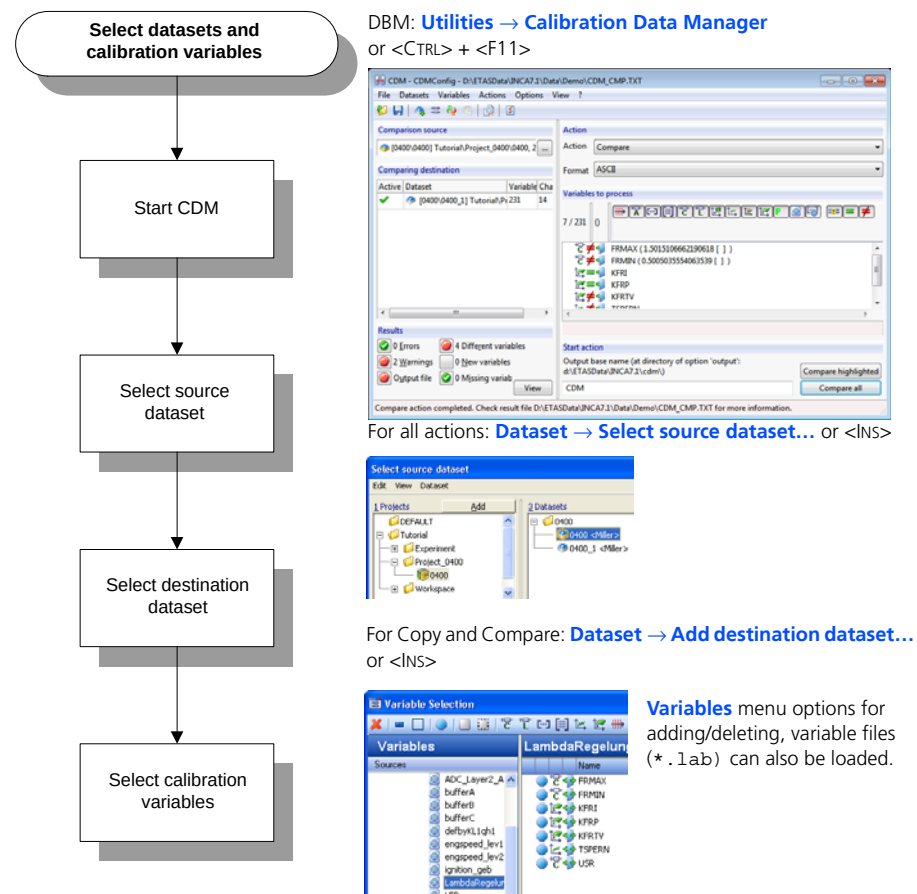


Fig. 5-14 Selecting the datasets and calibration variables

5.5.2 Executing the List, Compare, or Copy Action

The **List** action creates a table listing the data of the selected calibration variables in the source dataset. The **Compare** action compares the selected calibration variables from the source dataset with the comparison datasets and provides the different values for evaluation. The **Copy** action copies the selected calibration variables from the source dataset into the activated destination datasets. Only the **List** action provides you with data exchange formats (CVX, DCM, CDF V2.0 or PaCo) in addition to the ASCII, HTML or PDF output formats.

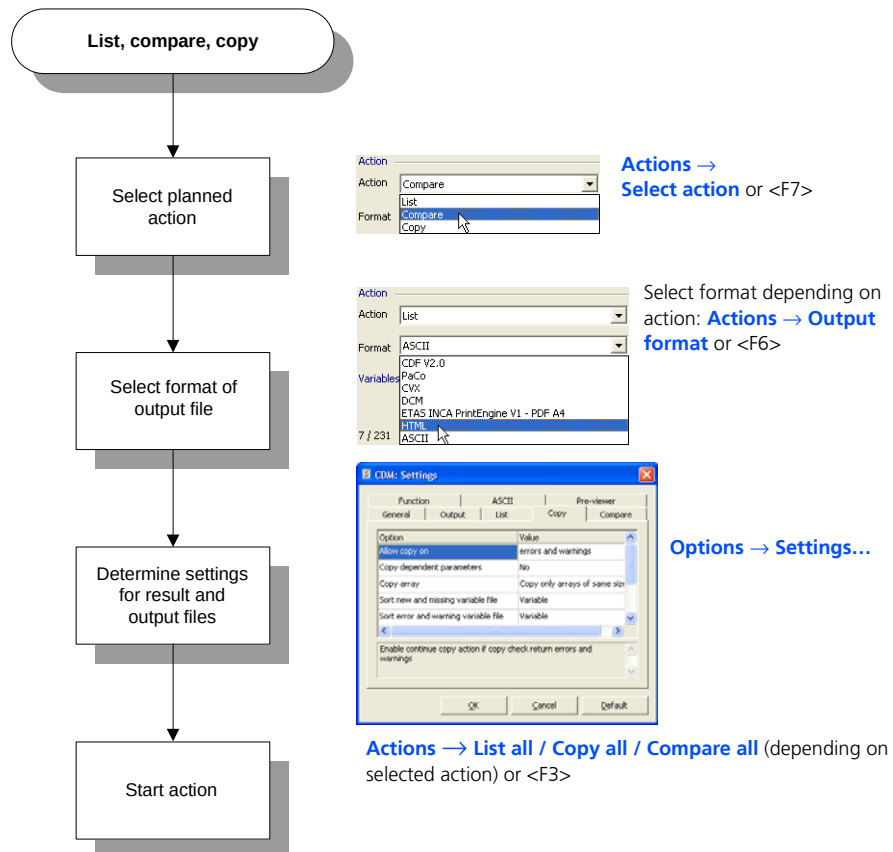


Fig. 5-15 Listing, comparing and copying

5.5.3 Analyzing Result and Output Files

Depending on the operation carried out, different result or output files are generated. The action selected automatically determines the file extension of the output files, which can be ***.txt**, ***.htm**, ***.pdf**, ***.csv**, ***.dcm**, ***.cdfx**, or ***.xml**. Apart from the standard output files ***_lst.***, ***_cpy.***, and ***_cmp.***, INCA may also generate additional files (***.lab**).

The result files (***.lab**) may contain various information items such as errors, differences or simple lists of calibration variables that can be used to select the calibration variables. Any errors that may have occurred during the action, are indicated in the "Results" field (red fields).

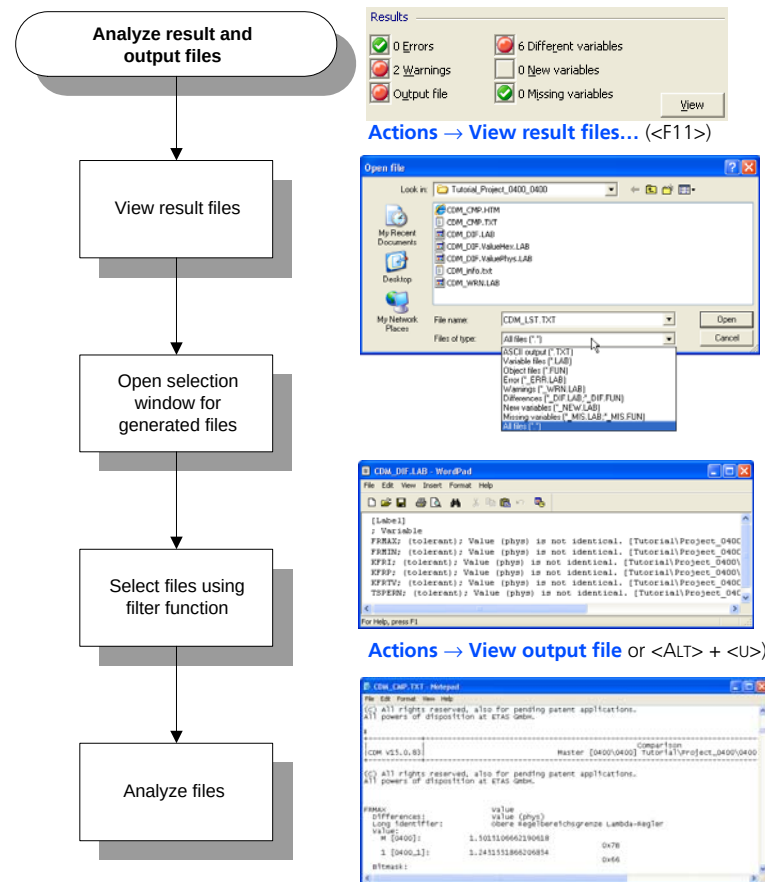


Fig. 5-16 Analyzing the result and output files

5.6 Data Processing and Data Exchange in the DBM

5.6.1 Working with Database Items

Depending on the selected database item, such as workspace, experiment, project, etc., the Database Manager (DBM) provides you with different processing functions.

It must be distinguished between general processing functions, which apply commonly to all database items in the left window ("Database objects" field), and processing functions that apply to specific database items. The latter can be used only after the corresponding items have been selected in the left window, and they affect only the items in the right window.

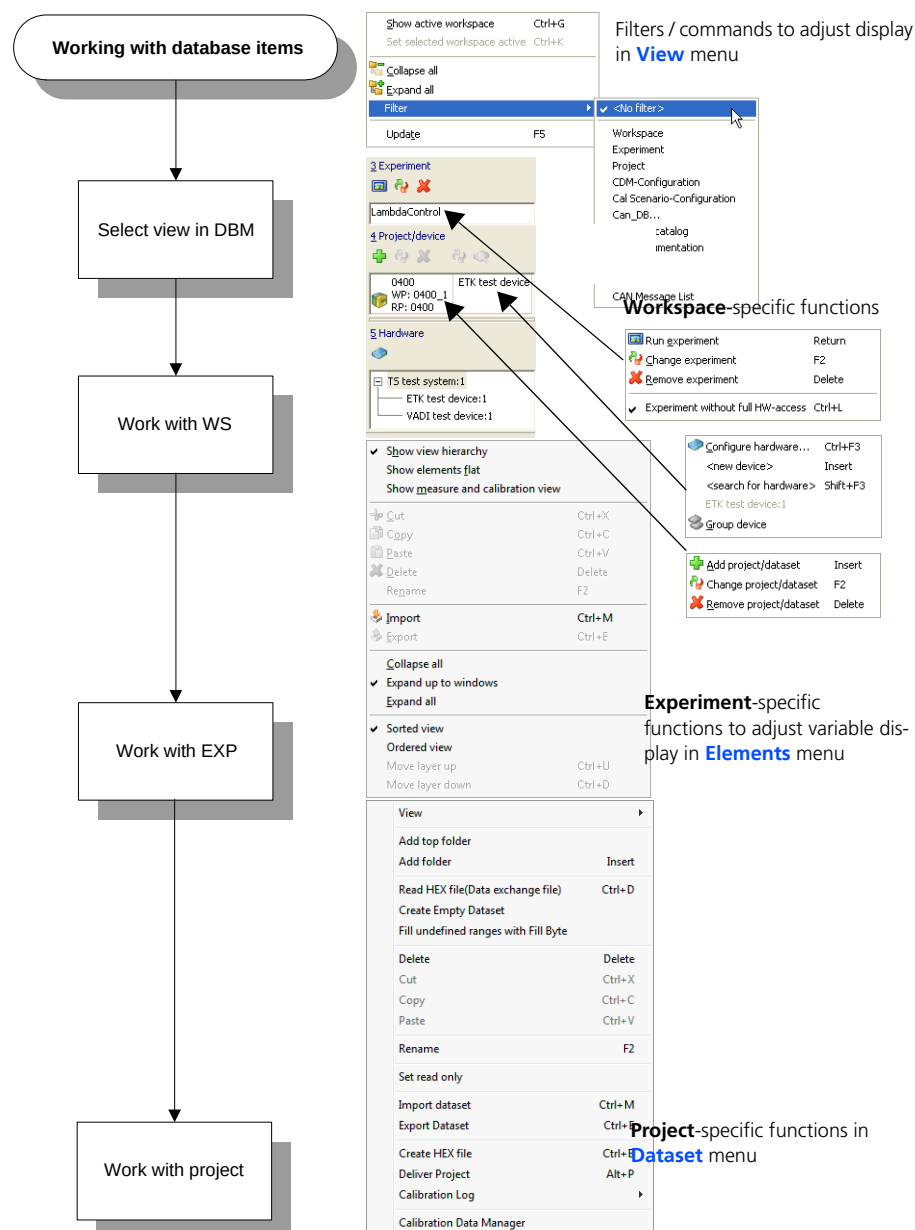


Fig. 5-17 Working with database items

5.6.2 Exchanging Data Using the Import/Export Function

You can use the Database Manager to export or import all or only selected items of a database. The Import/Export function is an invaluable tool when making use of existing experiments.

Consistent with the general management concept of dividing the DBM window into a left and right pane, you can use the Export/Import functions for the items in the "Database objects" field by choosing options in the **Edit** menu and for the datasets in the "Datasets" field by choosing the **Dataset** menu. Consequently, there are different export data, although the file extension (*.exp) is the same. To avoid conflicts, you should give these files distinctive file names indicating the particular item type.

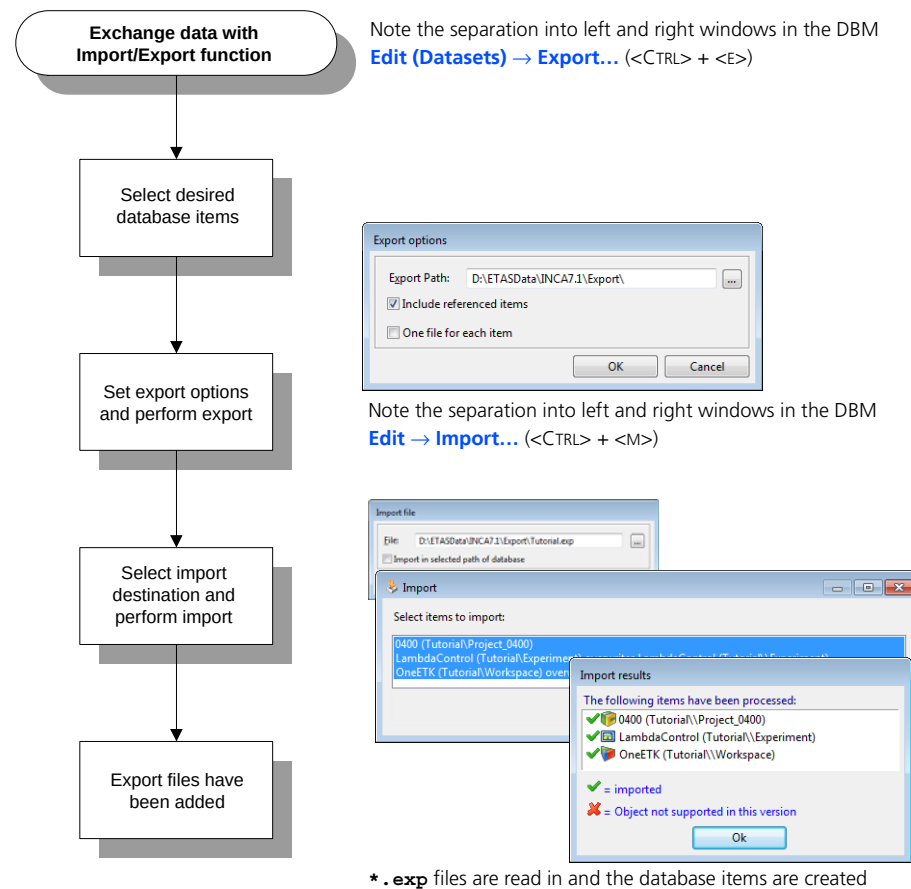


Fig. 5-18 Exchanging data by exporting and importing

5.6.3 Working with Databases

As mentioned before, INCA allows you to work with several databases. Apart from creating a new database, initially containing no items at all (see „Working with the Database Manager – Preparations“ on page 94), you can also use existing databases (e.g., from a co-worker) by loading one of them. Of course, working with databases also involves maintenance and making backup copies. INCA provides you with four functions for this purpose, some of which can be combined: Optimize, Convert, Repair, and Check.

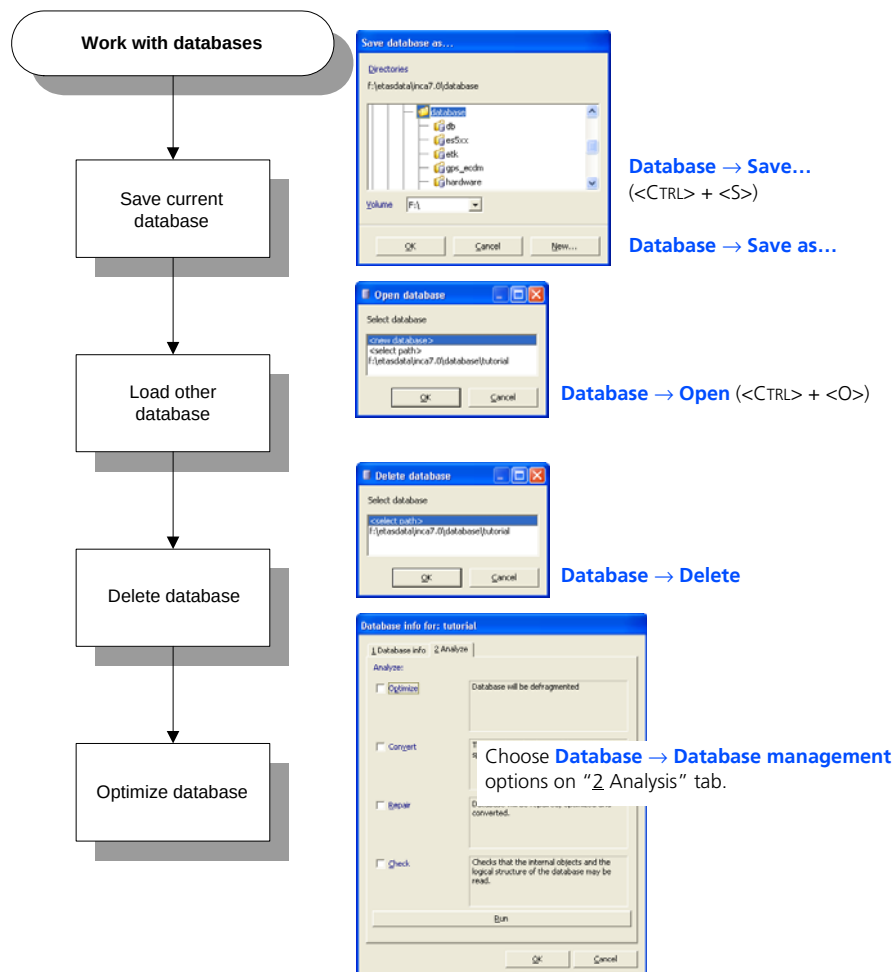


Fig. 5-19 Working with several databases

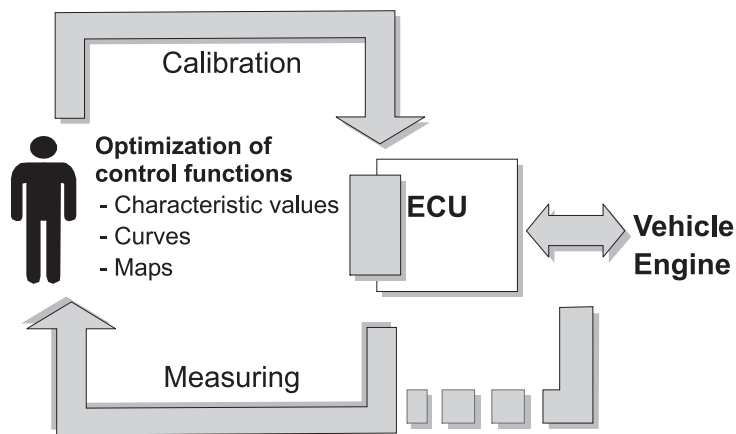
6 Understanding INCA

This section provides an introduction to the calibration work and describes the use, structure, and functionality of INCA.

6.1 Calibration Basics

The implementation of complex functions and control algorithms in microprocessor-based control units, and the tuning and optimization of these systems for various engine and vehicle types is referred to as "calibration". INCA aides you throughout this process, from setting up the hardware to evaluating the measurement results and managing the measured data.

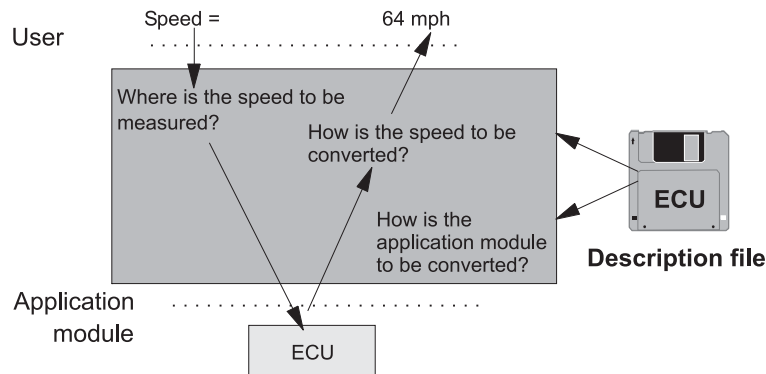
The objective of the tuning procedure is the vehicle-specific adaptation and optimization of the control functions of an electronic control unit. The control function algorithms are permanently stored in the control unit program; only the parameter values (maps, curves, and characteristic values) can be modified.



Information on the interfaces connecting to the control unit (hardware and software), the measurement values and parameters as well as the technical measurement capabilities of the measurement devices used (sampling rate, accuracy, etc.) are required to operate a calibration system.

During operation, the user interface software utilizes this information to carry out the actions desired by the user and to represent the parameters and measured signals accordingly.

The following steps need to be carried out to measure the vehicle speed from a control unit:



- Defining and parameterizing the control unit interface. How is the communication with the control unit implemented and what kind of parameters have to be used (baud rate, etc.)?
- Selecting the measured signal and communicating the hex address to be measured to the control unit. The control unit provides the measured value associated with this address.
- Representing the value provided by the control unit in physical units using a hex → physical conversion rule.

The calibration system must have address and conversion information for all measured signals and adjustable values as well as parameters which describe the control unit interface. Therefore, a standardized description file must be available for each new program running in the control unit.

Thus, each new software delivery includes at least one description file (*.a21) and one program file (*.hex).

6.1.1 Standard Interfaces Based on the ASAM-MCD Model

Standardized interfaces, which are independent of manufacturers and systems, have been established in the calibration tool environment. The work committee ("Arbeitskreis") for the Standardization of Automation and Measurement systems, with the work groups: Measurement, Calibration and Diagnosis, abbreviated **ASAM-MCD**, has standardized, among others, the following crucial calibration interfaces:

ASAM-MCD-1 provides the direct interface to the control unit.

- **ASAM-MCD-1a**
direct hardware interface to the control unit; hardware interface is the CAN bus (Controller Area Network) with CCP (Can Calibration Protocol) as the communication protocol.
- **ASAM-MCD-1b**
Software/driver interface between the calibration program on the PC and the connected calibration hardware.

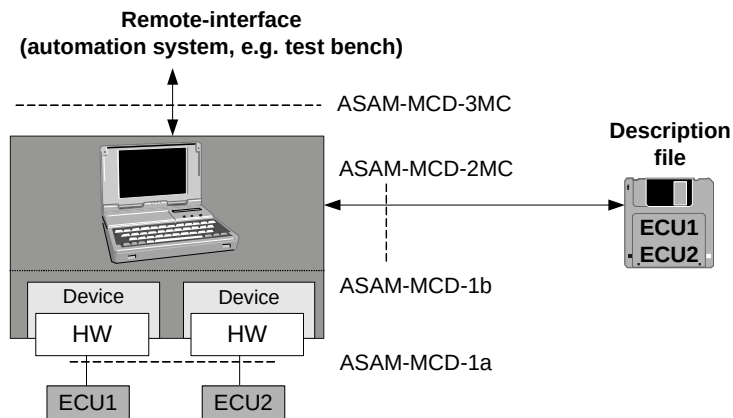
ASAM-MCD-2MC:

This is a file format used to describe the calibration variables and measured signals contained in the control unit software, and additional specific information designed to parameterize the calibration interface. You can use ASAM-MCD-

2MC to import the information required for this into a calibration system (A2L file). The file only contains address information and the data structure, however; the values are stored in the associated hex file.

ASAM-MCD-3MC:

This is a standard data exchange/communication protocol for remote control of a calibration system via Ethernet or the RS-232 interface. A test bench computer can work as the master to automatically optimize control unit parameters via the ASAM-MCD-3MC interface. For this purpose, the calibration system receives calibration commands via the interface and executes them in the control unit. When requested, control unit internal signals can also be measured and returned to the test bench computer.



For further information on ASAM-MCD-3MC, please refer to the ASAM-MCD-3MC documentation for INCA (see "Documentation for Tool Integration" on page 186).

6.1.2 Interfaces to the Control Unit - ASAM-MCD-1

The connection of a calibration system to a control unit can be parallel or serial.

- Parallel interface via ETK (emulator test probe) - the ETK is connected to all data and address bus lines in the control unit in parallel.
- Serial Interface

The control unit communicates with the application device via a standard serial interface and controls the engine, the gearbox or the brake system at the same time; a special application protocol is used for communication:

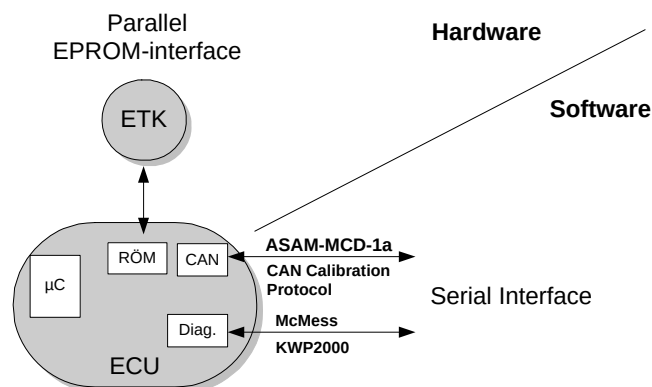
- Diagnostic interface K-Line (ISO9141-2) using KWP2000 or McMess as the communication protocol.
- CAN interface using CCP (CAN Calibration Protocol) as the communication protocol.
- XCP (eXtended Calibration Protocol) via Ethernet or USB

Additional memory (RAM) is required in the control unit, in which the calibration system performs the parameter calibration and from which the control unit reads the control parameters, both for the serial and the parallel application. In the case of the parallel application, this additional memory is implemented on the ETK.

ETK, CAN and K-Line differ in terms of performance when it comes to downloading data to the control unit and in terms of data acquisition.

High-End: ETK, very fast downloading of program and calibration data to the control unit and very fast data acquisition.

Medium and Low-End: CAN using CCP or K-Line with KWP2000, depending on the control unit implementation.



6.1.3 Calibration with ETK (Parallel Control Unit Interface)

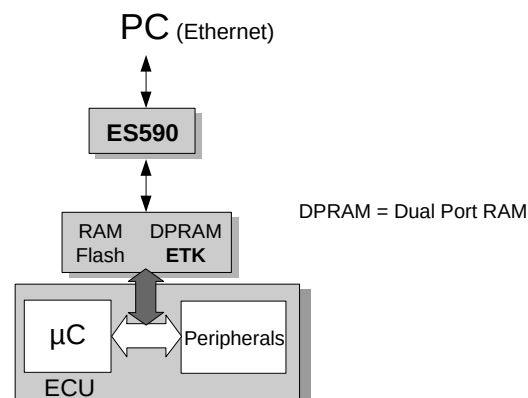
An emulator test probe (ETK) is installed in the control unit for the parallel application. The ETK is connected to all bus lines of the control unit processor and replaces either the control unit program and the calibration data or the calibration data only – depending on the control unit. Consequently, the calibration system can change control parameters *online*. Furthermore, the ETK provides RAM (so-called DPR = Dual Port RAM) to transfer measured data from the con-

trol unit to the calibration system. The control unit itself cannot recognize whether the program runs from the control unit flash (this is where the program is stored in the production control unit) or the ETK.

Only approximately 30 lines of additional program code are required for data acquisition via a table transferred from the calibration system to the control unit. The additional computing time required for this is negligible.

The ETK allows measured data to be acquired simultaneously from three measurement rasters (loops), which are also speed-synchronous. Using an interference-proof serial interface, the control parameters can be modified or read via the ES1000 (VME) calibration devices (see "Measurement and Calibration Hardware" on page 122). The ETK is the control unit-specific component of the memory emulation. Like the INCA PC software, all other calibration devices and cables used with the parallel application are the same for all projects. Depending on the micro-controller bus, clock frequency, memory size, etc., the ETK is available in different variants and will be adapted to the respective variant using programming on site. In addition to RAM, there is also an ETK flash on the ETK board. This is used to store the calibrated data on the ETK. This way, the control unit can be run immediately upon start and a calibration system for downloading the data is not required.

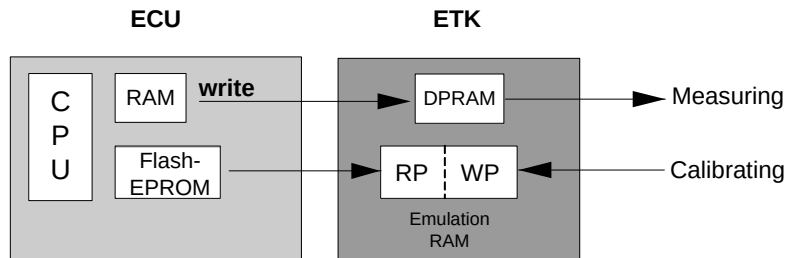
Don't confuse the ETK flash with the control unit flash. The ETK replaces the control unit flash (perhaps only partially, depending on the type) and provides a flash of its own which is used for data backup purposes.



The ETK can be installed on or in the control unit, so that the ETK and the control unit form a compact unit. This results in simple and interference-proof designs and considerably facilitates their handling.

ETKs are available in a large number of variants (physical dimensions, connections) for the most common micro-controllers. Typically, they are connected to the control unit module via specific adapters.

The functional principle of the ETK allows control parameters to be edited flexibly in the ETK RAM while the process is running.



In the ETK, the data to be applied is redundant, so that, for example, the response of an engine to modified data can be compared quickly to the response to the original data at the push of a button. One of the pages is called the Working Page (WP) because the calibration system can carry out changes on this page. The other page is the Reference Page (RP). The data on the reference page cannot be edited.

There are control units in which the ETK only manages the calibration data. Other control units also allow access to the code page so that the entire control unit program can be loaded into the ETK.

If only data are emulated in the ETK, the control unit accesses the data on the working page or reference page and reads the program from its own FLASH EPROM. In this case, a new program version first requires the control unit FLASH to be programmed and then data to be downloaded to the WP and RP.

Due to its parallel program and data access, fast downloading of data into the control unit and very fast data acquisition with negligible run-time load, the ETK interface is the most powerful interface.

6.1.4 Calibration via the Serial Interface

Calibration via the CAN, LIN and FlexRay and K-Line serial interfaces is implemented via communication protocols which must be available both in the control unit and the calibration device so that they can be used together.

The serial application is particularly important in situations, where due to the installation location of the control unit (e.g. directly in the gearbox or when using hybrid control units), the use of an ETK is not possible. A major benefit of the serial concepts is that little hardware is required, so that in most cases there is no need for mechanical and physical modifications. In most cases, the measured data rates are sufficient for a production intent application. INCA supports serial application via the following interfaces and protocols:

Protocol Interface	CCP ^a	XCP ^b	KWP2000 ^c	UDS ^d	McMess ^e	Monitoring
CAN	X	X	X	X		X
K-Line			X		X	
FlexRay		X				X
LIN						X
Ethernet		X				
USB		X				

a. CCP (CAN Calibration Protocol): protocol standardized on the CAN bus in accordance with ASAM-MCD-1a

b. XCP (eXtended Calibration Protocol): vendor- and interface-independent protocol for communication between calibration tools and control units, in accordance with ASAM-MCD-1

c. KWP2000 (Keyword Protocol 2000): communications protocol used for on-board vehicle diagnostic systems, specified in ISO 14230

d. UDS (Unified Diagnostic Services): diagnostic protocol, specified in ISO 14229-1

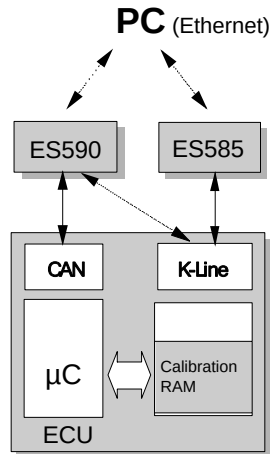
e. McMess: calibration and diagnostic protocol

Specifications of the CCP, KWP2000 and McMess implementations carried out in ETAS calibration devices can be obtained directly from ETAS.

Additional software has to be installed in the control unit for the application protocol. This means that the protocol puts constraints on the control unit in terms of computing time. Since in the case of the serial application, calibration and measurement activities are performed in addition to the actual control unit function, the download speed, the measured data rate, and user-friendliness are generally affected. Therefore, additional traffic on the bus caused by other control units might also have to be taken into consideration.

CCP allows the simultaneous acquisition of signals from several measurement rasters (which are also speed-synchronous). Using KWP2000 and McMess, the signals can only be acquired from one loop. In the case of KWP2000, this is a fixed time raster; McMess also provides speed-synchronous acquisition.

Since only data emulation takes place, you always have to program a new program version into the control unit flash before its data portion can be edited..

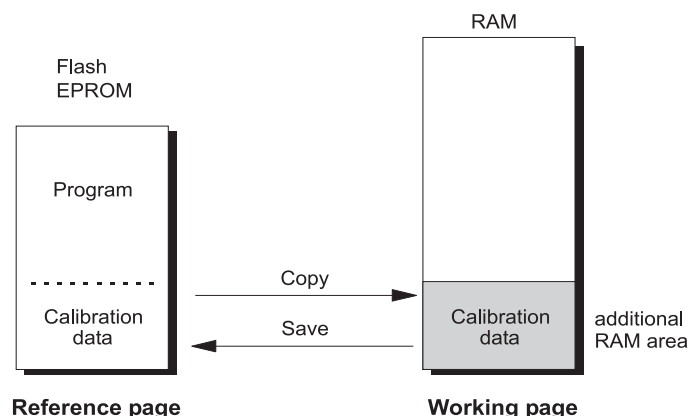


Basically, an emulation memory is used for the serial calibration as well. This memory, however, is smaller than that of the ETK and either contains all data (SERAM concept) or only a few characteristics (SERAP concept). In the case of SERAP, you generally have to program the control unit flash first so that it knows which values have to be applied initially. There are different methods to do this; for details, please contact ETAS.

An implementation could be like this: The control unit loads the data to be applied after its reset from the flash to the RAM and then reads the data from the RAM rather than from the flash.

The RAM is treated as the working page and the flash as the reference page. Any modifications carried out can be programmed back to the flash using either programming routines in the control unit or an external programming device. Switching between flash and working page is possible during operation.

Typically, the calibration RAM required can be implemented using an equipment variant of the production control unit with a larger external RAM. The additional RAM area will then be used only for calibration data.



The way INCA handles the data from the working page and reference page is described in Chapter "Data Management Concept for Working Page and Reference Page" on page 123.

6.1.5 Measurement and Calibration Hardware

One of the main focal points of the application activity is the customization and optimization of control unit parameters of the respective vehicle. All devices of the INCA family allow an optimization to be performed on the running vehicle. In the process, both the requirements of extreme climatic conditions and interference levels on the test bench and in the vehicle are taken into consideration. By its robustness, small size and sophisticated handling, the INCA hardware is optimally suited for daily use both in the test vehicle and at the test bench. Differences in the ETAS hardware are expandability (modular hardware system), connectivity with INCA (Centronics/Ethernet) and finally the applications supported by each hardware device.

For detailed information on the hardware supported by INCA, please refer to the ETAS internet pages at www.etas.com/inca_compatibility.

Note

The associated hardware components are described in separate operating instructions. Before connecting the hardware components, starting the system and working with INCA, be sure to thoroughly read, understand and follow these operating instructions.

6.2 INCA Basics

In order to understand INCA, you must be familiar with the terms and procedures introduced in the following sections. A basic knowledge of these terms is also required to understand the following chapters of this manual.

6.2.1 Data Management Concept for Working Page and Reference Page

Requirements for INCA

To prepare the work on the engine, gearbox or brake system, characteristic values, curves, maps, etc. are already edited individually *offline* at the office. The data are then stored on the hard disk of the PC so they can be loaded later into the vehicle at the push of a button.

The calibration data are often saved to the hard disk during online calibration in the running process, so that later on, they can be downloaded quickly to other control units/vehicles.

If the voltage at the control unit or the ETK drops, the data must be securely saved to the hard disk, so that you can keep working with the same data after the problem has been fixed.

For these reasons, INCA saves the data in a database on the hard disk. These data represent an accurate memory image of the control unit with the ETK or a serial calibration.

A2L and HEX File

Prerequisites for working with INCA are creating a memory image of the control unit and storing the information in the form of datasets in INCA. This requires the following files:

- The A2L description file (***.a2l**) contains the physical description of the data and/or parameters of the control unit program. It includes:
 - structural information
 - address ranges
 - memory size
 - address ranges of each measured signal and parameter
 - names of the measured signals and parameters
- The hex file (***.hex, *.s19**; Intel hex or Motorola format) contains the control unit program consisting of the code and the data. The contents of this file can be directly loaded into the control unit and executed by the respective processor.

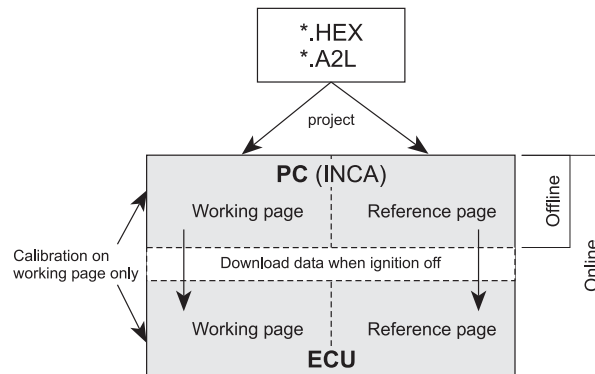
Both files are supplied by the software development process of the control unit manufacturer and need to be read into INCA.

Working Page and Reference Page

Regardless of whether you calibrate with an ETK (emulator test probe) or via a serial interface (e.g CCP on CAN, KWP2000 on K-Line, XCP on CAN, Ethernet or USB), an image of the corresponding working and reference pages of the control unit is created on the PC side (INCA), using the A2L and HEX files, and stored on the hard disk of the PC as working and reference datasets. This is done before downloading the files into the control unit.

The data on the reference page (reference dataset) cannot be edited directly by the user. The data is *always* applied on the working page (working dataset). For comparison purposes, the reference page contains the data version loaded originally. You can switch between working and reference pages at any time to check the different engine or gearbox responses.

This means that the user can switch between the working page and reference page even while the program is running, which allows for a direct and fast comparison of the different data versions of working page and reference page.



If the calibration hardware is connected, the calibration operations are performed in parallel on the working page in INCA *and* the working page in the control unit.

If the calibration hardware is not present, the calibration operations are performed on the working page in INCA only and can be written to the control unit later-on.

6.2.2 Project, Master, Working, and Reference Datasets

The database objects created from the A2L and HEX files are for INCA the foundation to the calibration of control units. They are stored in the database and presented to the user in the form of the following entities:

Project: The project is created by reading the project description file (A2L file) and saved as a database object in the database. It contains the description of all management information relevant to the calibration (addresses, filing methods, etc.) of a control unit program.

Master, Working, and Reference Datasets: When loading the first HEX file, the code portion is mapped to the selected project (transparent to the user). The data portion of this HEX file is stored as the so-called Master dataset in the database and it is write-protected. The reference and working datasets are usually only copies of the Master dataset.

During the calibration it is possible to write-protect a working dataset after it has been calibrated successfully. This will designate the "old" working dataset as the "new" reference dataset. A copy will be created automatically which will be used as the "new" working dataset.

For security reasons, when reading additional HEX files of a project, INCA checks whether their code portions match the code portion already read. If they match, it only stores the data portion as an additional dataset that can then be used as the reference or working dataset.

At the beginning of each working session, INCA compares the data and code of the created project or datasets with the data in the control unit. This is a way to check whether the data in the control unit differ from the data in INCA. If the data differ, the data applied are usually loaded from the hard drive to the control unit.

In this case, INCA automatically invokes the Memory Page Manager and suggests the download. It is also possible, however, to read the data from the control unit and save it to the hard drive.

6.2.3 Experiment

The Experiment is a database object which contains the settings for a particular calibration experiment or a particular measurement task.

An Experiment contains information as to which variables are represented in which windows. It also contains the settings in the individual measurement and calibration windows as well as the measurement control information (e.g., measured time, trigger condition). This information is stored in the database as a unit so that it can be reused.

6.2.4 The Workspace

A workspace is an object in which all information required for a specific measurement or calibration task is stored and consolidated. This gives the user central access to all data for maintenance and modification.

The workspace comprises:

- definition of the connected hardware (hardware configuration)
- reference to the project and the datasets
- reference to an experiment

Hardware Configuration as Part of the Workspace

As already mentioned, since a hardware configuration is a fixed part of a workspace, it is not stored in the database as an individual object. A hardware configuration is a data image of the actual measurement hardware and control unit interfaces and their parameterization in the software.

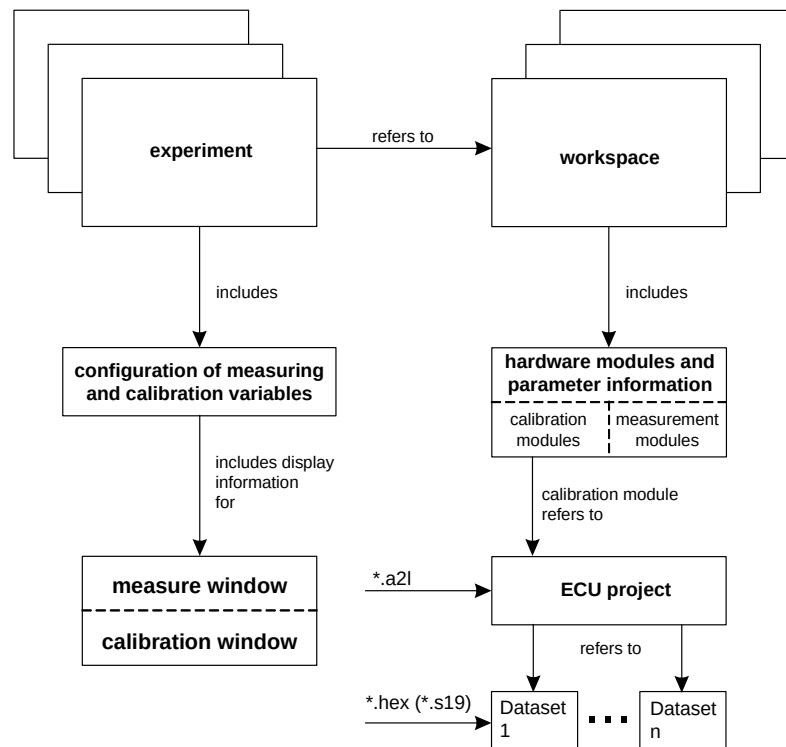
Before a calibration task can be carried out, you have to let the software know what kind of hardware is being used. This means, you have to specify, or the software has to recognize, which measurement hardware and control unit interface are connected to the PC. In addition to this, the user has to specify for each variable through which input port it is acquired.

For control unit interfaces, the project and the working and reference datasets have to be specified or created.

The information entered describes the hardware equipment and is stored in the workspace as the hardware configuration.

6.2.5 Relationships Between the Database Objects

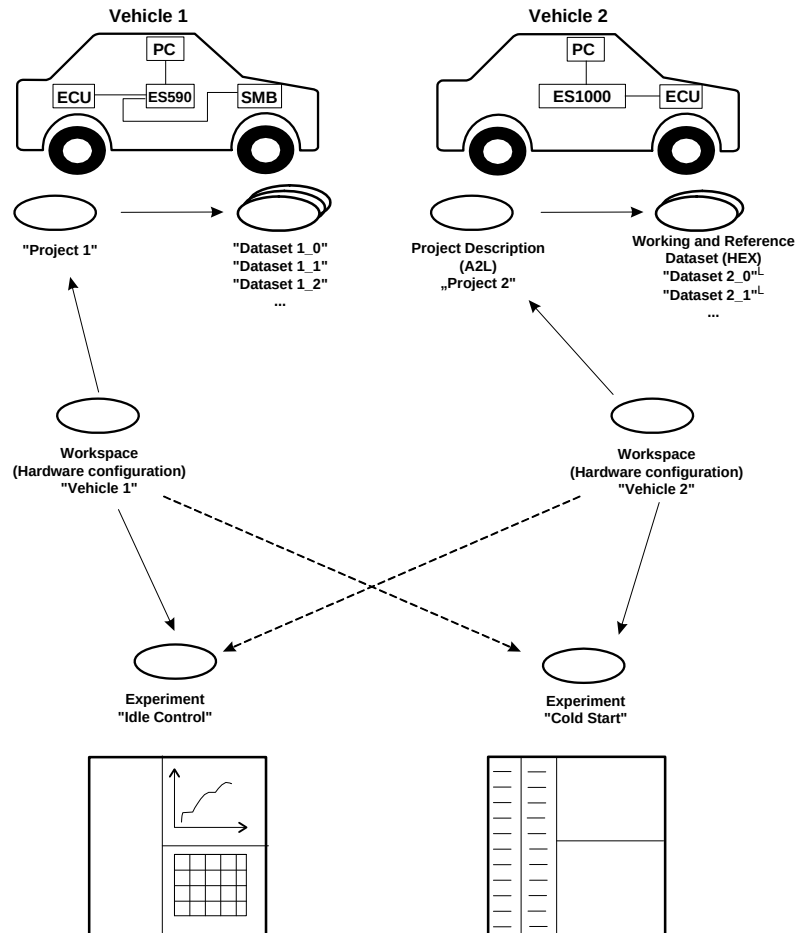
The terms introduced in the previous chapters, *experiment*, *project*, *master dataset*, *working dataset*, *reference dataset*, and *workspace*, are the names of database objects. The database objects have relationships with each other that can be modified, if necessary. The following figure is a simplified representation of these relationships.



The following example will make it easier to understand these relationships:

Calibration engineers have to apply different functions of a control unit software, e.g. cold starting or idling control. The functions are calibrated on the test bench or in the vehicle (or in different vehicles) using various hardware equipment. These locations provide control units with certain program versions whose data are to be optimized for the respective function.

An example is shown in the following figure. The database objects stored in the database of the calibration engineers' computer are represented as ellipses. The arrows between the database objects correspond to the database references.



The calibration engineer is supported by the appropriate Experiment corresponding to the various control unit functions to be calibrated.

The different hardware equipment on the test bench and in the individual vehicles is reflected in various workspaces with the individual hardware configurations.

The situation where a calibration engineer wants to apply cold starting first and then idling control on the same vehicle is supported by the fact that the two experiments can be loaded using the same workspace and thus the same hardware configuration. When loading an experiment from the hard disk, the hardware is parameterized to the main memory according to the data contained in the experiment (e.g., sampling time).

Because an experiment can be used with different workspaces, the same function can be calibrated both at the test bench and in the vehicle. The hardware is initialized as specified in the hardware configuration of the workspace, and configured according to the data that are downloaded from the dataset.

The same experiment can be used with different projects. This is under the condition, however, that the names of the variables in the control unit description file have not changed.

This example clearly shows that the approaches taken in INCA provide considerable benefits to the user and offer more flexibility. However, greater flexibility also means additional complexity. For example, when you delete an experiment, you have to check whether this configuration is referenced by other workspaces. The user is relieved from attending to these tasks thanks to a database which keeps track of and manages these references.

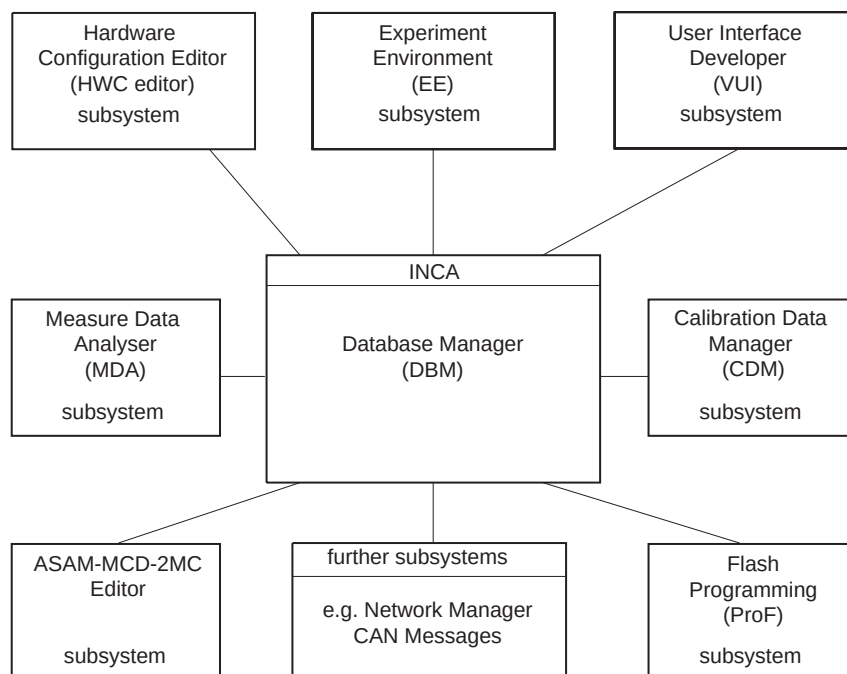
6.3 Product Structure from a User's Point of View

The following description of the modular structure provides an overview of INCA and its functionality:

INCA is designed according to a "building block concept". When needed, you can enhance INCA by so-called add-on modules (e.g., INCA-MIP, INCA-FLEXRAY). Any additional modules that you purchase later will integrate seamlessly into the accustomed operating concept. This principle allows individual customization of the entire system to your specific requirements.

6.3.1 Basic Software Functionality

The basic software is the outer framework that can be filled with exactly the "subsystems" (in the form of add-on modules) the user requires. Each subsystem supports a specific phase in the work process of a user and has its own user interface, i.e. its own window with menus and dialog boxes.



The following sections briefly describe the functionality of each of the depicted components.

6.3.2 The Database Manager (DBM)

The Database Manager is at the heart of INCA. This is where you start the individual subsystems. You can also manage users and, via an option dialog box, set user-specific options, for example, the definition of storage directories, preferences such as screen display options, or the start-up behavior of the software.

The main purpose of the DBM is, however, to systematically store all data in a database that are created during the calibration process (workspaces, experiments, projects, and datasets). The DBM allows you to manage the database objects from a comprehensible user interface. Similar to the Windows Explorer, you can create directories and subdirectories, move, copy, import, and export

individual items and also create entirely new databases. This means that you can organize your data in a similar fashion as you are accustomed to for the file system.

The different data types, which on the one hand are required for the measurement and calibration task and on the other hand are created or modified during the execution of the experiment, are stored in separate units. This has the advantage that the individual units can be reused and exchanged more easily. In addition to the various units, INCA also manages the existing references between the units. Using the database, the software “knows” these references and ensures the integrity of the data, making the user’s life easier.

6.3.3 Hardware Configuration Editor Subsystem (HWC Editor)

The Hardware Configuration Editor subsystem assists the user in preparing for the measurement and calibration tasks. The main purpose of this phase is to let the software know what kind of hardware (measuring hardware and control unit interface) is being used. Users can configure the hardware used in a workspace optimized for this purpose. This means they can indicate the hardware and the interfaces through which they are connected to the PC. They can also specify which measured signals and calibration parameters are acquired through each input. For the control unit interfaces, the user can specify the desired project and the working and reference datasets.

The information entered describes the hardware equipment and is stored in the selected workspace as the hardware configuration. Using the overall system, which has been prepared and parameterized, you can perform a variety of partial tasks (e.g., cold start optimization, idling optimization or measurement of different vehicle components) using the “Experiment Environment” subsystem.

Thanks to the reusability of the workspace database object, the hardware configuration defined in it can be used for various experiments by simply copying the workspace.

6.3.4 Experiment Environment (EE) Subsystem

The Experiment Environment subsystem mainly contains the functionality required to perform a measurement and calibration task and is used when preparing the individual partial tasks. The data required for a measurement and/or calibration task which is specific to this experiment can be stored in the database as an experiment.

A workspace refers to one specific experiment. An experiment, in turn, can be included in any number of workspaces.

The data contained in experiments include:

- measuring channels containing the settings specific to this experiment
- measured signals/parameters required for this experiment
- measuring and calibration windows required for this experiment

An experiment does not include any hardware-specific information. It also includes no information on the data and program version. It only references these data.

The functionality of this subsystem can be summarized in four groups:

- **Configuring**

The user can select the settings required for the respective partial task, i.e. select variables to be measured and calibrated, and set sampling rate, measuring range, and display parameters for each variable.

- **Measuring and recording**

You can customize the user interface (measurement and calibration windows) to the task at hand. The measuring control information (recorders used for different recordings, measuring times, trigger conditions, repeat triggers, etc.) is displayed and the recordings are carried out.

- **Calibrating control unit parameters**

You can modify characteristics graphically or numerically in a variety of different calibration windows (e.g., numerical editor, combined curve editor, etc.). You can also perform mathematical operations such as addition or multiplication with a factor. In INCA it is possible to perform measurement and calibration tasks simultaneously.

- **Managing and Copying Data Versions**

The user can load data versions into the control unit, copy them from the control unit to the hard disk and exchange them between different users.

6.3.5 Measured Data Analysis (MDA) Subsystem

Here, the measured data files created can be analyzed. Functions such as a measuring cursor, zooming in the x and y directions and superimposing of signals aid the user in this work. Individual settings for recurring tasks can be stored in configurations.

MDA is provided as a stand-alone program, which can be opened individually or from INCA.

6.3.6 Calibration Data Manager" (CDM) Subsystem

The values of individual or all parameters of a "source dataset" can be represented in tabular form.

- **Comparing**

Value changes of parameters between a source dataset and any number of other "comparison datasets" can be represented on a physical level.

- **Copying**

Individual or all parameters of a source dataset can be copied into a number of other "target datasets". Address changes or different conversion formulas are taken into account.

- **Listing**

The values of individual or all parameters of a "source dataset" can be represented in tabular form.

6.3.7 “User Interface Developer” (VUI) Subsystem

Similar to a visual programming environment, the User Interface Developer allows you to arrange any display and calibration elements in a window based on ergonomic considerations. Additional design elements such as labels, lines, rectangles, and split bars assist in dividing the surface into function areas. These screen layouts can then be integrated in the experiment environment of INCA.

6.3.8 “ASAM-MCD-2MC Editor” Subsystem

The ASAM-MCD-2MC Editor allows you to analyze and edit the items of existing projects. This means you can use the ASAM-MCD-2MC Editor to view and edit the calibration and measurement variables, functions, and other specific information for parameterizing the calibration interface that have been defined in the corresponding A2L file. This requires that you have created a project in INCA in the form of a project description file (*.a21).

Note

You can find a description of the ASAM-MCD-2MC standards and the possible parameters for the items in the Internet at <http://www.asam.de>.

7 General INCA Operation

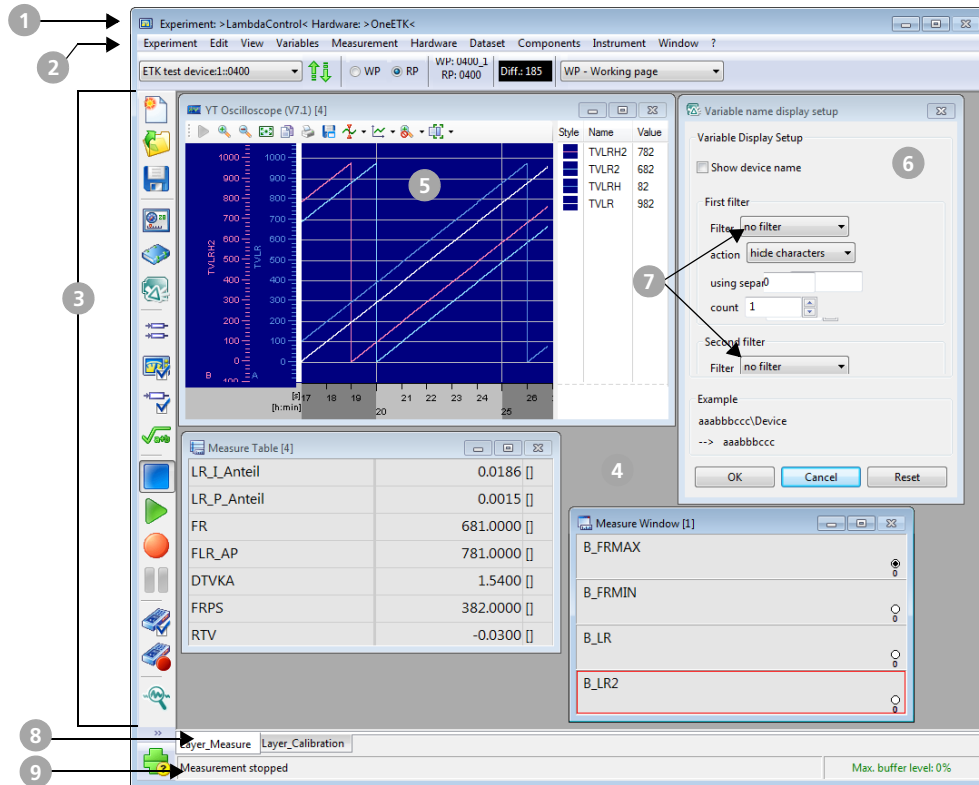
This chapter provides information on the window and menu structures, control options using the mouse and the keyboard, and the help system.

Be sure to read this chapter since some operational procedures are described only in this section. Although the presented techniques are standard Windows operations, they may not be familiar to the less experienced Windows user. They are therefore described here as a central summary.

Since INCA is mostly controlled from within the vehicle, the design of INCA put special emphasis on a simple and comprehensive operation by keyboard commands. For special features and deviations from the Windows keyboard conventions, refer to chapter "Operation Using the Keyboard" on page 138.

7.1 Window Structure

The INCA Window Elements



- Title bar (1)
- Menu bar (2)
- Toolbar (3)
- Window area (4)
- Instrument (5)
- Dialog box (6)
- Fields (7)
- Layer tabs (8)
- Status bar (9)

7.2 Toolbars

The most common commands are also available as buttons in a toolbar. In this way, a command can be executed simply with a click of the mouse.

Note

All commands that can be issued by clicking the specific buttons are also available in the corresponding menus.

All buttons located on the toolbar are mouse-sensitive. If you place the mouse cursor on a button and hold it for one second, a tooltip box is displayed right next to the selected button which displays the button function.



7.2.1 The Buttons in the Database Manager



1. Create new database
2. Open database
3. Save the database
4. Cut database item
5. Copy database item
6. Paste database item
7. Delete database item
8. Show active workspace
9. Collapse all branches in tree structure
10. Expand all branches in tree structure
11. Start data import
12. Start data export
13. Add workspace
14. Add experiment
15. Add project description file (*.a21, *.dbc, *.asp)
16. Add CDM configuration
17. Set up a user profile
18. Set whether experiment will be opened with or without full hardware access
19. Open Measure Data Analyzer
20. Start Calibration Data Manager
21. Start User Interface Developer
22. Call online help

7.2.2 The Buttons in the Hardware Configuration Window



1. Find connected hardware automatically
2. Initialize hardware
3. Enable/disable access to the control unit
4. Display hardware status
5. Add hardware manually
6. Remove selected hardware
7. Change project or working dataset
8. Read ECU identifier
9. Open the Memory Page Manager
10. Open the Experiment Environment with the current experiment
11. Change to the Database Manager window
12. Start the measuring monitor
13. Stop the measuring monitor
14. Perform an automatic sensor offset compensation

7.2.3 Buttons in the Experiment Environment



1. Create a new experiment
2. Open an existing experiment
3. Save an experiment
4. Open a user interface (*.vui).
5. Start the Hardware Configuration Editor
6. Go to the Database Manager
7. Open the variable selection dialog
8. Configure the display of the experiment
9. Configure the variables in the experiment
10. Define/manage calculated signals
11. Stop measurement
12. Start measure data display
13. Start measure data recording
14. Stop measure data recording
15. Configure measure data recording

16. Open the Recorder Manager
17. Open the Measure Data Analyzer
18. Edit the current user interface

7.3 Operation Using the Keyboard

Since most of the control unit software is calibrated in the vehicle, user-friendliness was of utmost importance in the INCA development process. Individual keys are preferred over the function keys <F1> to <F12>, which in turn take precedence over key combinations using <CTRL> and <ALT>. You can display a complete overview of the keyboard shortcuts currently used at any time by pressing <CTRL> + <F1>.

7.3.1 General Keyboard Operation

This table lists the primary keys and key combinations used to operate INCA. A complete list of all keyboard commands can be found in Chapter "Operation Using the Keyboard" on page 174.

Key	Function
Navigation	
<CTRL> + <F3>	Open Hardware Configuration Editor
<CTRL> + <F4>	Close current working window
<CTRL> + <F5>	Open Experiment Environment
<CTRL> + <F6>	Switch between open working windows
<CTRL> + <F7>	Move Database Manager to the foreground
<CTRL> + <F8>	Open UI Developer
<CTRL> + <F9>	Open ASAM-MCD-2MC Editor
<CTRL> + <F10>	Minimize/maximize current window
<CTRL> + <F11>	Open Calibration Data Manager
<CTRL> + <F12>	Open Measure Data Analyzer
Help	
<F1>	Invoke Help function
<SHIFT> + <F1>	Invoke "What's This" help
<CTRL> + <F1>	Show hotkey assignment
General Editing	
<F2>	Change to Edit mode (e.g., for table entries)
<SHIFT> + <F10>	Open shortcut menu for selected item (right mouse button)
<ALT>	Activates the main menu.
<ALT> + <F4>	Close active working window; if the DBM is active, INCA exits.
<ALT> + <F6>	Switch between open windows
<ALT> + <SPACE>	Open system menu of application window
<ALT> + <TAB>	Switch between open applications
<ALT> + <->	Open system menu of document window

Key	Function
<CTRL> + <A>	Select all items (e.g., in a list)
<CTRL> + <C>	Copy to clipboard
<CTRL> + <V>	Paste from clipboard
<CTRL> + <X>	Cut and put on clipboard
<CTRL> + <Y>	Redo last action
<CTRL> + <Z>	Undo last action
<DOWN ARROW> (↓)	Move cursor to table item or list item with arrow keys, <Down arrow> also opens active list box
<UP ARROW> (↑)	
<LEFT ARROW> (←)	
<RIGHT ARROW> (→)	
<ENTER>	Confirm input and quit input mode; expand or collapse branches
	Delete selected item
<ESC>	Cancel input, discard changes
<SPACE>	Select table or list item or deselect active selection
<TAB>	Move focus to next item (option) in a window (<Shift> + <Tab> moves focus in opposite direction)
<SHIFT>	Enable range selection, i.e., pressing the arrow keys while holding <Shift> key down allows selecting a table cell range.

7.3.2 Keyboard Operation Compliant With WINDOWS® Conventions

The general operation of INCA, such as navigating in menus or activating a certain windows, complies with the WINDOWS® conventions.

Pressing the underlined letter in a menu while holding down the <ALT> key activates the corresponding command. You can activate a subordinate menu command by pressing the underlined letter together with the <SHIFT> key.

For example, to open the **Edit** menu with a keyboard command, press the <ALT> + <E> key combination.

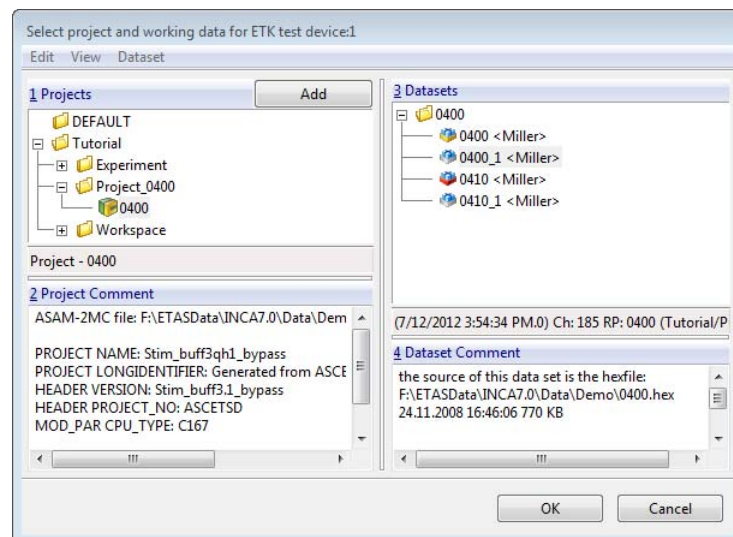
To switch to the next window or list box within the working windows, press the <TAB> key (in the order from top left to bottom right). You can use the underlined letter of the field or list label, while holding down the <Alt> key, to switch to the corresponding field or list box.

The arrow keys allow you to skip to the next item in list boxes. You can select multiple items by making your selection while pressing the <SHIFT> key.

Compliant with the WINDOWS® conventions, you can switch between the windows of different applications using the <ALT> + <TAB> key combination. In this sense, all subsystems of INCA, such as the Experiment Environment or UI Developer, are treated as individual applications.

Switching between the individual working windows of an application, e.g., in INCA between individual measure and calibration windows within the Experiment Environment, is done—pursuant to the Windows convention—by pressing the <Ctrl> + <TAB> key combination.

Within the working window, you can use the underlined number of the field or list label, while holding down the <ALT> key, to switch to the corresponding list box. For example, <ALT> + <3> would activate the "3 Datasets" list box.



7.4 Operation Using the Mouse

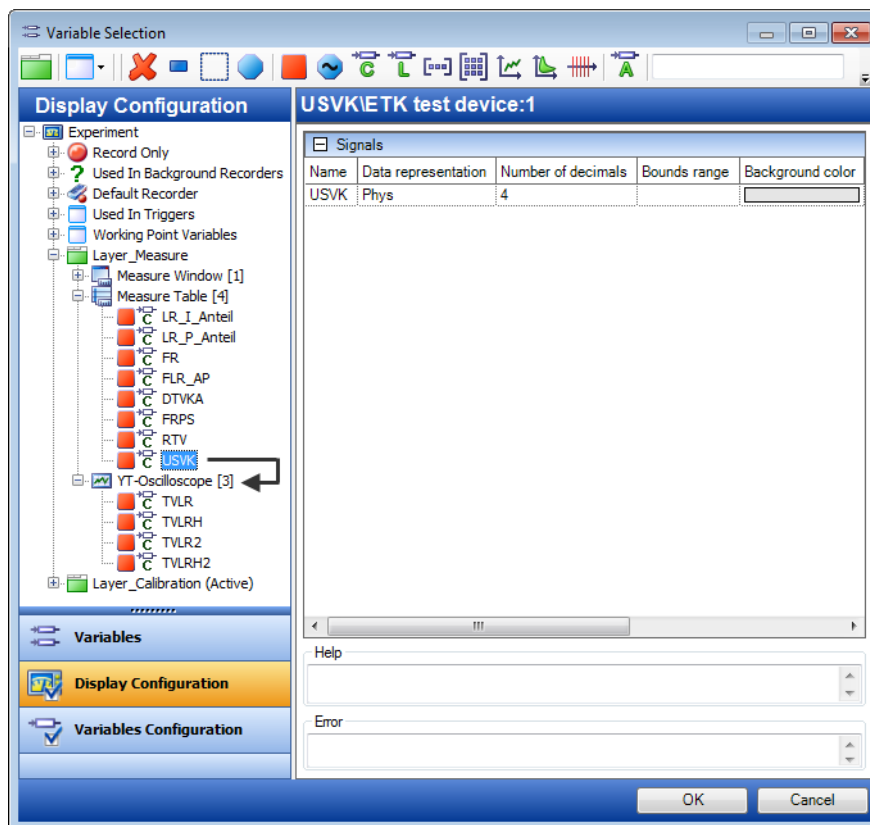
In the office or lab, you can use the mouse to operate INCA in a user-friendly way. The use of the mouse corresponds to WINDOWS conventions.

You can select multiple items by making your selection while pressing the <SHIFT> or <CTRL> key.

Right-clicking on the window elements opens context-sensitive pop-up (short-cut) menus.

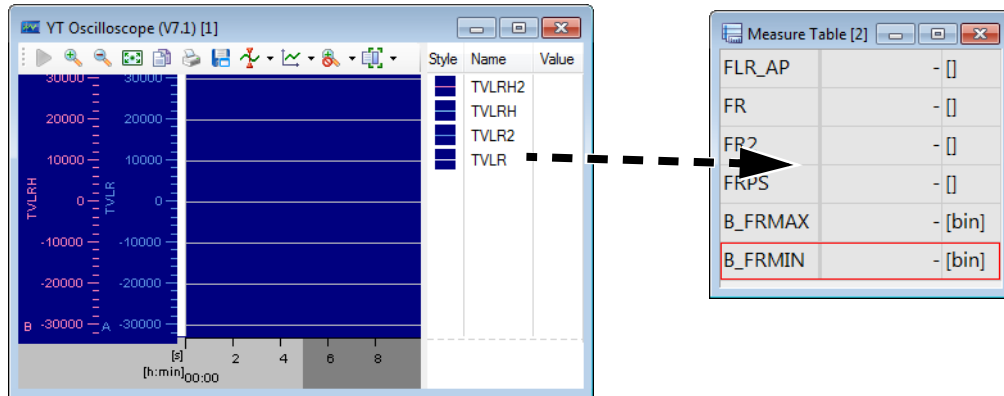
7.4.1 Drag & Drop

You can use Drag & Drop to move experiment elements in the Display configuration. For instance, if you want to move a variable from one window to another, just left-click on the variables, keep the button depressed and move the variable with the mouse to the target window..



Impossible commands, like copying a calibration to a measure window, will be ignored.

It is just as easy to move a variable from one window to another (left-click on the variables, keep the button depressed and move the variable with the mouse to the target window).



7.5 Hierarchy Trees

INCA often displays information such as the contents of a database in a hierarchical tree structure. In order to see all branches and the entire contents of such a tree structure, you have to expand or collapse the branches. INCA allows you to expand several branches automatically or to expand specific partial trees.

To expand several branches automatically:

- To expand or collapse all branches at once, you can use the **View → Expand all** or **View → Collapse all** menu option or the buttons in the toolbar (see "The Buttons in the Database Manager" on page 135).

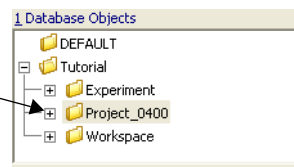
To expand individual partial trees:

- To open the desired branch, click with the mouse on the little "+" box next to this item or press the <+> key. Clicking a second time on the same little box or pressing the <-> key collapses the branch again.

Or

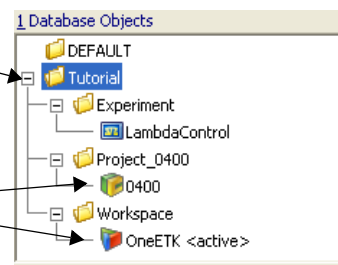
- Move the focus to the item with the <UP/DOWN ARROW> key and then press the <RIGHT ARROW> key to expand the branch. Collapse the branch by pressing the <LEFT ARROW> key.

A "+" box indicates a branch that can be expanded.



A "-" box indicates an expanded branch.

If there is no box, this is the end of the structure. Expansion is not possible here.



7.6 Editable Tables

Tables can be used to quickly modify the numerous INCA configurations which are often quite complex. You can directly change the items in the right column of these tables after selecting them with the mouse or the keyboard.

Option	Value
Zoom Mode	No Zoom
Legend Position	Right
Icon size in toolbar	Small (16x16)
Time Axis Span	10.0
Time Axis Background Color	
Time Axis Text Color	
Time Axis Background Color 2	
Timespan Overlap	0

Setting for zoom functionality

Depending on the type of items, you will use one of the following methods for editing:

- direct keyboard entry
- selection from a list box
- entry into a dialog box

To modify items by overwriting:

- Click on the item you want to modify.
- Or
- Use the arrow keys to move to the item.

Option	Value
Zoom Mode	No Zoom
Legend Position	Right
Icon size in toolbar	Small (16x16)
Time Axis Span	10.0
Time Axis Background Color	
Time Axis Text Color	
Time Axis Background Color 2	
Timespan Overlap	0

Time span of the axis in seconds [from 0.00001 seconds to 86,400,000 seconds, i.e. 10 microseconds to 1000 days]

- Enter the new value using the keyboard.
- Confirm your entry by pressing <ENTER>.

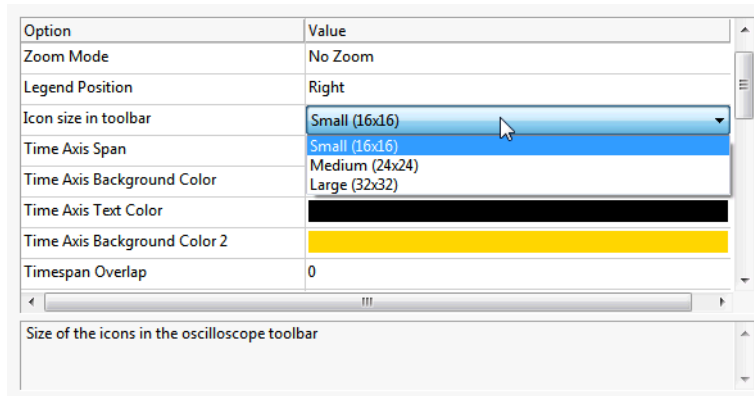
To select items from a list box:

- Use the mouse to select the item you want to modify and click again on the item.

Or

- Use the arrow keys to move to the item and press the <F2> key.

A list box with several items appears.



- Click on the desired item in the list box.

Or

- Use the arrow keys to move to the item.
- Confirm your selection by pressing <ENTER>.

To modify items via a dialog box:

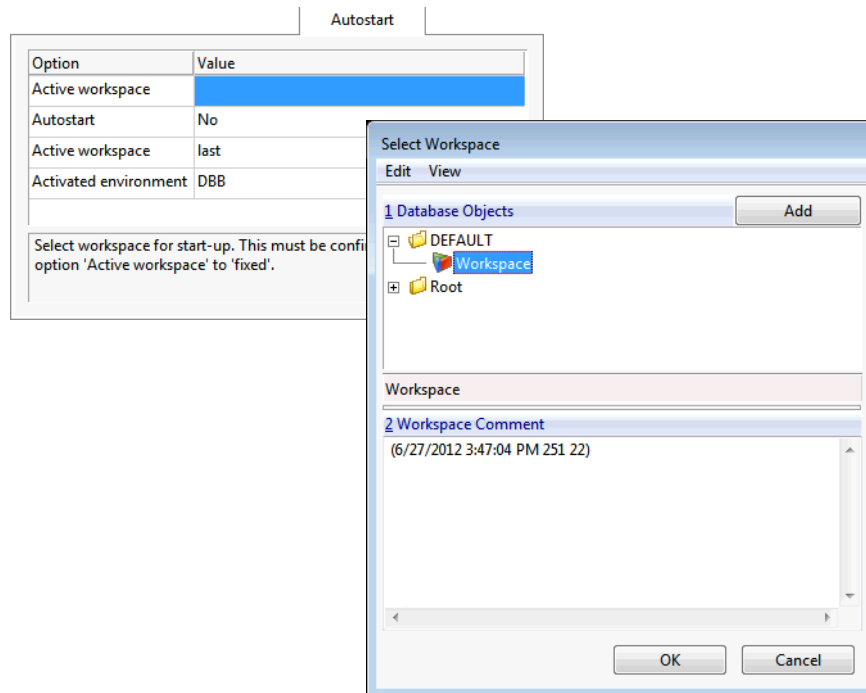
- Click on the item you want to modify.

Or

- Use the arrow keys to move to the item and press the <F2> key.

A dialog box opens.

- Make your selections.



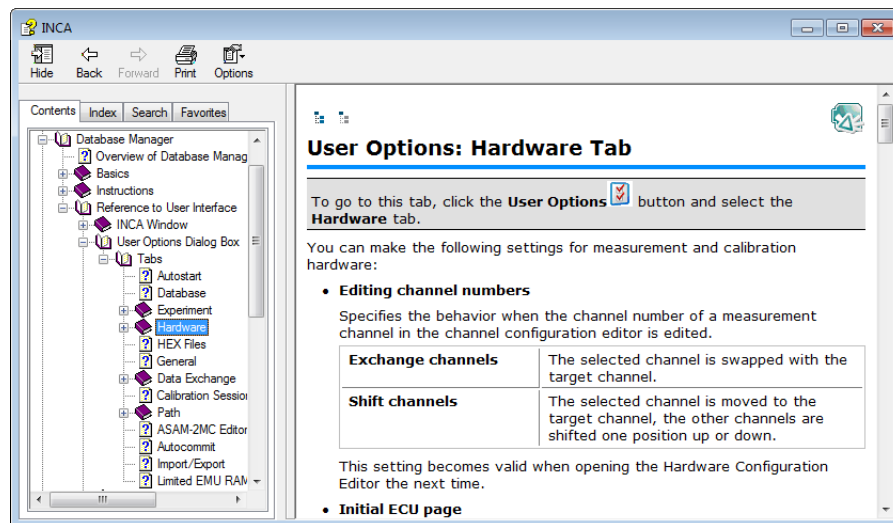
- Confirm your input by pressing <ENTER> or click the **OK** button.

7.7 Help Functions

7.7.1 INCA Online Help – Quick Guide

Choose **? → Help** to invoke the general help function of the corresponding module (e.g. the Experiment Environment). The <F1> key allows you to call context-sensitive help pertaining to the currently open working window.

If you want to browse the complete INCA online help click the **Full Help** button in the online help window or choose **? → Contents and Index**.

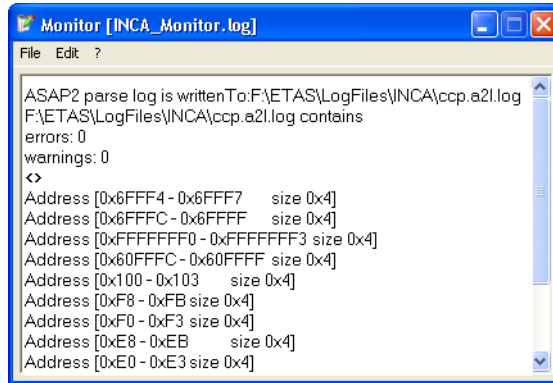


The four tabs of the help window provide you with the following options:

- The "Contents" tab allows you to browse the help topics by categories. The last chapter in the "Contents" tab of the INCA full help contains a glossary with explanations of important terms.
- The "Index" tab lists all index entries. Browse the entire list, or enter a search term to limit the scope of listing.
- The "Search" tab allows you to search for individual words or terms included in a help topic. Type a search string and let the help function list the entries it has found related to this term.
- You can use the "Favorites" tab to bookmark the most helpful pages in order to access them again from this tab.

7.7.2 Monitor Window

The monitor window is used to log the working steps performed by INCA. All actions, including errors and notifications, are logged. As soon as an event is logged, the monitor window is displayed in the foreground.



In addition to displaying information, the monitor window also provides the functionality of an editor.

- The display field of the monitor window can be freely edited. This way, your own remarks and comments can be added to the INCA messages.
- The INCA messages can be saved as text files along with your comments.
- Other INCA text files already stored can be loaded so that you can compare specific working steps.

7.7.3 Keyboard Assignment

You can display a complete overview of the keyboard shortcuts currently used at any time by pressing <CTRL> + <F1>.

7.7.4 Manual and Tutorial

The complete INCA manual as well as a tutorial with exercises is available in digital version and can be opened on the screen at any time.

Using the index, full text search, and hypertext links, you can find references fast and conveniently.

You can open the INCA manuals and tutorial via the help menu ([? → Manuals and Tutorial](#)) or via the INCA program group.

7.7.5 Video Tutorials

INCA offers video tutorials for presenting complex or new features and use cases which can best be shown by animations. The video tutorials are either linked from the online help or from an overview page which lists all available videos.

You can open the overview of the INCA video tutorial via the help menu ([? → Video Tutorials](#)).

8 Glossary

The chapter titled "Glossary" explains all technical terms used in the manual. The terms are listed in alphabetic order.

A2L file	Standard exchange format in ASCII format used for projects.
AML	ASAP2 meta language; the AML is defined in the ASAP2 Interface Specification Interface 2.
ASAM	Association for Standardization of Automation and Measuring Systems
ASAM MCD 3MC	ASAM MCD 3MC is an ASAM e.V. standard and describes an automation interface for measurement and calibration tools. The objective is to design an interface for MCD systems, which are used during the design, production, and maintenance phases of a vehicle's lifecycle. From a user view, the integration of control units (ECUs) shall be encapsulated by the MCD system, the user needs the access to the measurement and adjustment objects via identical methods. Version 1 of the ASAM MCD 3MC specification (ASAP3) used a procedural model to describe the interface, while version 2.0 and above describe an object oriented interface.
ASAP-1b	Software/driver interface between the calibration program on the PC and the connected calibration hardware.
ASAP2	This is a file format used to describe the calibration variables and measured signals contained in the control unit software, and additional specific information designed to parameterize the calibration interface. You can use ASAM-MCD-2MC to import the information required for this into a calibration system (A2L file). The file only contains address information and the data structure, however; the values are stored in the associated hex file.
ASAP3	ASAP3 MC V2.1.1 (ASAM MCD 3MC V2.1.1) is a specification that ASAM e.V took over from the former ASAP group. It describes a procedural application programming interface for measurement and calibration tools.
AUTOSAR	AUTomotive Open System ARchitecture; AUTOSAR files are description files in XML format (*.arxml) which describe the system, e.g. the system topology including data busses and clustering, the software components, and the hardware. INCA uses only the CAN bus descriptions from the description files.

Calculated Signals	Calculated signals are composed of several variables that are combined by means of an algorithm.
Calibration Data Manager	A window in which datasets or individual calibration data can be listed, copied, and compared.
Calibration elements	User interface elements used to represent and edit calibration variables.
Calibration hardware	Hardware modules belonging to the category of ECU interfaces: ETK, CCP, and K-Line.
Calibration variable	A control unit variable that can be modified by the user (characteristic value, curve, and map).
Calibration windows	INCA working window which can be used to modify calibration variables.
CAN-DB	Format for the description of data sent on the CAN bus (e.g. network nodes, signal values).
Characteristic curve	Two-dimensional calibration variable.
Characteristic map	Three-dimensional calibration variable.
Characteristic value	One-dimensional calibration variable (constant).
Characteristic variable	Generic term used for map, curve and characteristic value (see also "calibration variable").
Code	The executable code is the "actual" program excluding the data (contains the actual algorithms). The code is the program part that can be executed by the CPU.
Configuration dialog	Dialog box used to configure the individual measuring and calibration windows as well as the variables contained therein.
Data	The data are the variables of a program used for calibration.
Database Manager	Working environment in which the user can manage the data he/she created and which are stored in the database.
Dataset	Database item consisting of one set of application data, read in from files in the hex format (*.hex)
Data version	A data version is a particular revision (version) of the data.
Description file	Contains the physical description of the characteristics and measured signals in the control unit (names, addresses, conversion formulas, function assignments, etc.).
Display channel parameters	Parameters which can be set for a display channel. They do not affect the parameterization of the hardware device but only the visualization of the variables on the screen.

ECU interface	Calibration hardware which allows the control unit calibration to be performed (ETK, CCP, and K-Line).
Editor	See "Calibration window".
ETK	Emulator test probe. Calibration hardware used for a control unit. Directly accesses the control unit memory.
Experiment	Dataset containing the settings for a specific calibration experiment (channels, sampling time, representation, etc.).
Experiment Environment	Main working environment in which the user performs his/her measuring and calibration tasks.
FIBEX	FIBEX (Field Bus Exchange) is an exchange format based on XML schema which is used for complete descriptions of the in-vehicle communication network. FIBEX is defined for various network types (CAN, LIN, MOST, FlexRay) and contains information about the bus architecture, signals, node properties etc. INCA uses FIBEX files to align tools and ECUs with the communication scheme given by the FlexRay cluster, i.e. the FIBEX file defines the signals which can be monitored via INCA in the connected cluster, and it contains data for the configuration of the interface hardware. The FIBEX file format is standardized by ASAM (Association for Standardisation of Automation- and Measuring Systems).
FlexRay	FlexRay is a scalable and fault tolerant communication system for high-speed and deterministic data exchange. FlexRay's time-division multiplexing facilitates the design of modular or safety-related distributed systems. Its high bandwidth of 10 MBaud on two channels helps to cope with the high network load caused by the increasing amount of innovative electronic systems in modern vehicles. The communication system's specifications are released by the FlexRay consortium which is widely supported by vehicle manufacturers and suppliers worldwide.
Hardware configuration	Part of the workspace; contains the hardware used and the parameterization information.
Hardware Configuration Editor	Window used to configure the hardware.
Hardware module	Also used for the internal representation of the real hardware. Generic term used for measuring modules and ECU interfaces.
Hardware parameters	Parameters that can be set for a specific hardware.

HEX file	Exchange format of a program version; the hex file (*.hex, *.s19; Intel hex or Motorola format) contains the control unit program consisting of the code and data. The contents of this file can be directly loaded into the control unit and executed by the respective processor.
HSP	Hardware Service Pack; firmware update for ETAS hardware
Intel Hex	Exchange format used for program versions.
K-Line	Serial calibration interface to the control unit.
LAB file	A LAB file contains a list of variables. This list can be used to apply an already created variable compilation in other areas (e.g. additional experiments, Calibration Scenario Configurations, etc.). The files end in .lab.
LDF	LDF is as part of the LIN specification a file format used to describe an entire LIN cluster. The LDF format contains information regarding LIN nodes, frames and signal values with physical conversions. INCA uses LDF files to provide the measurable signals, to calculate physical values and for the configuration of the hardware interface.
LIN	The LIN bus (Local Interconnect Network) is a small and slow network system that is used as a low-cost sub-network of a CAN bus to integrate intelligent sensor devices or actuators in today's cars. The LIN specification is enforced by the LIN-consortium, see http://www.lin-subbus.org/ .
Master dataset	Read-only dataset that was created by reading the first HEX file. The code portion of the HEX file is stored in the project.
MDF	Measurement Data Format; MDF is a binary file format used for storing measured data from sensors, ECU and bus monitoring for post measurement processing. In addition to the plain measurement data, MDF also contains descriptive and customizable meta data within the same file. The format is organized in loosely coupled binary blocks to ensure high performance reading and writing. The measurement data is stored channel oriented and uses master channels for synchronization which can be time, angle, distance or simply index related. In contrast to other measure data formats, MDF supports none-equidistant and multiple sample rates per file, as used in the automotive area.
Measure channel parameter	Parameters which can be set for each channel of a measuring module.

Measure Data Analyzer	A window which can be used to analyze measurements.
Measured data	Term used for the data recorded during a measurement.
Measure elements	User interface elements used to represent measured signals.
Measure variable	A variable to be measured.
Measure variable catalog	Internal representation of the measured signal description file.
Measurement	During a measurement, the variables that are part of an experiment get sampled and displayed. A measurement is required for recordings. Depending on the selected recording mode, the default recorder or all recorders can be used for recording measurements.
Measurement setup	Consists of several measurement systems.
Measure variable description file	File in A2L format in which a certain set of measure variables is described.
Measure window	INCA working window which displays measured signals during a measurement.
Measuring	Capturing of data which are either displayed or stored, or both displayed and stored.
Measuring device	A measuring device can be used to interconnect several measuring modules to form a card group. A measuring device can be defined as a collection of measuring modules which are combined physically and logically to form a unit.
Measuring system	Hardware which the user perceives as self-contained devices. Includes at least one hardware module.
Memory Page Manager	Dialog box used to transfer the data versions between the working base pages, the hard disk, and the main memory, as well as between the control unit and the PC.
Memory pages	The memory of a control unit containing calibration hardware usually consists of two "pages" (reference and working page), each of which contains the entire control unit program including the data. The user can switch between these two pages. Usually, the reference page is read-only and contains the "reference version" already saved. The working page can be edited (i.e., modified) by the user.
Motorola S-Record	Exchange format used for program versions.

Parameter	Calibration object. Parameters have a value that can be calibrated. INCA knows several types of calibrations, e.g. scalar, curve, map, array, ascii string, ...
Process point	Current state of the control unit described using the values of the internal control unit measured signals. This results in the value (in curves) which the control unit currently accesses.
Program	A program consists of code and data and is executed as a unit by the CPU of the control unit.
Program version	Hex file containing the control unit program.
Project	Database item usually consisting of the project description file (*.a2l) and a program file (e.g. *.hex, *.s19). The program file can contain both code and data. The project is created by reading the project description file (A2L file) and saved as a database item in the database. It contains the description of all management information relevant to the calibration (addresses, filing methods, etc.) of a control unit program.
Project description file	The project description file (*.a2l) contains the physical description of the data and/or parameters of the control unit program. It includes: structure information, address ranges, memory size, address ranges of each measure and calibration variables, names of measure and calibration variables.
Raster	The raster describes the interval during which a measurement is recorded. Depending on the device the number of measurements which can be recorded per raster may be limited. This information is defined in the ASAP2 file.
Recorder	A recorder is used to record a measurement. In a recorder, the recording parameters, such as variables, the trigger conditions (trigger) and the output file are defined. Any number of recorders can be used for each measurement.
Recording mode	The two recording modes determine which recorders are used for recording measurements. In the "Visualization" mode, all enabled (background) recorders get started. In the "Recording" mode, the default recorder gets started in addition to the enabled background recorders. In order to actually start the recording, the trigger condition of the corresponding recorder must be fulfilled.
Reference dataset	A write-protected dataset. The data included are used as a reference and are, therefore, called reference data.

Reference page	Memory page in the ECU containing a specific data version. The data included are used as a reference and cannot be modified.
Trigger	A trigger is a condition that when fulfilled starts or stops an action. Common examples of triggers in INCA are start and stop recording triggers. Triggers are created, edited and managed in the Trigger Editor.
User Interface Developer	Window used to edit and create user interfaces.
User profile	A set of user-specific option settings.
Variable	Generic term used for calibration variables (CHARACTERISTIC according to ASAP2) and measured signals (MEASUREMENT according to ASAP2).
Variable selection dialog	Dialog box used to select the measure and calibration variables.
Variable User interface	Variable user interfaces in which measure and calibration windows can be designed and arranged according to the user's own requirements.
Window elements	Generic term for calibration and display elements.
Working dataset	Non-write-protected dataset. The data included can be modified and are, therefore, also called working data.
Working page	Memory page in the ECU containing a specific data version. The data included can be modified and are, therefore, called working data.
Workspace	A workspace is a database item in which all information required for a specific measurement or calibration task is stored and consolidated.
XCP	eXtended Calibration Protocol; XCP is a vendor- and interface-independent protocol for communication between calibration tools and control units. XCP was specified in the ASAM e.V.
XETK	Application hardware similar to ETK for the ECU access via standard protocol XCP on Ethernet.

9 Appendix A: Troubleshooting INCA Problems

This chapter gives some information of what you can do when problems arise during your work with INCA.

9.1 Support Function for Feedback to ETAS in Case of System Errors

While developing INCA, the functional safety of the program was of utmost importance. Should an error occur nevertheless, please forward the following information to ETAS:

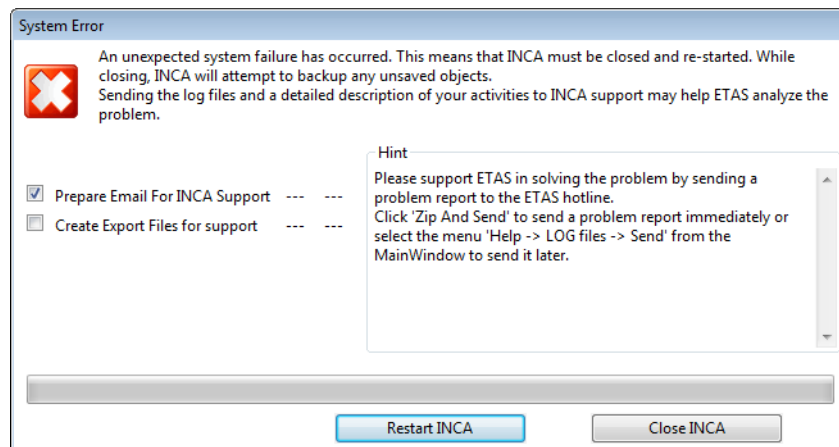
- Which step were you about to perform with INCA when the error occurred?
- What kind of error occurred (wrong function, system error or system crash)?

Note

It is important for the maintenance and further development of INCA that you inform ETAS about all errors occurring during a calibration. Please use the support function provided via the System Error window or via the INCA Help menu for this purpose.

9.1.1 Sending Problem Reports Using the System Error Window

If a critical system error occurs, the following window appears:



- **Checkbox Prepare Email For INCA Support**

If this checkbox is checked, an e-mail application will be launched and an e-mail opened. Enter a description of the observed problem and how it can be reproduced.

The log files saved by INCA, and optionally the export file, are automatically attached to the e-mail.

The e-mail is addressed to the INCA support address that is set in the INCA station options (set in the Database Manager under **Options → Station options → E-mail address for the ETAS hotline**). It will not be sent automatically; you must explicitly send it after having entered the required information.

- Checkbox **Create Export Files for Support**

This checkbox is disabled by default.

If this checkbox is checked, the active workspace including all referenced items (e.g. experiment, project and datasets) will be saved in binary format in an export file. This export file includes all log information that is required for analyzing the problem and will be added to the e-mail for the INCA support team. It will not be sent automatically.

Note

All information and data provided to ETAS will be treated as strictly confidential and will only be used internally by ETAS for solving the issue.

- Buttons **Restart INCA** and **Close INCA**

When you click one of these buttons, INCA prepares the e-mail to the INCA support team and creates export files for the support, if enabled. After that, INCA saves all unsaved changes in any database objects and creates backup items before closing INCA.

9.1.2 Sending Problem Reports Using Zip and Send

If you do not want to send a problem report immediately, you can also do that later.

To send a problem report using zip and send:

- Select the menu entry **? → Log Files → Zip**.

A dialog box opens in which you can enter a problem description. With the checkbox, you can select whether you want to send only log files or also an export of the current workspace including all referenced items (e.g. experiment, project, datasets).

INCA compresses the entire contents of the .log directory (all *.log files) including the textual description, plus optionally the INCA export file, into an archive file named

EtasLogFiles[01].zip in the
.\ETAS\LogFiles\ subdirectory.

For additional archive files, the file name is incremented automatically (up to [20]) to avoid that older archive files are immediately overwritten.

At the end of this process, you will be asked whether you want to send the ZIP archive to the INCA support address that is set in the INCA station options (set in the Database Manager under **Options → Station options → E-mail address for the ETAS hotline**).

- If you want to check the log files before sending them, you can select **? → Log Files → View**.

This command opens the INCA log directory in the Windows Explorer.

- If you want to send a previously prepared ZIP archive to the INCA support, select [?](#) → [Log Files](#) → [Send](#).

A new e-mail addressed to the INCA support is opened, the ZIP archive with the log files is already attached to the mail.

9.1.3 Using Extended Logging for the Reproduction of Problems

To be able to reproduce and analyze a reported problem, the ETAS support might need more detailed log files. However, as extended logging can reduce INCA's performance, it is disabled by default.

A description on how to enable extended logging is provided in a help file which you can open from the INCA Help menu in the Database Manager:

[?](#) → [Log files](#) → [Extended Logging](#).

Note

Only enable extended logging in INCA if this is requested by the ETAS Technical Support team.

9.2 General problems while working with INCA

9.2.1 INCA reacts only after substantial delays

Causes:

There is a large number of potential causes (e.g. virus scanner) which might slow down the INCA or system performance. These causes, as well as descriptions of how to minimize them, are described in detail in document [6]: "Performance Tips and Tricks", which you will find in the **Manuals** folder of your INCA installation.

9.2.2 Search for Ethernet Hardware fails in INCA

Cause: Add-On to support corresponding hardware not installed

The required hardware driver has probably not been installed if INCA is unable to detect hardware, although detected hardware is displayed in the task list of the IP manager.

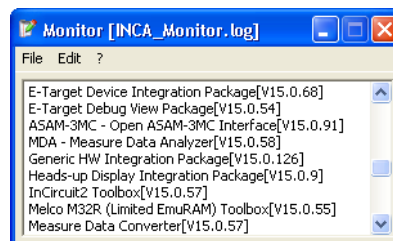
Some hardware modules are actually only supported in conjunction with certain INCA Add-Ons (e.g. the ES780 Heads Up Display with the HUD Integration add-on).

The same applies to interfaces that are only supported in conjunction with certain add-ons (e.g. the FlexRay interface of the ES520): those interfaces will only be found upon search for hardware if the corresponding add-on has been installed.

If you want to use these modules or interfaces with INCA, make sure that the Add-On was installed and registered on the used INCA version.

Check installation and licensing of INCA Add-Ons:

- In order to check if the Add-On is installed, select in the INCA Database Manager in the **?** menu the entry **Loaded packages**.
The installed packages are listed in the monitor window.
- Make sure that package entries with the name of the Add-On are included.



If this is not the case, the Add-On must still be installed.

- In order to check if the Add-On is licensed correctly, select in the INCA Database Manager **Tools** → **ETAS License Manager**.

- Make sure that a valid entry for the Add-On is listed in the license manager.

If this is not the case or if the status is “Not Available”, you will have to install the license.

More information on license management is available in the online help of the license manager.

9.3 Problems in the Experiment Environment

9.3.1 While using the Tool-API (COM-API) a wrong INCA-version is used

Cause: The Tool-API of another INCA version was registered last

If several INCA versions are installed on one PC the version installed last is always registered during the installation of the Tool-API (COM-API). This version will therefore be used with all available INCA installations.

In the registration you have to set manually the version of the Tool-API to be used.

Note

To be able to execute the following steps, you need read and write access to the registry folder HKEY_CLASSES_ROOT and all subfolders. Please contact your administrator if you do not have the required access rights.

Registering the used Tool-API version:

- Change in the INCA program directory of the desired INCA version to the `cebra` directory and double-click on the `reReg.bat` file.

Or

- Open a command window, change in the INCA program directory of the desired INCA version to the `cebra` directory and enter the following command:

```
regsrv32 incacom.dll
```

Note

To make sure that the expected INCA version is used while accessing INCA via the Tool-API, you can use the API method `GetCurrentVersion` to check the INCA version before continuing.

9.3.2 A warning message appears: "GDI Resources Exceeded"

Cause: Too many instruments are set up in the experiment

If the experiment contains too many instruments, it might not be possible to display all instruments, and a corresponding warning message will appear. You can optimize the usage of resources and thus avoid the problem in the following way:

1. Change measure windows and calibration windows into measure tables and scalar calibration tables.
2. Distribute the instruments to several layers. This saves resources because INCA does not have to display all instruments at once.
3. If the warning message "GDI Resources Exceeded" still appears after having reorganized the experiment and restarted INCA, you should remove individual instruments from the experiment until the message disappears.

9.4 Problems with Calculated Signals

- 9.4.1 While trying to select calculated signals, the error message appears: "Perl Interface: DllGetEtasDataPath() failed"
-

Cause: No write permission for the required registry folder

Calculated signals can be imported and exported by means of exchange files with the *.xcs extension. The perl interface is used for this purpose.

In order to use the perl interface correctly, you require read and write access to the HKEY_LOCAL_MACHINE\Software\ETAS registry folder.

10 Appendix B: Troubleshooting General Problems

This chapter gives some information of what you can do when problems arise that are not specific to an individual software or hardware product.

10.1 Problems and Solutions

10.1.1 Network Adapter cannot be selected via Network Manager

Cause: APIPA is disabled

The alternative mechanism for IP addressing (APIPA) is usually enabled on all Windows XP and Vista systems. Network security policies, however, may request the APIPA mechanism to be disabled. In this case, you cannot use a network adapter which is configured for DHCP to access ETAS hardware. The ETAS Network Manager displays a warning message.

The APIPA mechanism can be enabled by editing the Windows registry. This is permitted only to users who have administrator privileges. It should be done only in coordination with your network administrator.

To enable the APIPA mechanism:

- Open the Registry Editor:
 - Windows XP:
Click **Start** and then click **Run**. Enter `regedit` and click **OK**.
 - Windows Vista:
Click **Start**, enter `regedit` in the entry field, and push <ENTER>.

The registry editor is displayed.
- Open the folder `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\`
- Click **Edit** → **Find** to search for the key `IPAutoconfigurationEnabled`.

If you cannot find any instances of the registry key mentioned, the APIPA mechanism has not been disabled on your system, i.e. there is no need to enable it. Otherwise proceed with the following steps.

- Set the value of the key `IPAutoconfigurationEnabled` to 1 to enable the APIPA mechanism.
You may find several instances of this key in the Windows registry which either apply to the TCP/IP service in general or to a specific network adapter. You only need to change the value for the corresponding network adapter.
- Close the registry editor.
- Restart your workstation in order to make your changes take effect.

10.1.2 Search for Ethernet Hardware fails

There is a number of different causes which might lead to connection problems, most of them being based on inappropriate Windows or hardware settings. Usually, these can easily be modified, once they have been identified.

The following list of causes shall help you in finding the root cause of the problem and fixing it.

Cause: The versions of the Hardware and the ETAS MC Software are not compatible

If you are using ETAS hardware with ETAS MC software, you can use the ETAS HSP Update Tool to check the firmware version of your hardware:

- Make sure you use the ETAS HSP Update Tool with the latest HSP (Hardware Service Pack) version.
- Also use the HSP Update Tool to check whether the hardware is compatible with the MC software used.
- Make sure any additional drivers for that hardware are installed correctly.

You can get the required HSP from the ETAS internet pages under www.etas.com.

If you still cannot find the hardware using the HSP Update Tool, check whether the hardware offers a Web interface and whether you can find using this interface. Otherwise check whether one of the following causes and solutions might apply.

Cause: Personal Firewall blocks Communication

For a detailed description on problems caused by personal firewalls and possible solutions see "Personal Firewall blocks Communication" on page 166.

Cause: Client Software for Remote Access blocks Communication

PCs or notebooks which are used outside the ETAS hardware network sometimes use a client software for remote access which might block communication to the ETAS hardware. This can have the following causes:

- A firewall which is blocking Ethernet messages is being used (see „Cause: Personal Firewall blocks Communication“ on page 164)
- By mistake, the VPN client software used for tunneling filters messages. As an example, Cisco VPN clients with versions before V4.0.x in some cases erroneously filtered certain UDP broadcasts.

If this might be the case, please update the software of your VPN client.

Cause: ETAS Hardware hangs

Occasionally the ETAS hardware might hang. In this case switch the hardware off, then switch it on again to re-initialize it.

Cause: ETAS Hardware went into Sleep Mode

In order to save power, some ETAS devices will go to sleep mode if they do not see that they are connected to another device/computer.

To solve that, connect your Ethernet cable from your computer to the "HOST"/"Sync In" port on the device. After the device turns on, connect to the device using the web interface and change the settings so that the device stays always on. Consult the device's manual for details on how to do that.

Cause: Network Adapter temporarily has no IP Address

Whenever you switch from a DHCP company LAN to the ETAS hardware network, it takes at least 60 seconds until ETAS hardware can be found. This is caused by the operating system's switching from the DHCP protocol to APIPA, which is being used by the ETAS hardware.

Cause: ETAS Hardware had been connected to another Logical Network

If you use more than one PC or notebook for accessing the same ETAS hardware, the network adapters used must be configured to use the same logical network. If this is not possible, it is necessary to switch the ETAS hardware off and on again between different sessions (repowering).

Cause: Device driver for network card not in operation

It is possible that the device driver of a network card is not running. In this case you will have to deactivate and then reactivate the network card.

To deactivate and reactivate the network card:

- To deactivate the network card first select in the Windows start menu the following item:
 - Windows XP:
Control Panel → Network Connections
 - Windows Vista:
Control Panel → Network and Internet → Network and Sharing Center
- Right click on the used network adapter and select **Disable** in the context menu.
- In order to reactivate the network adapter right click on it again and select **Enable**.

Cause: Laptop power management deactivates the network card

The power management of a laptop computer can deactivate the network card. Therefore you should turn off power monitoring on the laptop.

To switch off power monitoring on the laptop:

- From the Windows Start Menu, select
 - Windows XP:
Control Panel → System.
Then select the **Hardware** tab and click on **Device Manager**.
 - Windows Vista:
Control Panel → System and Maintenance → Device Manager.
- In the Device Manager open the tree structure of the entry **Network Adapters**.

- Right click on the used network adapter and select **Properties** in the context menu.
- Select the **Power Management** tab and deactivate the **Allow the computer to turn off this device to save power** option.
- Select the **Advanced** tab. If the property **Auto-sense** is included, deactivate it also.
- Click **OK** to apply the settings.

Cause: Automatic disruption of network connection

It is possible after a certain period of time without data traffic that the network card automatically interrupts the Ethernet connection. This can be prevented by setting the registry key `autodisconnect`.

To set the registry key `autodisconnect`:

- Open the Registry Editor.
- Select under `HKEY_LOCAL_MACHINE\SYSTEM\ControlSet001\Services\lanmanserver\parameters` the Registry Key `autodisconnect` and change its value to `0xffffffff`.

10.1.3 Personal Firewall blocks Communication

Cause: Permissions given through the firewall block ETAS hardware

Personal firewalls may interfere with access to ETAS Ethernet hardware. The automatic search for hardware typically cannot find any Ethernet hardware at all, although the configuration parameters are correct.

Certain actions in ETAS products may lead to some trouble if the firewall is not properly parameterized, e.g. upon opening an experiment in ASCET or searching for hardware from within INCA or HSP.

If a firewall is blocking communication to ETAS hardware, you must either disable the firewall software while working with ETAS software, or the firewall must be configured to give the following permissions:

- Outgoing limited IP broadcasts via UDP (destination address 255.255.255.255) for destination ports 17099 or 18001
- Incoming limited IP broadcasts via UDP (destination IP 255.255.255.255, originating from source IP 0.0.0.0) for destination port 18001
- Directed IP broadcasts via UDP to the network configured for the ETAS application, destination ports 17099 or 18001
- Outgoing IP unicasts via UDP to any IP in network configured for the ETAS application, destination ports 17099 through 18020
- Incoming IP unicasts via UDP originating from any IP in the network configured for the ETAS application, source ports 17099 through 18020, destination ports 17099 through 18020

- Outgoing TCP/IP connections to the network configured for the ETAS application, destination ports 18001 through 18020

Note

The ports that have to be used in concrete use cases depend on the hardware used. For more precise information on the port numbers that can be used please refer to your hardware documentation.

The Windows operating systems come with a built-in personal firewall. On many other systems it is very common to have personal firewall software from third party vendors, such as Symantec, McAfee or BlackIce installed. The proceedings in configuring the ports might differ for each personal firewall software used. Therefore please refer to the user documentation of your personal firewall software for further details.

As an example for a firewall configuration, you will find below a description on how to configure the widely used Windows XP firewall if the hardware access is prohibited under Windows XP with Service Pack 2.

Solution using the Example of the Windows XP Firewall, Users with Administrator Privileges

Note

Please consult your IT responsible and/or check the IT security policies of your company before changing your firewall configuration and reconnecting the computer to the network!

If you have administrator privileges on your PC, the following dialog window opens if the firewall blocks an ETAS product.

**To unblock a product:**

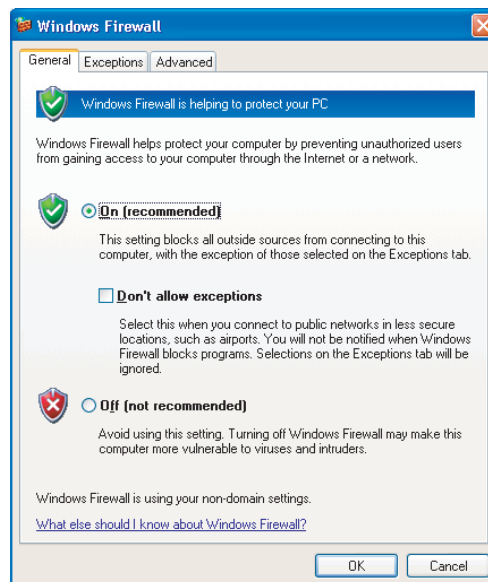
- In the "Windows Security Alert" dialog window, click on **Unblock**.

The firewall no longer blocks the ETAS product in question (in the example: ASCET). This decision survives a restart of the program, or even the PC.

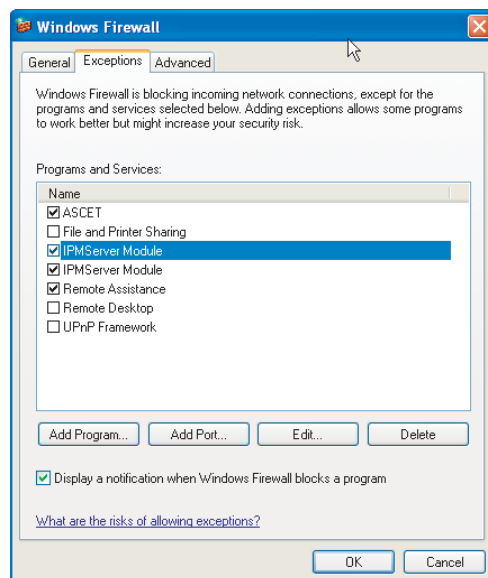
Instead of waiting for the "Windows Security Alert" dialog window, you can unblock ETAS products in advance.

To unblock ETAS products in the firewall control:

- From the Windows Start Menu, select **Settings** → **Control Panel**.
- In the control panel, double-click the **Windows Firewall** icon to open the "Windows Firewall" dialog window.

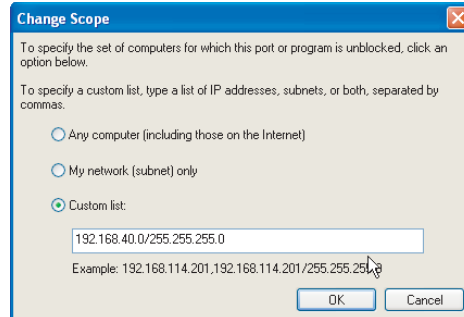


- In the "Windows Firewall" dialog window, open the "Exceptions" tab.



This tab lists the exceptions not blocked by the firewall. Use **Add Program** or **Edit** to add new programs, or edit existing ones.

- Make sure that the ETAS products and services you want to use are properly configured exceptions.
 - Open the "Change Setup" window.



- To ensure proper ETAS hardware access, make sure that at least the IP addresses 192.168.40.xxx are unblocked.
 - Close the "Change Setup" window with **OK**.
- Close the "Windows Firewall" dialog window with **OK**.

The firewall no longer blocks the ETAS product in question. This decision survives a restart of the PC.

Solution using the Example of the Windows XP Firewall, Users without Administrator Privileges

This section addresses users with restricted privileges, e.g., no system changes, write restrictions, local login.

Working with an ETAS software product requires "Write" and "Modify" privileges within the ETAS, ETASData, and ETAS temporary directories. Otherwise, an error message opens if the product is started, and a database is opened. In that case, no correct operation of the ETAS product is possible because the database file and some *.ini files are modified during operation.

The ETAS software has to be installed by an administrator anyway. It is recommended that the administrator assures that the ETAS program/processes are added to the list of the Windows XP firewall exceptions, and selected in that list, after the installation. If this is omitted, the following will happen:

- The "Windows Security Alert" window opens when one of the actions conflicting with a restrictive firewall configuration is executed.



To unblock a program (no Admin privileges):

- In the "Windows Security Alert" dialog window, activate the option **For this program, don't show this message again**.
- Click **OK** to close the window.

An administrator has to select the respective ETAS software in the "Exceptions" tab of the "Windows Firewall" dialog window to avoid further problems regarding hardware access with that ETAS product.

11 Appendix C: Reference Lists

The chapter titled "Reference Lists" contains information on directory structure and a list of all keyboard commands sorted by working windows.

11.1 Default Directories

11.1.1 Default – Storage Directories

- Calibration Log
ETASData\INCA7.1\CalibrationSessionLog
- Import and export files (*.exp)
ETASData\INCA7.1\Export
- Databases
ETASData\INCA7.1\Database
- HEX files (*.hex)
ETASData\INCA7.1\Data\Demo
- Project files (project description files, Hex files and ECU documentation files)
ETASData\INCA7.1\Data\Demo
- Protocol of configuration
ETASData\INCA7.1\Data\ConfigurationProtocol
- measure files (*.dat)
ETASData\INCA7.1\Measure
- Custom user interfaces (*.vui)
ETASData\INCA7.1\Data\Screen Layouts

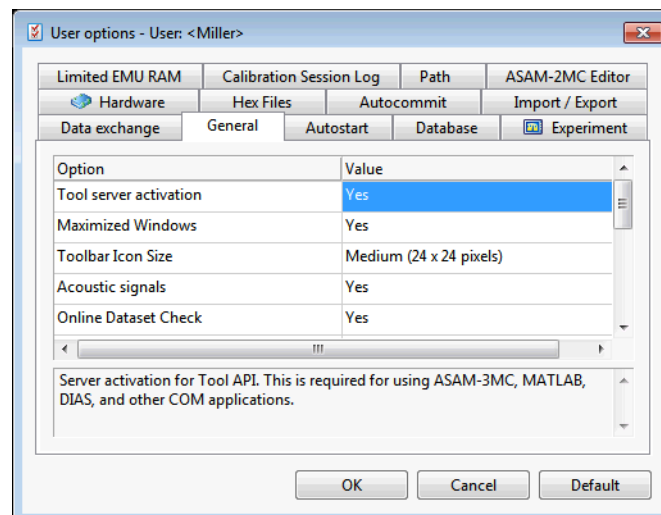
11.1.2 Change Storage Directories

You can use the **User Options** menu option to change the default storage directories of the various file types. Proceed as follows:

To change the default storage directories:

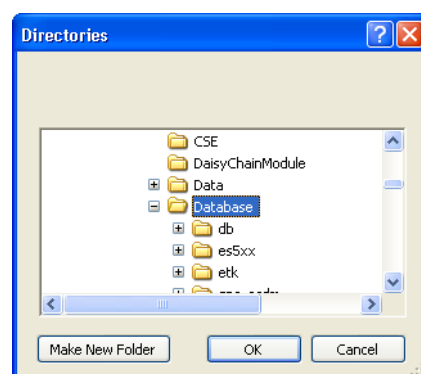
- In the INCA Database Manager select the menu command **Options → User options...**

The following dialog box appears:



- Select the **Path** tab if it is not displayed yet.
- In the table, click on the entry you want to change.

Another dialog box appears:



- Specify the directory that you want to use as the default directory for the selected data type.
- Click **OK**.

- Repeat these steps for each entry you want to change.

Note

The **New** button allows you to create new subdirectories to the selected top directory.

- When you are done, click **OK** in the first dialog box to accept the changes, or click **Cancel** if you do not want to accept the changes.

11.2 Operation Using the Keyboard

11.2.1 General Control Functions

Key	Function
Navigation	
<CTRL> + <F3>	Open Hardware Configuration Editor
<CTRL> + <F4>	Close the working window
<CTRL> + <F5>	Open Experiment Environment
<CTRL> + <F6>	Switch between open working windows
<CTRL> + <F7>	Switch Database Manager to foreground
<CTRL> + <F8>	Open UI Editor
<CTRL> + <F9>	Open ASAM-MCD2 Editor
<CTRL> + <F10>	Minimize/maximize current window
<CTRL> + <F11>	Open Calibration Data Manager
<CTRL> + <F12>	Open Measure Data Analyzer
Help	
<F1>	Invoke help
<SHIFT> + <F1>	Invoke "What's this" help
<CTRL> + <F1>	Show hotkey assignment
General Editing	
<F2>	Change to Edit mode (e.g., for table entries)
<SHIFT> + <F10>	Open shortcut menu for selected item (right mouse button)
<ALT>	Activate the main menu
<ALT> + <F4>	Close active working window; if the DBM is active, INCA exits.
<ALT> + <F6>	Switch between open windows
<ALT> + <SPACE>	Open system menu of application window
<ALT> + <TAB>	Switch between open applications
<ALT> + <->	Open system menu of document window
<CTRL> + <A>	Select all items (e.g., in a list)
<CTRL> + <C>	Copy to clipboard
<CTRL> + <V>	Paste from clipboard
<CTRL> + <X>	Cut and put on clipboard
<CTRL> + <Y>	Redo last action

Key	Function
<CTRL> + <Z>	Undo last action
<DOWN ARROW> (↓) <UP ARROW> (↑) <LEFT ARROW> (←) <RIGHT ARROW> (→)	Move cursor to table item or list item with arrow keys, <DOWN ARROW> also opens active list box
<ENTER>	Confirm input and quit input mode; expand or collapse branches
	Delete selected item
<ESC>	Cancel input, discard changes
<SPACE>	Select table or list item or deselect active selection
<TAB>	Move focus to next item (option) in a window (<SHIFT> + <TAB> moves focus in opposite direction)
<SHIFT>	Enable range selection, i.e., pressing the arrow keys while holding <SHIFT> key down allows selecting a table cell range.

11.2.2 Keyboard Commands in the Database Manager

The keyboard commands listed below merely supplement the commands already mentioned under "General Control Functions" on page 174.

Key	Function
<F2>	Rename selected item
	Change experiment (only when workspace has been selected and focus is on " <u>4</u> Experiment" field)
	Change project/dataset (only when workspace has been selected and focus is on " <u>5</u> Project/device" field)
	Open HWC (only when workspace has been selected and focus is on " <u>6</u> Hardware" field)
<F5>	Updates the database contents.
<SHIFT> + <F3>	Search for connected hardware (only when workspace has been selected and focus is on " <u>5</u> Project/device" field)
<CTRL> + 	Write HEX file (only when dataset has been selected)
<CTRL> + <D>	Read HEX file (only when ECU project has been selected)
<CTRL> + <E>	Activate the export function
<CTRL> + <G>	Set focus to active workspace
<CTRL> + <K>	Set selected workspace active
<CTRL> + <M>	Activate the import function
<CTRL> + <N>	Create new database

Key	Function
<CTRL> + <O>	Open an existing database or create a new database
<CTRL> + <P>	Create an ECU project (load A2L file)
<CTRL> + <S>	Save database
<CTRL> + <T>	Create new experiment
<CTRL> + <W>	Create new workspace
<INS>	Add project/dataset (only when workspace has been selected and focus is on "Project" item in "5 Project/device" field)
	Change hardware module (only when workspace has been selected and focus is on "Device" item in "5 Project/device" field)
	Add a new folder to the selected folder

11.2.3 Keyboard Commands in the Hardware Configuration Editor

The keyboard commands listed below merely supplement the commands already mentioned under "General Control Functions" on page 174.

Key	Function
<F3>	Initialize hardware
<SHIFT> + <F3>	Perform the automatic hardware detection
<SHIFT> + <F8>	Open Memory Page Manager
<ALT> + <F3>	Retrieve hardware status
<ALT> + <F8>	Change project and working dataset
<ALT> + <CTRL> + <U>	Enable/disable device group (alternatively also <ALTGR> + <U>)
<CTRL> + <D>	Read ECU ID (only for KWP2000)
<CTRL> + <U>	Group device or ungroup
<CTRL> + <SHIFT> + <UP ARROW>	Move selected hardware module up in list
<CTRL> + <SHIFT> + <DOWN ARROW>	Move selected hardware module down in list
<INS>	Insert hardware module
	Delete hardware module

11.2.4 Keyboard Commands in the Experiment Environment

The keyboard commands listed below merely supplement the commands already mentioned under "General Control Functions" on page 174.

Key	Function
<F3>	Initialize hardware
<F4>	Open dialog box to configure the experiment
<F5>	Release start trigger manually
<F8>	Switch between reference and working page
<F9>	1. Stop measure data visualization / measure recording 2. Release stop trigger manually
<F11>	Start measure data visualization
<F12>	1. Start measure data visualization / 2. measure recording
<SHIFT> + <F4>	Open variable selection dialog (also <INS>)
<SHIFT> + <F8>	Open Memory Page Manager
<SHIFT> + <PAUSE>	1. Stop measure data visualization (recording continues) 2. Start measure data visualization
<ALT> + <F1>	Show variable documentation (focus on variable)
<ALT> + <F3>	Retrieve hardware status
<ALT> + <G>	Activate list box with hardware modules
<ALT> + <T>	Configure layers
<ALT> + <LESS THAN>	Jump to layer on the left
<ALT> + <GREATER THAN>	Jump to layer on the right
<CTRL> + <K>	Define, during measured data recording, a label in the measured data file to which a comment can be attached
<CTRL> + <L>	Enable/disable calibration access to ECU
<CTRL> + <M>	Open Measure Data Converter
<CTRL> + <N>	Create new experiment
<CTRL> + <O>	Open an existing experiment
<CTRL> + <Q>	Configure measured data recording
<CTRL> + <S>	Save an experiment
<CTRL> + <T>	Activate device group
<PAUSE>	Stop measure recording (display continues)

Measure and Calibration Windows in General

The keyboard commands listed below equally apply to all measure and calibration windows. For specific keyboard commands for measure and calibration windows, please see further below.

Key	Function
<ALT> + <ENTER>	Retrieve display settings for the active window
<ALT> + <F9>	Move selected variable in a window one position down
<ALT> + <F10>	Move selected variable in a window one position up
<CTRL> + 	Display binary values in the active window
<CTRL> + <D>	Display decimal values in the active window
<CTRL> + <G>	Shows/hides the display grid
<CTRL> + <H>	Display hexadecimal values in the active window
<CTRL> + <I>	View information for selected variable
<CTRL> + <P>	Display physical values in the active window
<INS>	Insert variable in active window
<CTRL> + 	Remove variable from active window
<TAB>	Advance to next display field
<ARROW DOWN>	Select next variable in a window
<ARROW UP>	Select previous variable in a window

Calibration Windows

The keyboard commands listed below equally apply to all calibration windows and simply complement the commands listed above under “Measure and Calibration Windows in General”. For keyboard commands for specific calibration windows, please see further below.

Key	Function
<F6>	Decrement
<F7>	Increment
<SHIFT> + <F6>	Coarse decrement
<SHIFT> + <F7>	Coarse increment
<ALT> + <I>	Increment Y-axis value (Table Editor)
<ALT> + <J>	Decrement X-axis value (Table Editor or Curve Editor)
<ALT> + <K>	Increment X-axis value (Table Editor or Curve Editor)
<ALT> + <U>	Decrement Y-axis value (Table Editor)
<CTRL> + <Y>	Redo last action
<CTRL> + <Z>	Undo last action
<CTRL> + <W>	Show process point in Calibration Editor

Key	Function
<+>	Calculate total offset
<*>	Multiply with a factor
<=>	Enter value

The keyboard commands listed below are only available when the **Table Editor** is selected:

Key	Function
<CTRL> + <E>	Assign specific X-coordinate value to a point.
<CTRL> + <G>	Shift the current process point into the visible window area of the table editor
<CTRL> + <R>	Assign specific Y-coordinate value to a point.
<CTRL> + <U>	Reset Complete Map / Complete Curve to value of reference page
<CTRL> + <ALT> + <U>	Reset Selected Area of the Map / Curve to value of reference page

Measure Window

The keyboard commands listed below merely supplement the commands already mentioned under "Measure and Calibration Windows in General" on page 178.

The following keyboard commands are only available when a **YT Oscilloscope** or a **XY Oscilloscope** is selected:

Key	Function
<ALT> + <Q>	Change measure rate
<CTRL> + <Alt> + <Z>	Zoom to fit: match Y-axis scaling to current variables
<Ctrl> + <T>	Distribute selected variables to individual strips
<Ctrl> + <E>	Move selected variables to new strip
<Ctrl> + <ALT> + <T>	Use common strip for selected variables
<ALT> + <+> + <+>	Optimize window size
<Ctrl> + <R>	Add cursor
	Remove selected cursor
<Ctrl> + <ALT> + <R>	Remove all cursors
<PAUSE>	Continue/pause continuous display in the oscilloscope
<Ctrl> + <+>	Zoom in (on time axis)
<Ctrl> + <->	Zoom out (on time axis)

Key	Function
<Ctrl> + <ALT> + <P>	Display value in physical representation
<Ctrl> + <ALT> + <H>	Display value in hexadecimal representation
<Ctrl> + <ALT> + <D>	Display value in decimal representation
<Ctrl> + <ALT> + 	Display value in binary representation
<Ctrl> + <ALT> + <L>	Show/hide signal list
<Ctrl> + <W>	Show/hide selected signals in graphical display
<UP ARROW>	Scrolls up on the selected Y-axis
<DOWN ARROW>	Scrolls down on the selected Y-axis
<LEFT ARROW>	Scrolls back on the time axis (only in pause mode)
<RIGHT ARROW>	Scrolls forward on the time axis (only in pause mode)
<CTRL> + <LEFT ARROW>	Move left split bar to the left
<CTRL> + <RIGHT ARROW>	Move the left split bar to the right

The following keyboard commands are only available when the **Oscilloscope Window (outdated)**, **XY-Oscilloscope (outdated)** or the **XY-Plotter (outdated)** is selected:

Key	Function
<ALT> + 	Shows a list of signals used in the current window
<Ctrl> + <D>	Distribute the selected measure variables to separate display areas (for oscilloscope (outdated) only)
<ALT> + <J>	Match Y-axis scaling to current measure variables
<ALT> + <L>	Match Y-axis scaling to all measure variables
<ALT> + <Q>	Change acquisition rate
<ALT> + <Z>	Toggles the display of the sampling point markers on/off
<CTRL> + <LEFT ARROW>	Move left split bar to the left (Oscilloscope (outdated) and XY-Plotter (outdated) only)
<CTRL> + <RIGHT ARROW>	Move the left split bar to the right (Oscilloscope (outdated) and XY-Plotter (outdated) only)
<CTRL> + <R>	Enable/disable analysis mode (Oscilloscope (outdated) and XY-Plotter (outdated) only)

Key	Function
<CTRL> + <U>	Show/hide values in measure variable list (Oscilloscope (outdated) and XY-Plotter (outdated) only)
<CTRL> + <W>	Show/hide selected variable (oscilloscope (outdated) only)
<CTRL> + <Z>	Undo last scaling command

11.2.5 Keyboard Commands in the Memory Page Manager

Key	Function
<ALT> + 	Edit working dataset only
<ALT> + <CTRL> + <F>	Freeze working dataset (alternatively also <ALTGR> + <F>)
<CTRL> + <A>	Save working dataset under
<CTRL> + <R>	Edit reference dataset only

11.2.6 Keyboard Commands in the Variable Selection Dialog

Key	Function
<CTRL> + <A>	Select all variables in the data list (focus in the data list)
<CTRL> + <F>	Search for variables, rasters or functions (variables: focus in data list; rasters and functions: focus in sources list)
<CTRL> + <R>	Activate filter for calibration variables
<CTRL> + <L>	Select online adjustable variables
<CTRL> + <M>	Activate filter for measure variables
<CTRL> + <I>	Open information window for selected variable
<CTRL> + <N>	Add layer (focus in Display Configuration)
<F2>	Rename layer (focus on layer in Display Configuration)
	Delete selected item (focus in Display Configuration; only possible if item was added in current session)
<INS>	Add window (focus in Display Configuration)

11.2.7 Keyboard Commands in the Trigger Editor

Key	Function
<CTRL> + <A>	Select complete formula (focus in formula)
<CTRL> + <L>	Start calculator
<INS>	Insert variable (by means of Variable Selection Dialog)
<CTRL> + <R>	Replace variable (by means of Variable Selection Dialog; focus in formula)
<CTRL> + <I>	Add selected trigger definition to current formula
<CTRL> + <D>	Add selection to Snippets toolbox
<STRG> + <N>	Create new trigger
	Delete selected trigger(s)
<ENTER>	Edit selected trigger
<CTRL> + <M>	Import trigger
<CTRL> + <E>	Export all triggers
<CTRL> + <W>	Export selected triggers
<CTRL> + <T>	Set focus to trigger list
<CTRL> + 	Set focus to toolbox

11.2.8 Keyboard Commands in the Calibration Scenario Editor

The calibration scenario editors support the same keyboard commands as calibration windows in general and the table editor in particular (see „Calibration Windows“ on page 178). The keyboard commands listed below merely supplement those commands.

Key	Function
<INS>	Add variables (by means of Variable Selection Dialog)
	Delete variables in current scenario
<CTRL> + 	Delete selected variables completely in the scenario configuration
<CTRL> + <I>	Open information window for selected variable
<CTRL> + <9>	Toggle between currently active and previously active calibration scenario
<SHIFT> + <DOWN ARROW> (↓) / <UP ARROW> (↑) / <LEFT ARROW> (←) / <RIGHT ARROW> (→)	Extends / reduces the marked area

Key	Function
<SHIFT> + <CTRL> + <C>	Create new calibration history entry
<SHIFT> + <CTRL> + <E>	Edit calibration history entry
<SHIFT> + <CTRL> + 	Open calibration history

11.2.9 Keyboard Commands in the Manage Recorders Dialog

Key	Function
<CTRL> + <N>	Create new recorder (focus on recorder)
	Delete recorder (focus on recorder)
<INS>	Insert variables (by means of Variable Selection Dialog) (focus on recorder)
<CTRL> + <F>	Find a recorder by name (focus on recorder)
<CTRL> + <R>	Show/hide variables that are assigned to the recorder (focus on recorder)
<ALT> + <L>	Open the Recorder Event Viewer (focus on recorder)
<ENTER>	Open Recorder Configuration (focus on recorder)
<CTRL> + <Z>	Undelete recorder (focus on recorder)
<F2>	Rename recorder (focus on recorder)
<CTRL> + <C>	Copy recorder (focus on recorder)
<CTRL> + <V>	Paste recorder (focus on recorder)
<ALT> + <ENTER>	Open Recorder Manager Properties dialog
	Delete variable (focus on variable)
<INS>	Insert variables (by means of Variable Selection Dialog) (focus on variable)
<CTRL> + <C>	Copy variable (focus on variable)
<CTRL> + <V>	Paste variable (focus on variable)
<CTRL> + <I>	Open information window for selected variable (focus on variable)

11.3 Sounds for INCA Events

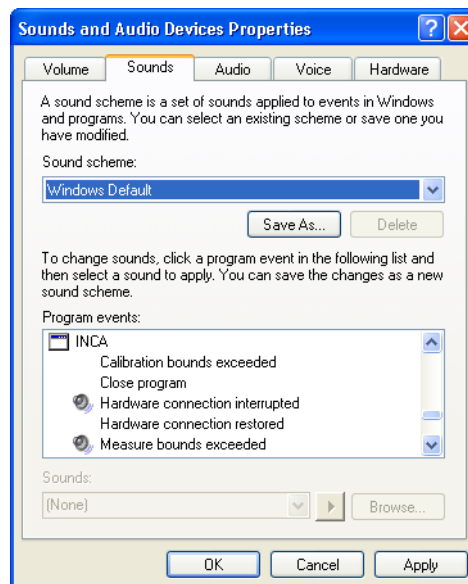
INCA can play sound notifications in case of certain events, e.g. when the hardware connection is disrupted or hard limits of measurement variables are exceeded during measurement.

Sounds can be configured for the following event types:

- Calibration bounds exceeded
- Close program
- Hardware connection interrupted
- Hardware connection restored
- Measure bounds exceeded
- Open program
- Recording finished
- Recording started
- Trigger occurred

To configure sounds for selected events:

1. Go to the Windows Control Panel.
2. Double-click on the entry **Sounds and Audio Devices**.
3. In the dialog box **Sounds and Audio Devices Properties**, go to the **Sounds** tab.
4. In the **Program events** list box, scroll down to the INCA items.



5. Select an event type you want to assign a sound to.

6. Select a sound from the **Sounds** list box.

Note

*You can also use customized sounds. To assign a customized sound clip, click the **Browse** button and select the corresponding wave file (*.wav).*

7. The event is marked with an icon  indicating that a sound is assigned.
8. Click **OK** to close the dialog and save the assignment of sounds.

12 Appendix D: Further Reading

Unless otherwise stated, the following additional documents are provided with the basic INCA installation and can be found in one of the INCA folders `Manual` or `Help`.

Further documents might be provided with INCA add-on products.

12.1 Documentation for Standard Users

- [1] INCA online help (available through INCA)
`help\incaEnglishUS.chm`¹
- [2] INCA Getting Started
`INCA_Manual_V7.1_EN.pdf`
- [3] INCA Tutorial
`INCA_Tutorial_V7.1_EN.pdf`
- [4] INCA Video Tutorials²
- [5] INCA Hot Key Assignment
`INCA_Keyboard_Layout_R7.1_EN.pdf`
- [6] INCA V7 Performance Tips and Tricks
`INCA_Performance_TipsAndTricks_EN.pdf`

12.2 Documentation for Special Use Cases

- [7] CDM Background — Functionality and Restrictions³
- [8] Serial ETK Calibration Concepts — Limited ETK Emulation RAM Toolbox for INCA
`INCA_LimitedEmuRAM_EN.pdf`
- [9] Serial Calibration with InCircuit2
`INCA_InCircuit2_EN.pdf`

12.3 Documentation for Tool Integration

- [10] INCA ASAM-ASAP3 Interface
`INCA_IF_ASAM-ASAP3_EN.pdf`
- [11] INCA ASAM-MCD-3MC V1.0.1 Interface
`INCA_IF_ASAM-MCD-3MC_V1.0.1_EN.pdf`
- [12] INCA ASAM-MCD-3MC V2.2 Interface
`INCA_IF_ASAM-MCD-3MC_V2.2_EN.pdf`
- [13] INCA Tool-API Documentation (online help)
`cebra\INCA Tool-API Documentation.chm`⁴

¹ The INCA online help is automatically installed together with INCA and can be accessed via the INCA **?** menu or by pushing F1.

² The INCA video tutorials can be installed together with INCA (optional) or viewed in the ETAS YouTube channel. You can access an overview and the videos themselves via the INCA **?** menu.

³ This document can be obtained from ETAS on request.

⁴ The Tool API documentation is automatically installed together with the Tool API component and can be accessed by double-clicking the help file.

12.4 Documentation for Suppliers

[14] ECU Document Interface (EDI) for INCA (PDF)¹

12.5 Specifications

[15] CVX (Calibration Values Exchange)²

[16] CDF (Calibration Data Format)³

¹ This document can be obtained from the ETAS download center (type: technical documentation)

² This document can be obtained from the ETAS download center (type: specification)

³ The CDF specification is available for download on the web pages of the ASAM Association for Standardization of Automation and Measuring Systems under www.asam.net

13 ETAS Contact Addresses

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ETAS Subsidiaries and Technical Support

For details of your local sales office as well as your local technical support team and product hotlines, take a look at the ETAS website:

ETAS subsidiaries WWW: www.etas.com/en/contact.php

ETAS technical support WWW: www.etas.com/en/hotlines.php

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